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MEET-COBRA: Machine-learning-based Energy-Efficient proacTive Coordination of handOver, Beamforming and Resource Allocation

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Authors

Mr. Bowen Xie
Submitting Author

 [ORCID](https://orcid.org/0009-0004-7332-8530)
<https://orcid.org/0009-0004-7332-8530>

Affiliations

- the Beijing National Research Center for Information Science and Technology, Department of Electronic Engineering, Tsinghua University, Beijing 100084, China

Prof. Sheng Zhou
Corresponding Author

Affiliations

- the Beijing National Research Center for Information Science and Technology, Department of Electronic Engineering, Tsinghua University, Beijing 100084, China

Prof. Zhisheng Niu

Affiliations

- the Beijing National Research Center for Information Science and Technology, Department of Electronic Engineering, Tsinghua University, Beijing 100084, China

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2. B. Xie, S. Chen, S. Zhou, Z. Niu, B. Galkin, and I. Dusparic, "Learningassisted user scheduling and beamforming for mmwave vehicular networks," IEEE Transactions on Vehicular Technology, vol. 73, no. 8, pp. 11262–11275, 2024.
3. H.-S. Park, H. Kim, C. Lee, and H. Lee, "Mobility management paradigm shift: from reactive to proactive handover using AI/ML," IEEE Netw., 2024.

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3. Y. Liu, W. Chen, and J. Lee, "Joint queue-aware and channel-aware scheduling for non-orthogonal multiple access," IEEE Transactions on Wireless Communications, vol. 21, no. 1, pp. 264–279, 2022.

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MEET-COBRA: Machine-learning-based Energy-Efficient proactive Coordination of handOver, Beamforming and Resource Allocation

Bowen Xie, Sheng Zhou, and Zhisheng Niu

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Abstract—Energy-efficient coordination of handover, beamforming and resource allocation (COBRA) in heterogeneous networks (HetNets) remains underexplored, particularly for high-mobility vehicular scenarios under latency constraints. We formulate a stochastic integer non-linear program (SINLP) for COBRA to minimize long-term system transmit power subject to a queue-length-based latency constraint. However, the NP-hardness of the resulting SINLP, together with the high overhead of acquiring timely and accurate per-link channel state information (CSI), makes online global COBRA optimization computationally prohibitive. To address this challenge, we propose the MEET-COBRA framework, which uses machine learning (ML) to predict next-frame beam-pair candidates and channel gains from low-overhead historical CSI measurements, and translates these predictions into per-link resource block (RB) demand estimates for globally optimized proactive handover. Meanwhile, each cell performs prediction-guided beamforming and queue-aware resource allocation to keep the realized link efficiencies consistent with the RB-demand estimates, thereby meeting the latency constraint with low transmit power. Simulation results show that, at a medium-to-high traffic load, MEET-COBRA lowers the average transmit power by about 49% and reduces the latency constraint violation probability from 9.84% to 0.16% compared with the Reactive-OBRA baseline, with only lightweight ML inference overhead.

Index Terms—Heterogeneous networks, proactive handover, beamforming, resource allocation, machine learning, energy efficiency.

I. INTRODUCTION

As modern vehicles evolve into intelligent mobile terminals, their efficient communication with infrastructure, other vehicles, and pedestrians becomes essential, which leads to the emergence of connected autonomous vehicles (CAVs) and Vehicle-to-Everything (V2X) communications [1]. CAV applications demand high throughput for perception and infotainment and low latency for driving safety. These communication quality of service (QoS) requirements are increasingly considered in 3GPP enhanced V2X service specifications and the broader new radio (NR)-V2X ecosystem [2], [3].

In practice, heterogeneous network (HetNet) deployments integrating macro and micro base stations (BSs) across different frequency bands provide a promising evolutionary path for NR-V2X systems to accommodate these requirements [4]. Typically, macro BSs operate in lower-frequency bands to ensure broad and reliable coverage, whereas micro BSs utilize higher-frequency bands to enhance network capacity [5]. Such HetNet deployments can also reduce system transmit power

through traffic-aware resource allocation. Realizing these benefits requires balancing transmit power against timely traffic service. Increasing resource usage can reduce delay but also raise power consumption [6].

High mobility makes this balance more difficult. Fast-moving vehicles trigger frequent handovers (HOs), increasing signaling overhead and degrading link performance [7], [8]. Conventional reactive HO mechanisms based on thresholds and time-to-trigger further incur preparation and interruption latency and increase the risk of HO failures and ping-pong events [9]. Meanwhile, beam training and tracking become more error-prone under rapid topology changes and intermittent blockage, leading to excessive alignment overhead and reliability loss, especially near cell edges [10], [11].

Recent studies have mainly explored three directions. First, proactive handover (PHO) prepares HO in advance by forecasting traffic flows, vehicle trajectories, or channel states, thereby reducing failures and ping-pong events [9], [12]. With machine learning (ML), historical channel state information (CSI) can be leveraged to predict future positions [13], channel conditions [14], and target BSs [15], which further enhances PHO. Second, ML-assisted beamforming (BF) exploits side information such as out-of-band CSI [11] and context information [16] to infer blockage and narrow beam search space while maintaining robustness and throughput [17]. Reinforcement learning can also adapt beam training and tracking policies [18]. Third, queue-aware resource allocation (RA) prioritizes users by urgency or deadlines and opportunistically harnesses favorable channels to improve throughput and robustness under power budgets [19], [20].

Several studies also coordinate two or more of these decisions. Khosravi et al. [21] use deep Q-learning to coordinate HO and beam tracking, accounting for beam-training overhead in the effective throughput. Wang et al. [22] combine multi-agent reinforcement learning for HO triggering with optimization of target BS and beam selection and bandwidth allocation to improve throughput and delay performance. Heydarian et al. [23] coordinate user association, hybrid BF, and queue-aware user scheduling on multiple timescales. However, these studies primarily target throughput and delay performance, and do not explicitly address the joint coordination of PHO, BF, and RA for reducing the system transmit power while meeting latency requirements.

PHO and BF jointly determine the serving link gain and beam-training overhead. In HetNets, these quantities also depend on cross-tier differences in carrier frequencies, antenna configurations, and propagation conditions. Together with differences in bandwidth and transmit power across tiers, these factors affect the number of resource blocks (RBs)

The authors are with the Beijing National Research Center for Information Science and Technology, Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: xbw22@mails.tsinghua.edu.cn; {sheng.zhou, niuzhs}@tsinghua.edu.cn).

required to serve a given traffic demand and the resulting power consumption, while the available RB budgets constrain service feasibility. Moreover, HO changes the traffic loads of the source and target BSs, which can alter their resource usage and intra-tier co-channel interference, thereby affecting the resource demands of other links. This motivates energy-efficient Coordination of handOver, Beamforming and Resource Allocation (COBRA) in vehicular HetNets.

This paper extends our earlier conference work on joint PHO and RA [13]. Unlike the conference version, this paper incorporates MIMO channels, beam-pair selection, and beam-training overhead into the joint optimization. We also change the ML prediction targets from vehicle positions to beam-pair indices and link gains. Moreover, the simulation dataset uses ray-traced microcell MIMO channels instead of the statistical channel model adopted in the conference version.

We consider a two-tier vehicular HetNet, and formulate a COBRA optimization problem to minimize the long-term system transmit power subject to a queue-length-based latency constraint. This problem is a two-timescale stochastic integer non-linear program (SINLP). The frame-level PHO decisions are tightly coupled with slot-level BF and RA; the channel states and traffic arrivals are time-varying and only causally observable; and the decision space grows exponentially with the numbers of BSs, users, RBs, and BF codebook sizes. The NP-hardness of the resulting SINLP, together with the high overhead of acquiring timely and accurate per-link CSI, makes online global COBRA optimization computationally prohibitive.

To address these challenges, we develop MEET-COBRA, a prediction-driven coordination framework that jointly optimizes PHO, BF, and RA via a two-timescale closed loop. The key idea is to leverage low-overhead historical CSI measurements to predict next-frame beam-pair candidates and channel gains, and translate these predictions into per-link RB-demand estimates for globally optimized frame-level PHO. At the slot level, each BS performs prediction-guided BF and queue-aware RA to keep the realized link efficiencies consistent with these RB-demand estimates, thereby meeting the latency constraint with low transmit power.

To evaluate the proposed MEET-COBRA framework, we build a simulation platform that combines realistic vehicular mobility traces generated by SUMO [24] with ray-tracing-based channel data produced by Sionna RT [25]. Simulation results show that MEET-COBRA effectively reduces system transmit power and latency constraint violations across different traffic loads.

The rest of the paper is organized as follows: Section II details the system model; Section III formulates the optimization problem; Section IV presents the MEET-COBRA framework; Section V reports simulation results; and Section VI concludes the paper. The main notation is summarized in Table I.

II. SYSTEM MODEL

We consider a vehicular communication scenario covered by a two-tier HetNet adopting a control-data-plane separation architecture. The network consists of one macro BS and

TABLE I
MAIN NOTATION

Symbol	Description
m, v	Indices of BSs and vehicles
n, x, i	Indices of slots, frames, and within-frame slots
$\mathcal{M}, \mathcal{V}$	Sets of BSs and active vehicles
$N_{\text{HO}}, \Delta t$	Number of slots per frame and slot duration (s)
$\tau_{\text{HO}}, L_{\text{HO}}$	HO interruption duration (s) and number of HO interruption slots
δ_v, s_v	Binary indicators of HO execution and data-service availability
$c_{m,v}$	HO-aware RB-demand correction factor
K_m, B_m, p_m	Number of available RBs, bandwidth per RB (Hz), and transmit power per RB (W)
$u_{m,v}$	Binary indicator of BS-vehicle association
$k_{m,v}$	Number of allocated RBs
$\mathbf{H}_{m,v}$	MIMO channel matrix
$M_{\text{T}}, M_{\text{R}}$	Numbers of transmit/receive antennas
$\mathcal{C}_{\text{T}}, \mathcal{C}_{\text{R}}$	Transmit/receive beam codebooks
$\mathbf{w}_{\text{T},m,v}, \mathbf{w}_{\text{R},m,v}$	Transmit precoding/receive combining vectors
$g_{m,v}, g_{\text{I},m',v,v'}$	Instantaneous desired-/interfering-link gains
$G_{m,v}, G_{\text{I},m,v}$	Frame-level desired-/interfering-link gains
$\tilde{\mathcal{P}}_{m,v}$	Predicted probability distribution over beam pairs
N_{P}	Number of pilots per frame
M_{P}	Number of candidate beam pairs
η	Beam-search overhead factor
q_v, a_v	Amounts of queued and newly arriving data (bit)
λ_v	Mean data arrival rate (bps)
r_v	Effective receive rate (bps)
τ_v^{th}	Latency threshold (s)
U	Probability of queue-length constraint violation
$\bar{P}$	Time-averaged system transmit power (W)

Time indices are omitted for brevity. Symbols with tildes denote predicted or estimated quantities, which are not listed separately when their unmarked counterparts are included.

M micro BSs. The macro BS serves as the control-plane anchor and makes frame-level PHO decisions, while the micro BSs execute the corresponding data-plane re-association and perform slot-level BF and RA. The BS set is denoted by $\mathcal{M} = \{0, 1, \dots, M\}$, where index $m = 0$ represents the macro BS and $m > 0$ represents a micro BS. BS- m is allocated K_m orthogonal RBs in the frequency band f_m , where each RB has a bandwidth of B_m and a per-RB transmit power of p_m . For simplicity, all micro BSs are assumed to have identical frequency configurations, i.e., $K_m = K_\mu, f_m = f_\mu$ and $B_m = B_\mu$ for all $m > 0$, while the macro BS has distinct parameters K_0, f_0, B_0 .

We model the system in discrete time, indexed by slots $n \in \{0, 1, \dots\}$ of duration Δt . At any time slot n , the set of vehicles is $\mathcal{V}(n)$, with cardinality $V(n) = |\mathcal{V}(n)|$. The data-plane connection between BS- m and vehicle- $v \in \mathcal{V}(n)$ is represented by a binary indicator $u_{m,v}(n)$, where $u_{m,v}(n) = 1$ signifies the selected data-plane association and $u_{m,v}(n) = 0$ otherwise. Each vehicle can be associated with at most one BS on the data plane at a time, which imposes the constraint:

$$\sum_{m=0}^M u_{m,v}(n) \leq 1, \quad \forall v \in \mathcal{V}(n). \quad (1)$$

Note that the control-plane connection to the macro BS is

always maintained and thus is not explicitly modeled.

The data-plane connection state $u_{m,v}(n)$ is managed by a central PHO server deployed at the macro BS and updated periodically over a time window of N_{HO} slots, which define a PHO frame. The handover configuration for the x -th frame is determined during the preceding $(x-1)$ -th frame and remains static throughout frame x . Therefore, we introduce a frame-based index. Let $\{\cdot\}^{[x]} \triangleq \{\cdot\}(xN_{\text{HO}})$ denote a variable at the start of the x -th frame, and let $\{\cdot\}^{[x]}(i) \triangleq \{\cdot\}(xN_{\text{HO}} + i)$ be the variable at the i -th slot within that frame, where $i \in \mathcal{N}_{\text{HO}} \triangleq \{0, 1, \dots, N_{\text{HO}} - 1\}$. Consequently, the connection state is constant within any frame, which is expressed as:

$$u_{m,v}^{[x]}(i) = u_{m,v}^{[x]}, \quad \forall i \in \mathcal{N}_{\text{HO}}. \quad (2)$$

We further assume the set of vehicles is static throughout each frame, denoted as $\mathcal{V}^{[x]}$.

After a PHO decision is made, the signaling exchanges required for HO preparation are assumed to finish before the current frame ends, without occupying data-service resources. A vehicle switching BS is assumed to experience a data-service interruption of duration $\tau_{\text{HO}} = L_{\text{HO}}\Delta t$ caused by HO execution at the start of the next frame, where L_{HO} is a nonnegative integer less than N_{HO} . We assume successful HO execution for all PHO decisions. The macro control-plane connection and periodic CSI probing remain available during this data-plane HO interruption.

Let $\delta_v^{[x]} = 1$ if vehicle- v changes its serving BS from frame- $(x-1)$ to frame- x , and $\delta_v^{[x]} = 0$ otherwise. Its slot-level data-service availability is

$$s_v^{[x]}(i) = 1 - \delta_v^{[x]} \mathbf{1}\{i < L_{\text{HO}}\}, \quad (3)$$

where $\mathbf{1}\{\cdot\}$ is the indicator function.

A. Channel Model

We specify the channel models used to calculate the instantaneous desired-link gains $g_{m,v}^{[x]}(i)$ for both macro-to-vehicle ($m = 0$) and micro-to-vehicle ($m > 0$) links, as well as the interfering-link gains when applicable.

1) *Channel model for macro BS:* The macro BS operates in a lower-frequency band (e.g., the microwave band) to provide wide-area coverage. The frame-averaged channel gain between the macro BS and vehicle- v during the x -th frame, denoted as $G_{0,v}^{[x]}$, is modeled as

$$G_{0,v}^{[x]} = \alpha \left(d_{0,v}^{[x]}\right)^{-\beta} 10^{S_v^{[x]}/10}, \quad (4)$$

where α is a power gain factor, β is the path loss exponent, and $d_{m,v}^{[x]}$ is the distance between BS- m and vehicle- v in the x -th frame. The term $S_v^{[x]}$ captures the shadow fading effect, which is modeled as an independent and identically distributed (i.i.d.) zero-mean Gaussian random variable with variance σ_S^2 .

Based on the frame-averaged channel gain $G_{0,v}^{[x]}$, the instantaneous channel gain $g_{0,v}^{[x]}(i)$ in the i -th time slot of the x -th frame is modeled by Rician fading as

$$g_{0,v}^{[x]}(i) = G_{0,v}^{[x]} \left\| \sqrt{\frac{K_R}{K_R + 1}} + \sqrt{\frac{1}{K_R + 1}} h_{0,v}^{[x]}(i) \right\|^2, \quad (5)$$

where K_R is the Rician K-factor and $h_{m,v}^{[x]}(i) \sim \mathcal{CN}(0, 1)$ is an i.i.d. standard complex Gaussian random variable.

2) *Channel model for micro BS:* Each micro BS operates in a higher-frequency band (e.g., the mmWave band) and is equipped with a uniform linear array (ULA) of M_T antennas, while each vehicle has M_R antennas at this band. For the micro-to-vehicle links, we adopt a geometric stochastic MIMO channel model [26]. The channel between micro BS- m and vehicle- v during the x -th frame is represented by the frame-averaged channel matrix $\mathbf{H}_{m,v}^{[x]} \in \mathbb{C}^{M_R \times M_T}$, which is modeled as a sum of L distinct propagation paths:

$$\mathbf{H}_{m,v}^{[x]} = \sqrt{M_T M_R} \sum_{l=1}^L \gamma_l \mathbf{a}_R(\theta_l) \mathbf{a}_T(\psi_l)^\dagger, \quad (6)$$

where γ_l , θ_l , and ψ_l denote the complex impulse response, angle of arrival (AoA), and angle of departure (AoD) of the l -th path, respectively. The vectors $\mathbf{a}_R(\cdot) \in \mathbb{C}^{M_R}$ and $\mathbf{a}_T(\cdot) \in \mathbb{C}^{M_T}$ are the receive and transmit array response vectors. For the ULAs, the array response vector is given by

$$[\mathbf{a}(\phi)]_i = \frac{1}{\sqrt{N}} e^{-j \frac{2\pi d}{\lambda} i \sin \phi}, \quad i = 0, \dots, N-1, \quad (7)$$

where N is the number of antennas (M_T or M_R), d is the inter-element spacing (typically $d = \lambda/2$), λ is the carrier wavelength, and ϕ is the AoA or AoD (θ_l or ψ_l). The instantaneous channel matrix in the i -th slot of frame- x is modeled by applying small-scale Rician fading as

$$\mathbf{H}_{m,v}^{[x]}(i) = \left(\sqrt{\frac{K_R}{K_R + 1}} + \sqrt{\frac{1}{K_R + 1}} h_{m,v}^{[x]}(i) \right) \cdot \mathbf{H}_{m,v}^{[x]}. \quad (8)$$

Beamforming is employed for the MIMO links by selecting precoding and combining vectors from predefined codebooks. Specifically, we use discrete Fourier transform (DFT) codebooks. A DFT codebook of size N consists of N orthogonal vectors, $\mathcal{C} = \{\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{N-1}\}$, where each codeword $\mathbf{c}_n \in \mathbb{C}^{N \times 1}$ corresponds to a spatial direction. The i -th element of the n -th codeword is

$$[\mathbf{c}_n]_i = \frac{1}{\sqrt{N}} e^{-j \frac{2\pi n i}{N}}, \quad \forall n, i \in \{0, 1, \dots, N-1\}. \quad (9)$$

The transmit codebook $\mathcal{C}_T$ is a DFT codebook of size M_T , and the receive codebook $\mathcal{C}_R$ is a DFT codebook of size M_R .

When micro BS- m serves vehicle- v in the x -th frame, it selects a transmit precoding vector $\mathbf{w}_{T,m,v}^{[x]}(i) \in \mathcal{C}_T$, and vehicle- v selects a receive combining vector $\mathbf{w}_{R,m,v}^{[x]}(i) \in \mathcal{C}_R$. The resulting instantaneous desired-link channel gain is

$$g_{m,v}^{[x]}(i) = \left| \left(\mathbf{w}_{R,m,v}^{[x]}(i) \right)^\dagger \mathbf{H}_{m,v}^{[x]}(i) \mathbf{w}_{T,m,v}^{[x]}(i) \right|^2. \quad (10)$$

When another micro BS- m' serves vehicle- v' , the interfering-link gain from BS- m' to vehicle- v is

$$g_{I,m',v,v'}^{[x]}(i) = \left| \left(\mathbf{w}_{R,m',v'}^{[x]}(i) \right)^\dagger \mathbf{H}_{m',v}^{[x]}(i) \mathbf{w}_{T,m',v'}^{[x]}(i) \right|^2. \quad (11)$$

B. Resource Allocation Model

Let $k_{m,v}^{[x]}(i) \in \{0, 1, \dots, K_m\}$ denote the number of RBs allocated by BS- m to vehicle- v in the i -th time slot of the x -th frame. Since RBs can only be allocated to connected users without HO interruption, this variable must satisfy

$$k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} s_v^{[x]}(i) k_{m,v}^{[x]}(i), \quad \forall i \in \mathcal{N}_{\text{HO}}. \quad (12)$$

Each BS employs orthogonal frequency-division multiple access (OFDMA). Accordingly, the number of RBs utilized by BS- m , $k_m^{[x]}(i)$, is constrained by its total RB capacity K_m :

$$k_m^{[x]}(i) \triangleq \sum_{v \in \mathcal{V}^{[x]}} k_{m,v}^{[x]}(i) \leq K_m. \quad (13)$$

The instantaneous transmit power of BS- m is then given by

$$P_m^{[x]}(i) = p_m k_m^{[x]}(i). \quad (14)$$

The time-averaged system transmit power is the sum of the average power of all BSs:

$$\begin{aligned} \bar{P} &= \sum_{m=0}^M \bar{P}_m = \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{x=0}^{X-1} \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} \sum_{m=0}^M P_m^{[x]}(i) \\ &= \lim_{X \rightarrow \infty} \frac{1}{X N_{\text{HO}}} \sum_{x=0}^{X-1} \sum_{i \in \mathcal{N}_{\text{HO}}} \sum_{v \in \mathcal{V}^{[x]}} \sum_{m=0}^M p_m k_{m,v}^{[x]}(i). \end{aligned} \quad (15)$$

For a link with $k_{m,v}^{[x]}(i) > 0$, the signal-to-interference-plus-noise ratio (SINR) is modeled as

$$\text{SINR}_{m,v}^{[x]}(i) = \frac{p_{m,v}^{[x]}(i) g_{m,v}^{[x]}(i)}{F_m N_0 W_{m,v}^{[x]}(i) + I_{m,v}^{[x]}(i)}, \quad (16)$$

where $p_{m,v}^{[x]}(i) = k_{m,v}^{[x]}(i) p_m$ is the power allocated to the vehicle, $W_{m,v}^{[x]}(i) = k_{m,v}^{[x]}(i) B_m$ is the allocated bandwidth, N_0 is the noise power spectral density, F_m is the noise figure, and $I_{m,v}^{[x]}(i)$ is the inter-cell interference. For the macro BS, $I_{0,v}^{[x]}(i) = 0$ since its frequency band is orthogonal to those of micro BSs. Given the allocated RB counts, each micro BS assigns mutually disjoint RB sets to its associated vehicles uniformly at random, independently of the other micro BSs. The expected number of overlapping RBs between links (m, v) and (m', v') , with $m' \neq m$, is therefore $k_{m,v}^{[x]}(i) k_{m',v'}^{[x]}(i) / K_\mu$. Accordingly, the interference power averaged over the random RB assignments is given by

$$\begin{aligned} I_{m,v}^{[x]}(i) &= \begin{cases} \frac{k_{m,v}^{[x]}(i)}{K_\mu} \sum_{\substack{m' \neq 0, m \\ v' \in \mathcal{V}^{[x]}}} p_{m'} k_{m',v'}^{[x]}(i) g_{1,m',v,v'}^{[x]}(i), & m > 0, \\ 0, & m = 0. \end{cases} \end{aligned} \quad (17)$$

Under this average interference approximation, the link SINR is approximated as

$$\begin{aligned} \text{SINR}_{m,v}^{[x]}(i) &= \begin{cases} \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m + \sum_{\substack{m' \neq 0, m \\ v' \in \mathcal{V}^{[x]}}} \frac{p_{m'} k_{m',v'}^{[x]}(i)}{K_\mu} g_{1,m',v,v'}^{[x]}(i)}, & m > 0, \\ \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m}, & m = 0. \end{cases} \end{aligned} \quad (18)$$

The corresponding link rate is modeled using the Shannon formula as

$$R_{m,v}^{[x]}(i) = W_{m,v}^{[x]}(i) \log_2 \left(1 + \text{SINR}_{m,v}^{[x]}(i) \right). \quad (19)$$

When $k_{m,v}^{[x]}(i) = 0$, no data are transmitted on the link, and thus $R_{m,v}^{[x]}(i) = 0$. Considering the beam-searching overhead, the instantaneous receive rate of vehicle- v is given by

$$r_v^{[x]}(i) = \sum_{m=0}^M (1 - \eta_{m,v}^{[x]}(i)) R_{m,v}^{[x]}(i), \quad (20)$$

where $\eta_{m,v}^{[x]}(i) \in [0, 1)$ represents the throughput loss factor resulting from beam alignment. Specifically,

$$\eta_{m,v}^{[x]}(i) = \eta N_{m,v}^{[x]}(i), \quad (21)$$

with $N_{m,v}^{[x]}(i)$ denoting the number of beam-search attempts conducted by BS- m and vehicle- v in the i -th slot of frame- x , and η denoting the fraction of OFDM symbols in a slot required to probe one beam pair. Besides, the frame-averaged receive rate of vehicle- v is

$$\bar{r}_v^{[x]} \triangleq \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} r_v^{[x]}(i), \quad (22)$$

C. Traffic Model

We model the data traffic for each vehicle- v with a backlog queue, where $q_v(n)$ denotes its length at the beginning of time slot n . The queue length evolves as

$$q_v(n+1) = \max \{q_v(n) - r_v(n) \Delta t, 0\} + a_v(n), \quad (23)$$

where $a_v(n)$ is the amount of data arriving for vehicle- v during time slot- n . This model assumes that the existing backlog $q_v(n)$ is served first at a rate of $r_v(n)$, and new arrivals $a_v(n)$ are enqueued for service in subsequent slots. We denote the average data arrival rate for vehicle- v as λ_v , thus we have $\mathbb{E}[a_v(n)] = \lambda_v \Delta t$. We consider Poisson arrivals that are independent across vehicles and slots, with $a_v(n) \sim \text{Poisson}(\lambda_v \Delta t)$. This model does not capture temporally correlated traffic bursts.

Queueing delay is an important component of end-to-end latency, alongside transmission, propagation, and processing delays. We focus on queueing delay to assess how HO, BF, and RA jointly affect traffic accumulation through the available service rates. Motivated by *Little's law* [27], we define $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. Requiring this

proxy not to exceed the latency threshold τ_v^{th} of vehicle- v gives the queue-length-based latency constraint:

$$q_v(n) \leq \lambda_v \tau_v^{\text{th}}. \quad (24)$$

To evaluate the system performance with respect to this latency constraint, we define the latency constraint violation probability U following the approach in [28] as:

$$U = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \frac{\sum_{v \in \mathcal{V}(n)} U_v(n)}{V(n)}, \quad (25)$$

where $U_v(n) = 1$ if $q_v(n) > \lambda_v \tau_v^{\text{th}}$, and $U_v(n) = 0$ otherwise. Specifically, U measures the probability that the queueing-delay proxy $\tau_v(n)$ exceeds its threshold τ_v^{th} .

III. PROBLEM FORMULATION

Our goal is to minimize the long-term system transmit power while satisfying the latency-constrained QoS requirements for all vehicles. This joint optimization of HO, BF and RA over different time scales is formulated as the following problem

$$\begin{aligned} \mathbf{P1}: \quad & \min_{\mathbf{u}, \mathbf{k}, \mathbf{W}} \bar{P} \\ \text{s.t.} \quad & \text{C1: } q_v^{[x]}(i) \leq \lambda_v \tau_v^{\text{th}}, \\ & \text{C2: } \sum_{m=0}^M u_{m,v}^{[x]} \leq 1, \\ & \text{C3: } \sum_{v \in \mathcal{V}^{[x]}} k_{m,v}^{[x]}(i) \leq K_m, \\ & \text{C4: } k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} s_v^{[x]}(i) k_{m,v}^{[x]}(i), \\ & \text{C5: } u_{m,v}^{[x]} \in \{0, 1\}, \\ & \text{C6: } k_{m,v}^{[x]}(i) \in \{0, 1, \dots, K_m\}, \\ & \text{C7: } \mathbf{w}_{T,m,v}^{[x]}(i) \in \mathcal{C}_T, \mathbf{w}_{R,m,v}^{[x]}(i) \in \mathcal{C}_R. \end{aligned} \quad (26)$$

In **P1**, the decision variables are the sets of data-plane connection states $\mathbf{u} \triangleq \{u_{m,v}^{[x]}\}$, RB allocation numbers $\mathbf{k} \triangleq \{k_{m,v}^{[x]}(i)\}$, and BF vectors $\mathbf{W} \triangleq \{\mathbf{w}_{T,m,v}^{[x]}(i), \mathbf{w}_{R,m,v}^{[x]}(i)\}$. In our system, $\mathbf{u}$ is decided per frame, while $\mathbf{k}$ and $\mathbf{W}$ are decided per time slot. They are subject to several constraints. C1 is the queue-length-based latency constraint equivalent to Eq.(24). The HO constraints C2 and C5 enforce that each vehicle connects to at most one BS in the data-plane. The RA constraints C3, C4, and C6 limit the total RBs per BS, ensure RBs are only assigned to connected vehicles without HO interruption, and define the integer nature of $k_{m,v}^{[x]}(i)$. Finally, C7 restricts the BF vectors to be chosen from the predefined finite codebooks, which makes the BF decision a combinatorial integer optimization component.

P1 is a two-timescale SINLP. Obtaining its optimal *online* solution is computationally prohibitive due to the following challenges:

1. **Stochastic and causally observed dynamics:** Future system states, including the vehicle set $\mathcal{V}^{[x]}$, the channel matrices $\mathbf{H}_{m,v}^{[x]}(i)$, and the traffic arrivals $a_v(n)$, are time-varying and only causally observable. Moreover, current

decisions have lasting impacts on future power consumption and latency performance. Therefore, the long-horizon online optimization of **P1** is fundamentally challenging.

2. **Combinatorial decision space and NP-hardness:** All decision variables in **P1** are discrete and jointly define a high-dimensional decision space over BS-vehicle association, beam-pair selection, and RB allocation, further complicated by inter-cell interference coupling through the SINR expressions. Consequently, **P1** is an NP-hard integer nonconvex optimization problem whose feasible set grows exponentially with the numbers of BSs, vehicles, RBs, and BF codebook sizes, rendering exhaustive search or exact global optimization impractical.
3. **Coupled two-timescale optimization:** Frame-level HO decisions determine the association topology and constrain the slot-level BF and RA decisions within that frame, while the chosen BF vectors directly affect the desired-link gains, thereby influencing RB allocation and queue evolution. Such cross-timescale dependencies imply that optimizing PHO, BF, and RA in isolation can lead to persistently suboptimal performance, and call for low-complexity joint coordination mechanisms.
4. **Prohibitive CSI measurement overhead:** An online global optimizer would require timely and accurate per-link CSI for all BS-vehicle links and beam-pair options to evaluate channel gains and interference. In high-mobility vehicular scenarios, acquiring such per-link CSI via frequent channel measurements, beam training, and feedback incurs excessive overhead and latency, thus impractical for realization.

In summary, the NP-hardness of **P1**, together with stochastic dynamics, two-timescale coupling, and the prohibitive overhead of real-time per-link CSI acquisition, makes the online global optimization for **P1** computationally prohibitive.

IV. ALGORITHM DESIGN

To solve **P1** with an efficient and computationally feasible online approach, we propose a novel algorithm framework named Machine-learning-based Energy-Efficient proactive COBRA (MEET-COBRA). It leverages ML-predicted beam-pair candidates and channel gains to enable proactive, two-timescale COBRA optimization. The central PHO server determines the next-frame PHO configuration globally, while each BS performs slot-level BF and RA locally to serve its associated vehicles. The following subsections introduce the ML-based prediction module, the *predictive, frame-level* PHO algorithm, the *real-time, slot-level* BF and RA algorithms, and the overall MEET-COBRA procedure.

A. ML-Based Prediction

A neural network (NN) model is developed to predict, for each vehicle, the next-frame best beam-pair indices, their corresponding desired-link gains, and interfering-link gains to all micro BSs. To reduce CSI feedback overhead and avoid centralized NN inference latency that grows with the number of vehicles, each vehicle performs NN inference locally based on its historical CSI measurements obtained from

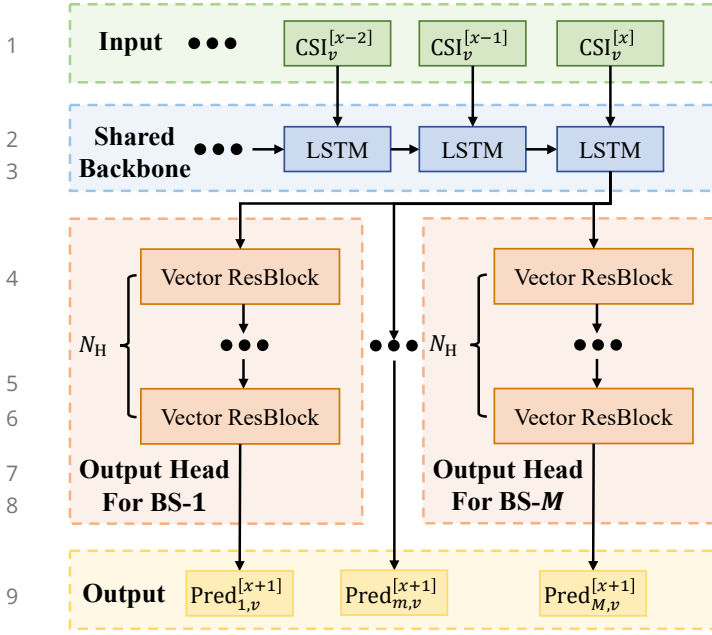


Fig. 1. Neural network architecture for predicting the next-frame best beam-pair index, its corresponding desired-link gain, and the interfering-link gain.

a low-overhead pilot probing procedure, and reports only the prediction results to the macro BS. The site-specific NN model is shared among vehicles by the macro BS over the control channel. Deployment in a different propagation environment or BS layout requires retraining the prediction models using channel data from the target site. As shown in Fig. 1, the NN consists of an input preprocessing layer, a shared feature-extraction backbone, and M micro-BS-specific output heads.

1) *Input Layer*: The NN input is a sequence of historical CSI measurements of vehicle- v . In each frame, the CSI is measured by a low-overhead pilot probing mechanism based on *pilot superposition*. Specifically, at the beginning of frame- x , all micro BSs simultaneously transmit a series of N_P pilot beams uniformly selected from the transmit codebook $\mathcal{C}_T$. In the p -th pilot probing ($p = 1, \dots, N_P$), the received signal at vehicle- v is the superposition of pilots from all micro BSs, i.e.,

$$\mathbf{y}_v^{[x]}(p) = \sum_{m=1}^M \mathbf{H}_{m,v}^{[x]} \mathbf{w}_{T,m}(p) s(p) + \mathbf{n}_v^{[x]}(p), \quad (27)$$

where $\mathbf{w}_{T,m}(p) \in \mathcal{C}_T$ is the transmit beam used by micro BS- m in the p -th probing, $s(p)$ is the pilot symbol, and $\mathbf{n}_v^{[x]}(p)$ is the received noise. Compared with exhaustive per-BS per-beam pilot sweeping with overhead $O(MM_T)$, pilot superposition reduces the measurement overhead to $O(N_P)$. Accordingly, the CSI measured by vehicle- v in frame- x is defined as

$$\mathbf{CSI}_v^{[x]} \triangleq [\mathbf{y}_v^{[x]}(1)^T, \mathbf{y}_v^{[x]}(2)^T, \dots, \mathbf{y}_v^{[x]}(N_P)^T]^T \in \mathbb{C}^{M_R N_P}, \quad (28)$$

where $\mathbf{y}_v^{[x]}(p) \in \mathbb{C}^{M_R}$ is given in (27). Before being fed into the NN, $\mathbf{CSI}_v^{[x]}$ is converted to a real-valued feature vector by normalizing amplitudes (in dB) and phases, and concatenating

them as $\mathbf{z}_{in,v}^{[x]} \in \mathbb{R}^{2M_R N_P}$. This preprocessing operation is denoted as f_{prep} :

$$\mathbf{z}_{in,v}^{[x]} = f_{\text{prep}}(\mathbf{CSI}_v^{[x]}). \quad (29)$$

2) *Shared Backbone*: The shared backbone processes the historical CSI sequence of each vehicle to extract a compact representation of its mobility pattern and surrounding radio environment. It consists of a Long Short-Term Memory (LSTM) unit [29]. The LSTM unit, with N_C hidden cells, captures the temporal dependencies in the CSI by maintaining a hidden state $\mathbf{h}_v \in \mathbb{R}^{N_C}$ and a cell state $\mathbf{c}_v \in \mathbb{R}^{N_C}$. In frame- x , the LSTM unit updates these states based on the current CSI feature vector $\mathbf{z}_{in,v}^{[x]}$ and the previous states ($\mathbf{h}_v^{[x-1]}, \mathbf{c}_v^{[x-1]}$):

$$(\mathbf{h}_v^{[x]}, \mathbf{c}_v^{[x]}) = f_{\text{LSTM}}(\mathbf{z}_{in,v}^{[x]}, \mathbf{h}_v^{[x-1]}, \mathbf{c}_v^{[x-1]}; \boldsymbol{\theta}_{\text{LSTM}}), \quad (30)$$

where f_{LSTM} denotes the LSTM transformations and $\boldsymbol{\theta}_{\text{LSTM}}$ denotes the corresponding weights. The states are retained across frames for each vehicle, so online inference processes only the current CSI feature vector.

3) *Per-BS Output Heads*: The backbone hidden state $\mathbf{h}_v^{[x]}$ is fed into M micro-BS-specific output heads, one for each micro BS:

$$\text{Pred}_{m,v}^{[x+1]} = f_{\text{hd}}(\mathbf{h}_v^{[x]}; \boldsymbol{\theta}_{\text{hd},m}), \quad \forall m \in \{1, 2, \dots, M\}, \quad (31)$$

where f_{hd} represents the common head structure, $\boldsymbol{\theta}_{\text{hd},m}$ denotes the weights of the m -th head, and $\text{Pred}_{m,v}^{[x+1]}$ is the prediction for the link between micro BS- m and vehicle- v in frame- $(x+1)$. Each head comprises N_H stacked vector ResBlock units [30] with feature dimension N_C , followed by a task-specific final layer. For beam-pair prediction, the final layer has dimension $M_T M_R$, whereas for channel-gain prediction it has dimension 1.

4) *Output Prediction*: Beam-pair and channel-gain prediction are implemented using independently trained NN models with a common backbone architecture and task-specific output layers:

1. **Beam Prediction Model**: Beam-pair prediction is formulated as a multi-class classification task. The final layer of each output head employs a softmax activation function to generate a predicted probability distribution $\tilde{\mathcal{P}}_{m,v}^{[x+1]}$ over the $M_T M_R$ beam pairs. The model is trained with a cross-entropy loss, using a one-hot label whose non-zero entry indicates the index of the optimal beam pair that maximizes the frame-averaged desired-link gain in frame $(x+1)$.

2. **Gain Prediction Models**: Channel-gain prediction is formulated as a regression task. We train two separate NN models with the same architecture but independent weights, one for the optimal desired-link gain and the other for the beam-averaged interfering-link gain. Each model has one output head per micro BS. In the desired-link gain model, each head outputs a scalar $\tilde{G}_{m,v}^{[x+1]}$ to approximate the maximum frame-averaged desired-link gain

$$G_{\text{opt},m,v}^{[x+1]} \triangleq \max_{\mathbf{w}_{T,m,v}^{[x+1]}, \mathbf{w}_{R,m,v}^{[x+1]}} G_{m,v}^{[x+1]}, \quad (32)$$

where $G_{m,v}^{[x+1]}$ is obtained by averaging the instantaneous desired-link gains in Eq. (10) over all slots in frame $(x+1)$. The interfering-link gain in (11) depends on the receive beam of the desired link and the transmit beam used by the interfering BS. To obtain a prediction target independent of these beam selections, we adopt the beam-averaged surrogate

$$\tilde{g}_{1,m',v}^{[x]}(i) \triangleq \frac{1}{M_T M_R} \|\mathbf{H}_{m',v}^{[x]}(i)\|_F^2, \quad (33)$$

which equals the effective gain uniformly averaged over all beam pairs in the transmit and receive DFT codebooks. Each head of the interfering-link gain model outputs $\tilde{G}_{1,m',v}^{[x+1]}$, with training label $G_{1,m',v}^{[x+1]}$ defined as the frame average of $\tilde{g}_{1,m',v}^{[x+1]}(i)$. Both models are trained using a mean squared error (MSE) loss function, with predictions and labels represented in normalized dB.

For the macro BS ($m = 0$), whose low-frequency link does not involve beam alignment or inter-cell interference, only the frame-averaged desired-link gain $\tilde{G}_{0,v}^{[x+1]}$ is required. Owing to the slow variation of the macro-channel, this gain is approximated by the channel gain estimated from control-plane measurements in the current frame, without NN-based prediction.

B. Generalized Assignment-based Proactive Handover

This subsection proposes a generalized assignment-based proactive handover (GAP-HO) algorithm, detailed in Alg. 1, which determines the data-plane connection state for the upcoming frame. Its primary objective is to minimize the per-frame HetNet transmit power while satisfying both the latency-constrained QoS requirements of vehicles and the RB-capacity constraints of BSs.

To ensure queue stability, the expected frame-averaged receive rate $\mathbb{E}[\bar{r}_v^{[x]}]$ should be no less than the data arrival rate λ_v . The following theorem establishes a lower bound on the frame-averaged number of RBs required to satisfy this condition in the interference-free case.

Theorem 1. Consider vehicle- v associated with BS- m in the x -th frame. Given $G_{m,v}^{[x]}$, λ_v , B_m , p_m , F_m and N_0 , the frame-averaged number of allocated RBs, denoted by $\bar{k}_{m,v}^{[x]}$, required to ensure $\mathbb{E}[\bar{r}_v^{[x]}] \geq \lambda_v$ satisfies

$$\bar{k}_{m,v}^{[x]} \geq \lambda_v \left[B_m \log_2 \left(1 + \frac{p_m G_{m,v}^{[x]}}{F_m N_0 B_m} \right) \right]^{-1}, \quad (34)$$

where $\bar{k}_{m,v}^{[x]} \triangleq N_{\text{HO}}^{-1} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i)$.

Proof. Since vehicle- v is associated with BS- m in frame- x , we have $u_{m,v}^{[x]} = 1$ and $u_{m',v}^{[x]} = 0$ for all $m' \neq m$. Combining (12), (19), (20), and (22), the frame-averaged receive rate is

$$\bar{r}_v^{[x]} = \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i) B_m \log_2 \left(1 + \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m} \right). \quad (35)$$

For any constant $C > 0$, the function $f(z) = \log_2(1 + Cz)$ is concave on $z > 0$. By Jensen's inequality,

$$\mathbb{E} \left[\log_2 \left(1 + \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m} \right) \right] \leq \log_2 \left(1 + \frac{p_m \mathbb{E}[g_{m,v}^{[x]}(i)]}{F_m N_0 B_m} \right). \quad (36)$$

Moreover, by (10), $\mathbb{E}[g_{m,v}^{[x]}(i)] = G_{m,v}^{[x]}$. Taking expectation of $\bar{r}_v^{[x]}$ and using (36) yield

$$\begin{aligned} \mathbb{E}[\bar{r}_v^{[x]}] &= \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i) B_m \mathbb{E} \left[\log_2 \left(1 + \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m} \right) \right] \\ &\leq \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i) B_m \log_2 \left(1 + \frac{p_m G_{m,v}^{[x]}}{F_m N_0 B_m} \right) \\ &= \bar{k}_{m,v}^{[x]} B_m \log_2 \left(1 + \frac{p_m G_{m,v}^{[x]}}{F_m N_0 B_m} \right). \end{aligned} \quad (37)$$

Combining with $\mathbb{E}[\bar{r}_v^{[x]}] \geq \lambda_v$, we obtain the lower bound in (34). This completes the proof. $\square$

Based on Theorem 1, we now construct frame-averaged RB-demand estimates for each BS-vehicle pair- (m, v) in frame- $(x+1)$. For the macro BS ($m = 0$), whose low-frequency link is free of inter-cell interference and beam-alignment overhead, the required frame-averaged number of RBs for vehicle- v is estimated as

$$\tilde{k}_{0,v}^{[x+1]} = \lambda_v \left[B_0 \log_2 \left(1 + \frac{p_0 \tilde{G}_{0,v}^{[x+1]}}{F_0 N_0 B_0} \right) \right]^{-1}, \quad (38)$$

where $\tilde{G}_{0,v}^{[x+1]}$ is estimated from control-plane measurements in the current frame.

For each micro BS ($m > 0$), the lower bound in Theorem 1 is extended to account for inter-cell interference and beam-alignment overhead, while enhancing robustness against occasional large NN prediction errors. Given the NN predictions of the frame-averaged desired-link gains and interfering-link gain surrogates for frame- $(x+1)$, each gain is smoothed by weighted averaging with the corresponding gain measured in frame- x (when available); if no measurement exists, the NN prediction is used directly. With a slight abuse of notation, we still denote these robustness-enhanced gain estimates by $\tilde{G}_{m,v}^{[x+1]}$ and $\tilde{G}_{1,m',v}^{[x+1]}$. The effective SINR for vehicle- v served by BS- m in frame- $(x+1)$ is then approximated as

$$\widetilde{\text{SINR}}_{m,v}^{[x+1]} = \frac{p_m \tilde{G}_{m,v}^{[x+1]}}{F_m N_0 B_m + \sum_{m' \neq 0, m} \frac{\tilde{k}_{m'}^{[x+1]}}{K_\mu} p_{m'} \tilde{G}_{1,m',v}^{[x+1]}}, \quad (39)$$

where $\tilde{k}_{m'}^{[x+1]}$ denotes the estimated frame-averaged RB usage of BS- m' in frame- $(x+1)$ and K_μ is the RB capacity of each micro BS. To capture the beam-training overhead, we conservatively assume that at most M_P beam-search attempts are performed per slot, so that the effective data-transmission fraction per slot is at least $(1 - \eta M_P)$. Therefore, a conservative estimate of the required frame-averaged number of RBs is

$$\tilde{k}_{m,v}^{[x+1]} = \frac{\lambda_v}{(1 - \eta M_P) B_m \log_2(1 + \widetilde{\text{SINR}}_{m,v}^{[x+1]})}, \quad m > 0. \quad (40)$$

Algorithm 1 Generalized Assignment-based Proactive Handover (GAP-HO)

Input: $\tilde{G}_{m,v}^{[x+1]}, \tilde{G}_{1,m,v}^{[x+1]}, \lambda_v, p_m, B_m, F_m, K_m, N_0, K_\mu, N_{\text{iter}}, M_P, \eta, \zeta, u_{m,v}^{[x]}, L_{\text{HO}}, N_{\text{HO}};$

Output: $u_{m,v}^{[x+1]};$

- 1: **Initialize** $\tilde{k}_m^{[x+1]} \leftarrow K_\mu$ for all $m > 0$;
- 2: Compute $c_{m,v}^{[x+1]}$ for all $m \in \mathcal{M}$ and $v \in \mathcal{V}^{[x]}$ by (45);
- 3: Compute $\tilde{k}_{0,v}^{[x+1]}$ for all $v \in \mathcal{V}^{[x]}$ by (38);
- 4: **for** $n_{\text{iter}} = 1$ **to** N_{iter} **do**
- 5: Compute $\tilde{k}_{m,v}^{[x+1]}$ for each $m > 0$ and $v \in \mathcal{V}^{[x]}$ by (40);
- 6: Form **P2** with $\{\tilde{k}_{m,v}^{[x+1]}\}$ and $\{c_{m,v}^{[x+1]}\}$;
- 7: **while** continuous relaxation of **P2** is infeasible **do**
- 8: Multiply the right-hand side of (42) by ζ ;
- 9: **end while**
- 10: Apply the GAP approximation algorithm in [31] to solve **P2** and obtain a PHO schedule $\hat{u}_{m,v}^{[x+1]}$;
- 11: Compute the estimated per-BS RB usage by $\hat{k}_m^{[x+1]} = \min\{K_m, \sum_{v \in \mathcal{V}^{[x]}} \tilde{k}_{m,v}^{[x+1]} \hat{u}_{m,v}^{[x+1]}\}$ for all m ;
- 12: $\tilde{k}_m^{[x+1]} \leftarrow \hat{k}_m^{[x+1]}$ for all m ;
- 13: **end for**
- 14: **while** $\hat{u}_{m,v}^{[x+1]}$ is not feasible **and** there exists a feasible reassignment **do**
- 15: Execute the greedy feasibility-repair reassignment that yields the minimum power increase subject to (42);
- 16: **end while**
- 17: $u_{m,v}^{[x+1]} \leftarrow \hat{u}_{m,v}^{[x+1]}$;
- 18: **return**

Since the interference term in (39) depends on the unknown per-BS RB usages $\{\tilde{k}_m^{[x+1]}\}$, we jointly compute the RB demands $\{\tilde{k}_{m,v}^{[x+1]}\}$ and the BS-vehicle associations $\{u_{m,v}^{[x+1]}\}$ via a fixed-point iteration over N_{iter} rounds, as summarized in Alg. 1. Specifically, we initialize $\tilde{k}_m^{[x+1]} = K_\mu$ for all $m > 0$, evaluate $\tilde{k}_{m,v}^{[x+1]}$ using (39) and (40), and then determine $u_{m,v}^{[x+1]}$ by solving the generalized assignment problem (GAP):

$$\mathbf{P2:} \quad \min_{u_{m,v}^{[x+1]}} \sum_{m=0}^M \sum_{v \in \mathcal{V}^{[x]}} p_m \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \quad (41)$$

$$\text{s.t.} \quad \sum_{v \in \mathcal{V}^{[x]}} c_{m,v}^{[x+1]} \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \leq K_m, \quad (42)$$

$$\sum_{m=0}^M u_{m,v}^{[x+1]} = 1, \quad (43)$$

$$u_{m,v}^{[x+1]} \in \{0, 1\}, \quad (44)$$

where

$$c_{m,v}^{[x+1]} = \left[1 - (1 - u_{m,v}^{[x]}) \frac{L_{\text{HO}}}{N_{\text{HO}}} \right]^{-1}. \quad (45)$$

The factor $c_{m,v}^{[x+1]}$ accounts for the influence of HO interruption on the RB-capacity constraint. A vehicle switching BS receives no traffic data during the first L_{HO} slots of the next frame. It therefore requires more RBs per slot during the remaining $(N_{\text{HO}} - L_{\text{HO}})$ slots to serve the data accumulated during the HO interruption together with new traffic arrivals.

Since **P2** is NP-hard, we apply the polynomial-time approximation algorithm in [31], which applies a min-cost-matching-based rounding to the continuous relaxation solution of GAP, to obtain a near-optimal solution $\hat{u}_{m,v}^{[x+1]}$; if the continuous relaxation of **P2** is infeasible, we progressively relax (42) via the ζ -scaling procedure in Alg. 1 until feasibility is reached. Denoting $V \triangleq |\mathcal{V}^{[x]}|$, the computational complexity of this approximation algorithm is $O((M+V)V \log V)$. Moreover, as proved in [31], if **P2** is feasible and the optimal objective value is P^* , $\hat{u}_{m,v}^{[x+1]}$ satisfies an objective value no larger than P^* , while the RB-capacity constraints may be violated by at most a factor of two, i.e., $\sum_{v \in \mathcal{V}^{[x]}} c_{m,v}^{[x+1]} \tilde{k}_{m,v}^{[x+1]} \hat{u}_{m,v}^{[x+1]} \leq 2K_m$. We then update $\tilde{k}_m^{[x+1]} = \min\{K_m, \sum_{v \in \mathcal{V}^{[x]}} \tilde{k}_{m,v}^{[x+1]} \hat{u}_{m,v}^{[x+1]}\}$ and continue the fixed-point iteration until N_{iter} rounds are completed. Finally, to strictly satisfy (42), a greedy feasibility-repair reassignment procedure offloads some vehicles from overloaded BSs (if exist) to underloaded BSs with minimum power increase; in the worst case, this reassignment incurs $O(MV^2)$ complexity. Therefore, the computational complexity of GAP-HO is $O(N_{\text{iter}}(M+V)V \log V + MV^2)$.

C. Predictive Early-Termination Beamforming

The predictive early-termination beamforming (PET-BF) algorithm in Alg. 2 is executed at the beginning of each time slot to select beam-pairs for the MIMO links between each micro BS and its associated vehicles. PET-BF exploits the NN-based predictions $\tilde{P}_{m,v}^{[x]}$ and $\tilde{G}_{m,v}^{[x]}$, obtained in the previous frame as described in Section IV-A, to guide a fast beam-training procedure in each slot of the current frame.

For each micro BS- m and its associated vehicle- v , PET-BF first selects the top- M_P beam-pairs with the highest probabilities from the predicted distribution $\tilde{P}_{m,v}^{[x+1]}$ over $M_T M_R$ candidates. This selection can be implemented via partial selection in $O(M_T M_R \log M_P)$ time. PET-BF then probes these candidates sequentially in descending order and applies an early-termination rule. Once the measured channel gain g_{meas} of a tested beam-pair exceeds the threshold $\tilde{G}_{m,v}^{[x]} - G_{\text{th}}$ in dB, that beam-pair is immediately selected and the search stops. If none of the M_P candidates satisfies the condition, the beam-pair with the largest measured gain among the probed candidates is selected. As a result, the probing-and-comparison cost is upper-bounded by $O(M_P)$ per link per slot.

This predict-then-refine strategy reduces the beam-training overhead $\eta_{m,v}^{[x]}(i)$ while tracking the instantaneous channel variations $g_{m,v}^{[x]}(i)$, thereby enabling robust and low-latency MIMO links for the considered V2X applications.

D. Opportunistic Tiered-Rounding Resource Allocation

For slot-level RA, we adopt an opportunistic tiered-rounding resource allocation (OTR-RA) algorithm, summarized in Alg. 3. In each slot, after PET-BF, each BS- m executes OTR-RA to allocate its RBs among the associated vehicles without HO interruption under the current channel conditions. The key idea is to exploit instantaneous RB efficiency for opportunistic scheduling, while using a queue-aware tiered-rounding rule for RB allocation.

Algorithm 2 Predictive Early-Termination Beamforming (PET-BF)

Input: $\tilde{\mathcal{P}}_{m,v}^{[x]}, \tilde{G}_{m,v}^{[x]}, M_P, G_{th}, \eta;$
Output: $w_{Tm,v}^{[x]}(i), w_{Rm,v}^{[x]}(i), g_{m,v}^{[x]}(i), \eta_{m,v}^{[x]}(i)$

- 1: Construct candidate beam-pair index set $\mathcal{W}$ by selecting the M_P largest entries of $\tilde{\mathcal{P}}_{m,v}^{[x]}$;
- 2: **Initialize** $w_{best} \leftarrow \emptyset, g_{m,v}^{[x]}(i) \leftarrow -\infty$ and $\eta_{m,v}^{[x]}(i) \leftarrow 0$;
- 3: **for all** $w \in \mathcal{W}$ (ordered by probability) **do**
- 4: Probe beam-pair w to measure the instantaneous channel gain in dB: $g_{meas} \leftarrow \text{MeasureBeamGain}_{dB}(m, v, w)$;
- 5: $\eta_{m,v}^{[x]}(i) \leftarrow \eta_{m,v}^{[x]}(i) + \eta$;
- 6: **if** $g_{meas} > g_{m,v}^{[x]}(i)$ **then**
- 7: $g_{m,v}^{[x]}(i) \leftarrow g_{meas}; w_{best} \leftarrow w$;
- 8: **end if**
- 9: **if** $g_{meas} > \tilde{G}_{m,v}^{[x]} - G_{th}$ **then**
- 10: **break**;
- 11: **end if**
- 12: **end for**
- 13: Map w_{best} to the corresponding $w_{Tm,v}^{[x]}(i)$ and $w_{Rm,v}^{[x]}(i)$;
- 14: **return**

Algorithm 3 Opportunistic Tiered-Rounding Resource Allocation (OTR-RA)

Input: $u_{m,v}^{[x]}, s_v^{[x]}(i), g_{m,v}^{[x]}(i), \tilde{G}_{I,m',v}^{[x]}, \eta_{m,v}^{[x]}(i), \tilde{k}_{m'}^{[x]}, q_v^{[x]}(i), K_m, B_m, \Delta t, p_m, F_m, N_0, \lambda_v, \tau_v^{th};$
Output: $k_{m,v}^{[x]}(i);$

- 1: **Initialize** $k_{m,v}^{[x]}(i) \leftarrow 0$ for all $v \in \mathcal{V}^{[x]}, \bar{K}_m \leftarrow K_m$ and $\mathcal{V}_m^{[x]}(i) \leftarrow \{v \in \mathcal{V}^{[x]} \mid u_{m,v}^{[x]} s_v^{[x]}(i) = 1\}$;
- 2: Compute $\Gamma_{m,v}^{[x]}(i)$ for all $v \in \mathcal{V}_m^{[x]}(i)$ by (46) and (47);
- 3: Sort $\mathcal{V}_m^{[x]}(i)$ in descending order of $\Gamma_{m,v}^{[x]}(i)$ to obtain $\mathcal{S}_m^{[x]}$;
- 4: **for all** $v \in \mathcal{S}_m^{[x]}$ **do**
- 5: Compute $k_{m,v}^{[x]}(i)$ by (48) and (49);
- 6: $\bar{K}_m \leftarrow \bar{K}_m - k_{m,v}^{[x]}(i)$;
- 7: **end for**
- 8: **return**

BS- m estimates the slot-level SINR using the measured desired-link gain, the predicted frame-averaged interfering-link gain, and the estimated RB usages obtained from GAP-HO:

$$\widetilde{\text{SINR}}_{m,v}^{[x]}(i) = \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m + \sum_{m' \neq m} \frac{\tilde{k}_{m'}^{[x]}}{\bar{K}_m} p_{m'} \tilde{G}_{I,m',v}^{[x]}}. \quad (46)$$

Here, $\tilde{G}_{I,m',v}^{[x]}$ is predicted in frame $(x-1)$ and used throughout frame- x . The corresponding RB efficiency is

$$\Gamma_{m,v}^{[x]}(i) \triangleq B_m \log_2(1 + \widetilde{\text{SINR}}_{m,v}^{[x]}(i)). \quad (47)$$

After computing $\Gamma_{m,v}^{[x]}(i)$ for all available associated vehicles, BS- m sorts them in descending order of $\Gamma_{m,v}^{[x]}(i)$ so that links with higher instantaneous RB efficiency are served first. The SINR and RB-efficiency evaluation requires $O(MV_m^{[x]})$ operations due to the summation over interfering micro BSs for each associated vehicle, and the sorting costs $O(V_m^{[x]} \log V_m^{[x]})$.

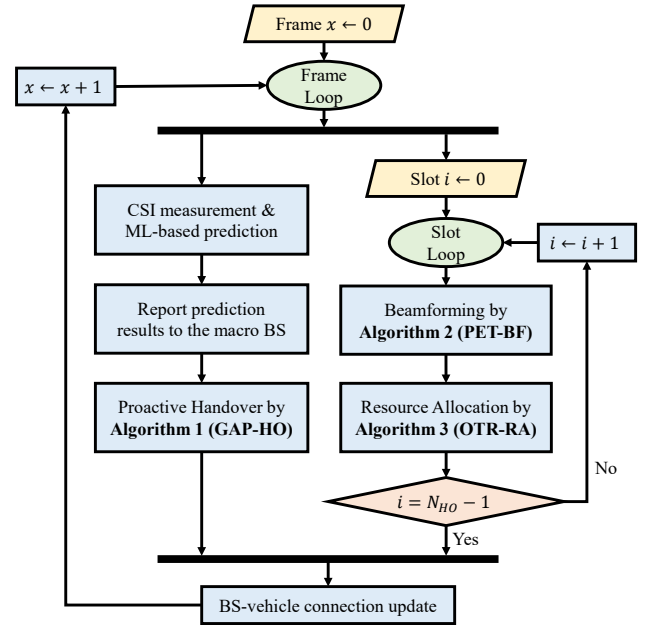


Fig. 2. Flowchart of the MEET-COBRA framework.

Given this ordered list, OTR-RA allocates RBs sequentially. For each vehicle- v , an ideal continuous RB demand to clear its backlog in slot- i is

$$\bar{k}_{m,v}^{[x]}(i) = \frac{q_v^{[x]}(i)}{(1 - \eta_{m,v}^{[x]}(i)) \Delta t \Gamma_{m,v}^{[x]}(i)}, \quad (48)$$

which is then mapped to an integer RA decision $k_{m,v}^{[x]}(i)$ via a queue-aware tiered-rounding rule, subject to the remaining RB budget $\bar{K}_m$:

$$k_{m,v}^{[x]}(i) = \begin{cases} \min\{\lceil \bar{k}_{m,v}^{[x]}(i) \rceil, \bar{K}_m\}, & \text{if } q_v^{[x]}(i) > \frac{\lambda_v \tau_v^{th}}{2}, \\ \min\{\lfloor \bar{k}_{m,v}^{[x]}(i) \rfloor, \bar{K}_m\}, & \text{otherwise.} \end{cases} \quad (49)$$

Then the budget updates as $\bar{K}_m \leftarrow \bar{K}_m - k_{m,v}^{[x]}(i)$. This tiered-rounding strategy avoids systematic RB waste that would occur if all $\bar{k}_{m,v}^{[x]}(i)$ were rounded up, while also preventing high-urgency users with $\bar{k}_{m,v}^{[x]}(i) < 1$ from repeatedly receiving no RBs under a purely rounding-down rule. The process continues until all the available associated vehicles have been scheduled or the RB budget is exhausted. Since each vehicle is processed at most once, the sequential allocation and budget updates cost $O(V_m^{[x]})$. Therefore, the overall per-slot computational complexity of OTR-RA at BS- m is $O(MV_m^{[x]} + V_m^{[x]} \log V_m^{[x]})$, making it a low-complexity RA scheme that balances RB efficiency and queue urgency under per-BS RB-capacity constraints.

E. MEET-COBRA

The proposed MEET-COBRA framework integrates the above modules into a unified two-timescale design, as illustrated in Fig. 2. Frame-level PHO decisions are computed

centrally at the macro BS, while slot-level BF and RA are executed locally at each BS.

At the beginning of frame- x , each vehicle performs local NN inference based on its historical CSI measurements to generate next-frame predictions $\{\hat{\mathcal{P}}_{m,v}^{[x+1]}, \hat{G}_{m,v}^{[x+1]}, \hat{G}_{I,m,v}^{[x+1]}\}$. Each vehicle reports only the top- M_P beam-pair indices in descending order of predicted probability and the two gain predictions for each micro BS to the macro BS through the control-plane uplink. Using these predictions, the macro BS executes GAP-HO to compute the PHO decisions $u_{m,v}^{[x+1]}$ for the next frame. The macro BS also forwards the interfering-link gain predictions and estimated RB usages required by OTR-RA to the corresponding serving BSs before frame- $(x+1)$ begins.

Within each slot- i of frame- x , each micro BS- m first applies PET-BF to probe the predicted top- M_P beam-pairs with early termination, thereby obtaining BF vectors $\{\mathbf{w}_{T,m,v}^{[x]}, \mathbf{w}_{R,m,v}^{[x]}(i)\}$ and instantaneous link gains $g_{m,v}^{[x]}(i)$ with limited beam-training overhead $\eta_{m,v}^{[x]}(i)$. After that, each BS applies OTR-RA to prioritize users by their RB efficiency $\Gamma_{m,v}^{[x]}(i)$ and perform queue-aware tiered-rounding to obtain integer RB allocations $k_{m,v}^{[x]}(i)$ under the RB-capacity constraints.

Remarkably, ML-based prediction serves as the coupling mechanism that aligns the modeling assumptions across HO, BF, and RA. The gain predictions enable GAP-HO to estimate the next-frame RB demands of each BS-vehicle pair and to solve **P2** based on a forward-looking characterization of the network state, while the beam-pair predictions guide PET-BF to realize high-quality beam alignment under reduced beam-training overhead. This improves the consistency between the link gains assumed in the RB-demand estimation of GAP-HO and the link gains realized during data transmission. Based on these realized link gains, OTR-RA further exploits opportunistic scheduling and queue-aware tiered rounding to enhance latency performance and reduce RB wastage, thereby narrowing the gap between the RB demands predicted by GAP-HO and the realized long-term RB usage. Consequently, MEET-COBRA provides a low-complexity online solution that effectively addresses the original problem **P1**.

Prediction errors can also affect this coordination. With other inputs fixed, overestimating the desired-link gain or underestimating the interfering-link gains lowers the estimated RB demand in (40), whereas errors in the opposite direction raise it. These changes affect the estimated power costs and RB requirements in GAP-HO, and can change BS-vehicle associations. In PET-BF, beam prediction errors affect the candidate beam-pair list and the search order, while desired-link gain errors affect the early-termination threshold. Both types of error can affect the gain of the selected beam pair and the beam-search overhead. OTR-RA uses measured desired-link gains, but errors in predicted interfering-link gains and estimated BS RB usages can affect RB-efficiency estimates, user ordering, and RB allocations. These effects can change transmit power and received rates, thereby influencing queue evolution and the violation probability U .

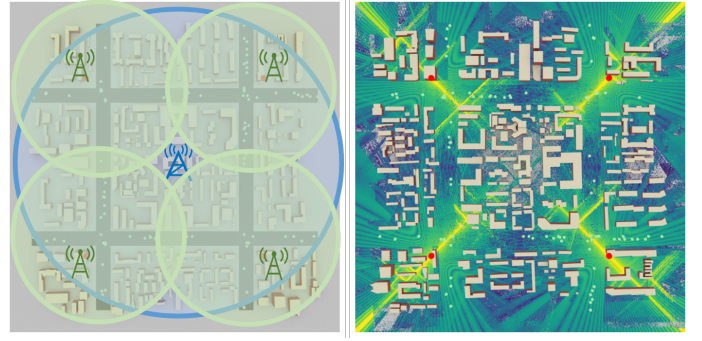


Fig. 3. Simulation scenario of the considered two-tier HetNet. Left: road layout, buildings, and locations and coverage regions of the macro BS (blue) and four micro BSs (green). Right: example Sionna RT ray-tracing-based received-power map at 28 GHz with vertically polarized antennas and micro-BS ULAs oriented toward the area center.

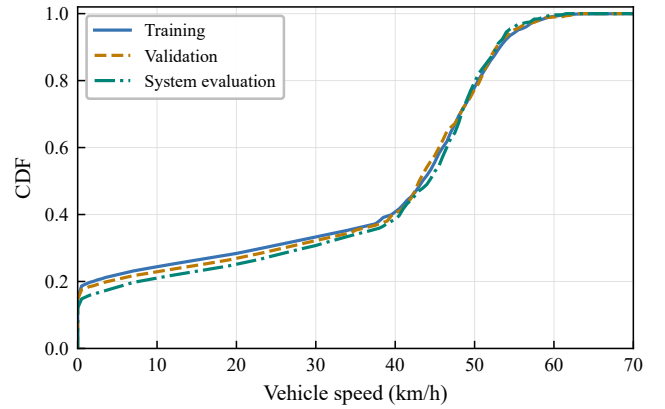


Fig. 4. Vehicle-speed CDFs over vehicle-frame samples from the training, validation, and system evaluation datasets.

V. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed MEET-COBRA framework using a simulation platform built on the SUMO [24] and the Sionna RT [25] simulators. We first describe the simulation setup and parameter configuration, and then present numerical results that compare MEET-COBRA with several baseline schemes in terms of power consumption and latency performance.

A. Simulation Setup

We consider a $1000 \text{ m} \times 1000 \text{ m}$ urban HetNet scenario shown in Fig. 3, where one macro BS is deployed at the center $(0, 0)$ and four micro BSs are located at $(\pm 300, \pm 300) \text{ m}$. The road topology consists of four intersections centered at $(\pm 250, \pm 250) \text{ m}$, and each road segment is bidirectional with three lanes per direction. Each micro BS employs a ULA oriented toward the scenario center. All antennas are vertically polarized. SUMO generates the vehicle trajectories, which are used by Sionna RT to obtain the time-varying MIMO channel matrices between vehicles and micro BSs. The large-scale channel gains between vehicles and the macro BS are generated using the 3GPP TR 38.901 urban macro model [32].

TABLE II
SIMULATION PARAMETERS

Global simulation parameters		Macro- and micro-BS parameters		
Parameter	Value	Parameter	Macro BS	Micro BS
Slot duration Δt	1 ms	Carrier frequency f_m	2.8 GHz	28 GHz
PHO frame length N_{HO}	100 slots	Number of RBs K_m	133	66
Pilots per frame N_P	8	RB bandwidth B_m	0.36 MHz	1.44 MHz
Rician factor K_R	3	Per-RB transmit power p_m	1 W	0.2 W
Beam-search overhead η	2/112	Noise figure F_m	5 dB	10 dB
Latency threshold τ_v^{th}	20 ms	Channel simulation	3GPP TR 38.901 UMa	Ray-tracing by Sionna RT [25]
HO interruption duration τ_{HO}	10 ms	Antenna number (Tx&Rx)	1&1	32&8
Number of GAP-HO iterations N_{iter}	2	Number of BSs	1	4
RB capacity relaxation factor ζ	1.1	Beamforming codebooks	Not applicable	DFT
Noise power spectral density N_0	-174 dBm/Hz	Number of candidate beam pairs M_P	Not applicable	5
System simulation duration per run	30 s	PET-BF gain margin G_{th}	Not applicable	5 dB

Fig. 4 shows the cumulative distribution functions (CDFs) of vehicle speed in the training, validation, and system evaluation datasets. The respective mean speeds are 33.56, 33.95, and 34.78 km/h. Samples with zero speed account for 14.97%, 14.64%, and 12.07% of the corresponding vehicle-frame observations. Vehicle speeds in the system evaluation dataset range from 0 to 61.53 km/h.

For the system-level performance comparisons in Section V-D, we vary the traffic arrival rate $\lambda_v = \lambda$ over $\{1, 3, \dots, 35\}$ Mbps and set $\tau_v^{\text{th}} = 20$ ms for all vehicles. For all schemes, receive rates are evaluated using actual RB overlaps and the selected transmit and receive beams. We set $N_{\text{iter}} = 2$ to balance performance and computational cost, and $\zeta = 1.1$ for 10% increments in the RB capacity bounds when the continuous relaxation of **P2** is infeasible. We use $M_P = 5$ and $G_{\text{th}} = 5$ dB to balance coverage of the optimal beam pair against beam search overhead, which occupies at most $\eta M_P = 8.93\%$ of each slot. Unless otherwise stated, the simulations use the parameter values listed in Table II.

B. Performance of ML-Based Prediction

We evaluate the NN models in Section IV-A for beam-pair and gain prediction. We set $N_C = 128$ for the LSTM unit and the vector ResBlock, with $N_H = 4$ vector ResBlocks per output head. The dataset is built from Sionna RT-generated channel traces spanning 6000 frames and contains 646,747 labeled samples from 706 vehicles. Each sample consists of the current-frame CSI of a vehicle, together with labels of the optimal beam-pair indices and the desired-link and beam-averaged interfering-link gains for the next frame. The dataset is randomly partitioned at the vehicle level, with 70% of the vehicles in the training set and the remaining 30% in the validation set.

All models are trained offline in two stages of 100 epochs each, using AdamW [33] with a weight decay of 10^{-4} . In the first stage, the models are randomly initialized and trained using the input windows of one to ten consecutive CSI frames at a learning rate of 10^{-3} . In the second stage, the models are initialized from the best first-stage checkpoints for stateful fine-tuning. This stage uses an initial learning rate of 10^{-4} and truncated backpropagation through time [34] with a truncation length of ten frames. Recurrent states $\mathbf{h}_v$ and $\mathbf{c}_v$ are carried across frames and reset only at the start of a new trajectory.

TABLE III
NN PARAMETER COUNTS AND INFERENCE FLOPS

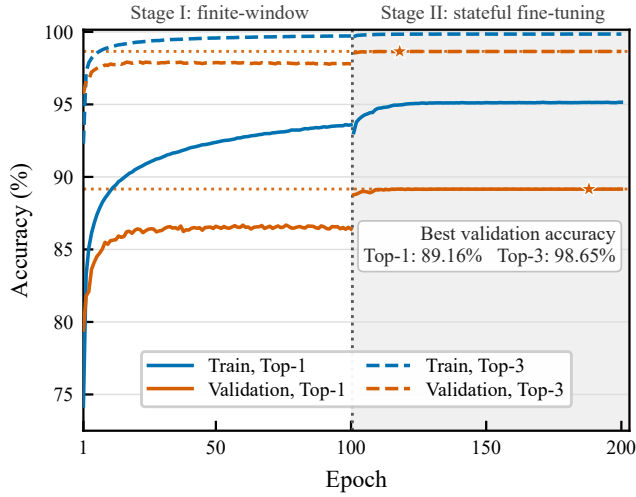
Predictor	Parameters	Matrix FLOPs (10^6)
Beam-pair	1,060,864	2.10
Desired-link gain	533,524	1.05
Interfering-link gain	533,524	1.05
Total	2,127,912	4.20

Matrix FLOPs count LSTM and linear-layer matrix operations, with two FLOPs per multiply-accumulate.

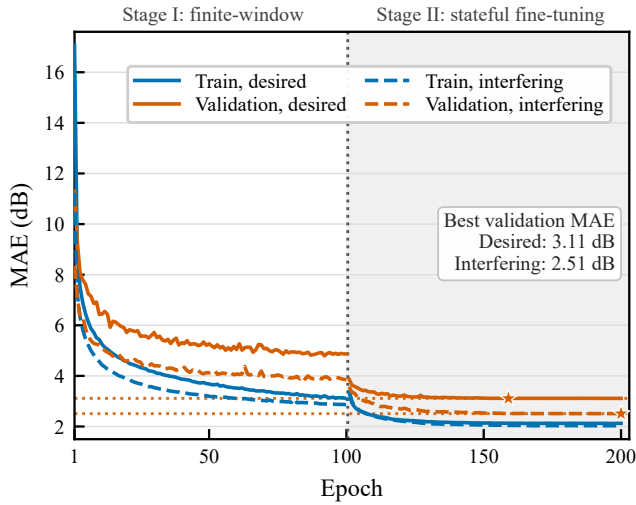
Both training and validation use finite input windows in the first stage and process samples in time order along each vehicle trajectory in the second. In both stages, the beam checkpoint is selected by the highest top-1 validation accuracy and each gain checkpoint by the lowest validation MAE. The selected second-stage checkpoints are used in all subsequent evaluations.

Fig. 5 illustrates the training and validation performance of the beam and gain prediction models. Stateful fine-tuning increases the best top-1 validation accuracy of beam prediction from 86.68% to 89.16% and achieves a best top-3 validation accuracy of 98.65%. For the selected beam checkpoint, the validation top- M_P accuracy exceeds 99% for $M_P \geq 4$ and 99.9% for $M_P \geq 17$, indicating that a short candidate list contains the optimal beam pair with high probability. Stateful fine-tuning also reduces the best validation MAEs of desired-link and beam-averaged interfering-link gain prediction from 4.79 and 3.85 dB to 3.11 and 2.51 dB, respectively. These results show improved beam and gain prediction after stateful fine-tuning. We further assess the selected models across four speed groups of $[0, 1)$, $[1, 20)$, $[20, 40)$, and ≥ 40 km/h on the system evaluation dataset. From the lowest to the highest speed group, the desired-link gain MAEs are 1.13, 2.31, 3.15, and 3.73 dB, and the corresponding interfering-link gain MAEs are 1.19, 1.79, 2.32, and 3.07 dB. Top-5 beam prediction accuracy exceeds 99% in all four groups.

Table III summarizes the NN parameter counts and matrix FLOPs for single-vehicle stateful inference using the current-frame CSI and recurrent states. The three models require 8.51 MB of 32-bit floating-point (FP32) parameter storage. With FP32 inference on an AMD EPYC 7742 CPU using one thread, their combined inference latency has a median



(a) Beam prediction model.



(b) Gain prediction models.

Fig. 5. Training and validation performance of the proposed NN models: (a) beam prediction model (top-1 and top-3 accuracies); (b) optimal desired-link and interfering-link gain prediction models (MAE). The vertical dotted line separates finite-window training (epochs 1–100) and stateful fine-tuning (epochs 101–200). Stars mark the best validation value of each metric.

of 5.38 ms and a 95th percentile of 5.72 ms. With $M = 4$, $M_P = 5$, and $M_T M_R = 256$, each vehicle reports, for each micro-BS link, M_P beam-pair indices in descending order of predicted probability, using 8 bits per index, together with one FP32 desired-link gain and one FP32 interfering-link gain. The reporting overhead is $M(8M_P + 2 \times 32) = 416$ bits per vehicle per frame (4.16 kbps). At the same reporting interval and 32-bit precision per real CSI component, this reduces reporting overhead by factors of 157.5 and 9.85 relative to full MIMO-CSI reporting and centralized inference using current-frame superposed-pilot observations, respectively.

C. Benchmark Schemes

To assess the performance of MEET-COBRA, we compare it with the following benchmark schemes:

1) Oracle-MC: MEET-COBRA with perfect beam and gain knowledge, where all ML predictions are replaced by

their ground-truth values from the simulator. It quantifies the performance loss due to prediction errors.

2) Reactive-OBRA: A reactive HetNet baseline without ML prediction or proactive coordination. Using current channel gains, it assigns vehicles sequentially to the BS with the lowest estimated transmit power cost among those with sufficient remaining RB capacity, falling back to the macro BS if none is available. BF uses a random opportunistic search inspired by [35]: after HO it probes M_P beam-pairs and keeps randomly probing thereafter, updating the serving beam only when a better one is found. RA estimates RB efficiency assuming full-load interfering micro BSs and applies stochastic rounding for integer RB allocation.

3) Ablations: *w/o GAP-HO*, *w/o PET-BF*, and *w/o OTR-RA*, where the corresponding module in MEET-COBRA is replaced by its Reactive-OBRA counterpart, while keeping the other two modules unchanged.

4) O-MAPPO-adapted: This baseline adapts the O-MAPPO framework in [22], which combines reinforcement learning with target BS optimization. A shared actor trained by multi-agent proximal policy optimization evaluates whether each vehicle should trigger HO after every 10 m of travel. For vehicles with active HO triggers, a mixed-integer optimizer selects target BSs using estimated RB demands, BS loads, and transmit power costs, with penalties for aggregate demands exceeding BS capacities. The actor and optimizer use the same predicted desired-link and interfering-link gains as MEET-COBRA. Association changes take effect in the next frame. Upon HO to a micro BS, hierarchical beam search starts after the HO interruption. It first probes all combinations of eight transmit and two receive sectors, then all combinations of four transmit and four receive DFT beams within the selected sectors, requiring 32 probes in total. In subsequent slots, local beam tracking probes the current beam pair and four adjacent pairs differing in only one beam index, and then retains the pair with the highest measured gain. Each BS applies OTR-RA in every slot.

5) MTS-GS-HBF-adapted: This baseline adapts the multi-timescale association and hybrid BF framework in [23]. Gale-Shapley matching updates BS–vehicle associations every five frames using the same predicted channel gains as MEET-COBRA. Matching preferences account for estimated rates, BS loads, queue backlogs, and energy and HO costs, while association decisions consider estimated RB demands and BS capacities. We replace the original hybrid BF with the hierarchical beam search and local tracking used in O-MAPPO-adapted. Unlike O-MAPPO-adapted, hierarchical beam search is repeated in the first available slot of every frame, after the HO interruption when applicable, followed by local beam tracking in the remaining slots. Each BS applies OTR-RA in every slot.

D. Performance Comparison

This subsection compares MEET-COBRA with the benchmark schemes in terms of transmit power and latency performance. At each data arrival rate, we run all schemes over the same 30 s mobility trace with three random seeds. Each

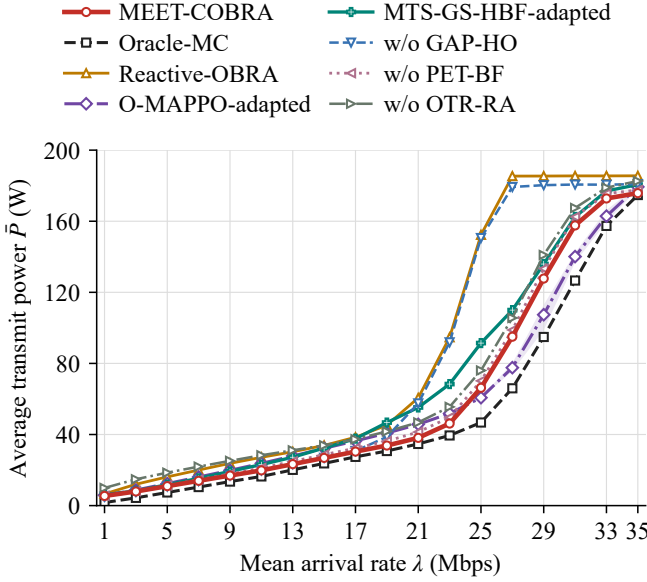


Fig. 6. Average system transmit power $\bar{P}$ versus the data arrival rate λ .

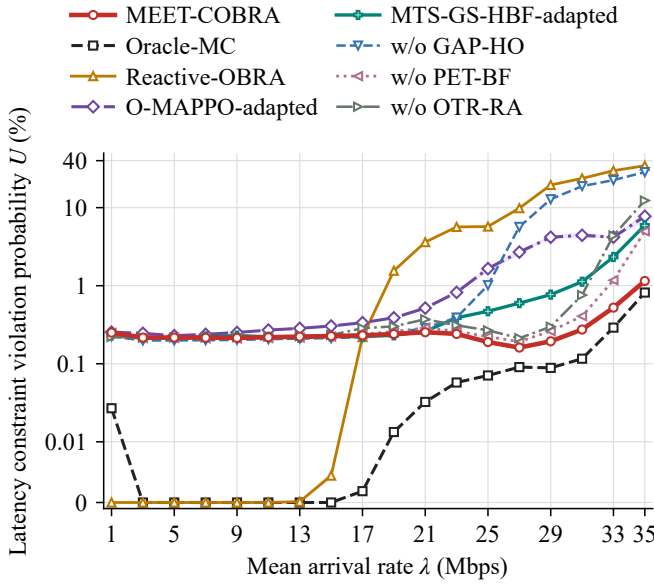


Fig. 7. Latency constraint violation probability U versus the data arrival rate λ . The vertical scale is linear below 0.01% and logarithmic above it.

seed generates a different set of traffic arrivals and small-scale fading samples, which are shared by all schemes. Curves show the mean of the three runs, and shaded bands span their minimum and maximum values.

Figs. 6 and 7 compare the average system transmit power $\bar{P}$ and the latency constraint violation probability U , respectively, at different data arrival rates λ . Under light loads, the transmit power of MEET-COBRA increases approximately linearly with λ , as additional traffic can largely be accommodated by allocating more RBs without changing serving links. At $\lambda = 19$ Mbps, MEET-COBRA uses 33.80 W with $U = 0.24\%$,

compared with 30.71 W and 0.013% for Oracle-MC. This relatively small power gap suggests that prediction errors have a limited impact on transmit power when RB resources are sufficient. The power gap widens at 27 Mbps, where the two schemes use 95.03 and 66.00 W, respectively. This larger power gap suggests that accurate RB-demand estimates become more important for coordinating PHO and RA when RB resources are limited.

Compared with Reactive-OBRA, MEET-COBRA achieves much lower transmit power and fewer latency violations under heavy traffic. At $\lambda = 27$ Mbps, Reactive-OBRA approaches the full-load power ceiling, using 185.39 W with $U = 9.84\%$, whereas MEET-COBRA uses 95.03 W with $U = 0.16\%$. These correspond to reductions of 48.7% in transmit power and 98.4% in violation probability. Further traffic increases mainly raise U for Reactive-OBRA because its RB resources are already nearly fully utilized. MEET-COBRA maintains a low violation probability over a wider range of data arrival rates, although U also increases as its RB resources approach full utilization. From 33 to 35 Mbps, its power increases only from 172.81 to 175.80 W, while U rises from 0.52% to 1.15%. Nevertheless, at 35 Mbps, the violation probability of MEET-COBRA remains well below the 34.46% observed for Reactive-OBRA.

Compared with O-MAPPO-adapted, MEET-COBRA uses less power over 1–23 Mbps and incurs fewer violations at every evaluated load. Over 25–33 Mbps, O-MAPPO-adapted uses less power but incurs substantially more violations. At 29 Mbps, O-MAPPO-adapted uses 107.40 W with $U = 4.18\%$, compared with 127.75 W and 0.19% for MEET-COBRA.

Compared with MTS-GS-HBF-adapted, MEET-COBRA has lower mean power at every evaluated load and fewer violations at $\lambda \geq 21$ Mbps. MTS-GS-HBF-adapted achieves slightly fewer violations at several lower loads. Its largest advantage in U occurs at 1 Mbps, with 0.23% compared with 0.25% for MEET-COBRA. At 29 Mbps, MTS-GS-HBF-adapted uses 135.91 W with $U = 0.77\%$, both higher than the corresponding values of 127.75 W and 0.19% for MEET-COBRA. MTS-GS-HBF-adapted updates associations only every five frames, which can delay traffic redistribution when link conditions and queue backlogs change. Its periodic hierarchical beam search and local tracking also require more probes than PET-BF in MEET-COBRA. The additional probing leaves less time for data transmission, increasing the RB demand for a given link gain and amount of data to transmit.

Fig. 8 reports L_{90} and L_{99} , the 90th and 99th percentiles of the queueing-delay proxy $\tau_v(n)$ over vehicle-slot samples within each run. As the arrival rate increases, L_{99} can rise sharply while L_{90} remains small. The mean L_{99} first exceeds 20 ms at 35 Mbps for MEET-COBRA, compared with 19 Mbps for Reactive-OBRA, 25 Mbps for O-MAPPO-adapted and w/o GAP-HO, 31 Mbps for MTS-GS-HBF-adapted, and 33 Mbps for the other two ablations. Oracle-MC remains below this threshold throughout the evaluated range. At 33 Mbps, MEET-COBRA and MTS-GS-HBF-adapted have similar L_{90} values (1.24 and 1.19 ms), but their L_{99} values are 5.00 and 267.54 ms, respectively. When evaluated separately

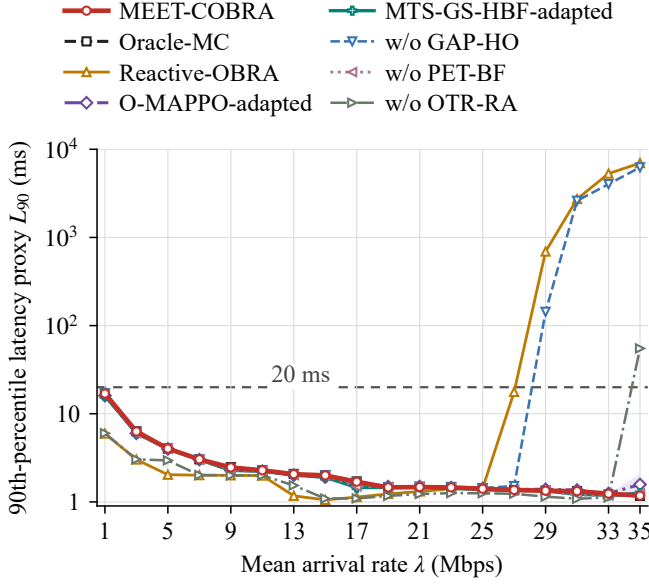
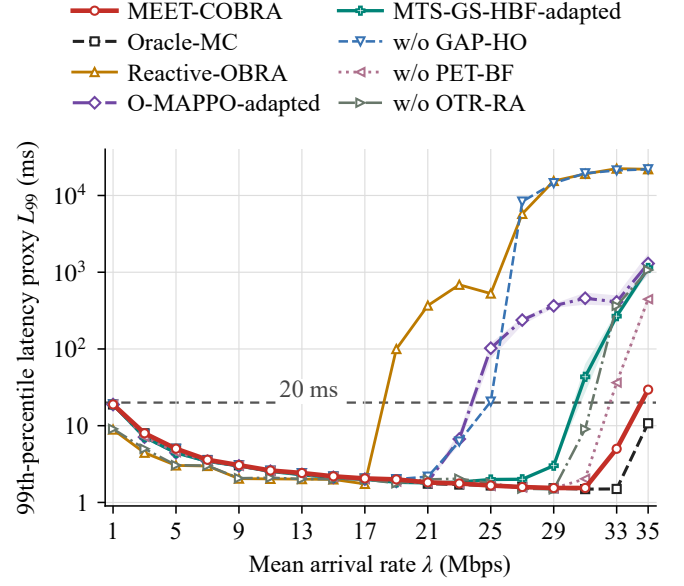
(a) L_{90} (90th percentile).(b) L_{99} (99th percentile).

Fig. 8. The 90th and 99th percentiles of the queueing-delay proxy $\tau_v(n)$ versus the data arrival rate λ , where the horizontal dashed lines denote the latency threshold $\tau_v^{\text{th}} = 20$ ms.

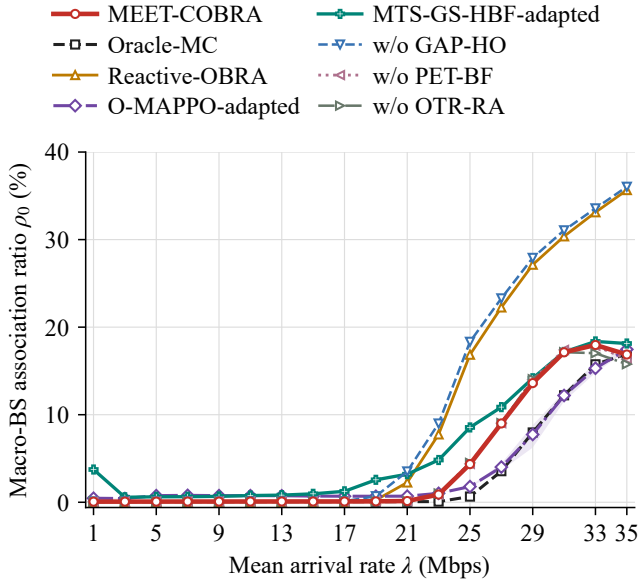


Fig. 9. Percentage of vehicles associated with the macro BS (BS-0) versus the data arrival rate λ .

for the speed groups $[0, 1)$, $[1, 20)$, $[20, 40)$, and ≥ 40 km/h, MEET-COBRA maintains a mean L_{99} below 20 ms in every group at all evaluated arrival rates up to 33 Mbps. At 35 Mbps, the corresponding values are 1.49, 11.19, 28.37, and 48.02 ms in increasing speed-group order. These results indicate that MEET-COBRA maintains low latency across the observed speed groups over most evaluated loads, while larger delays arise in the higher-speed groups at the highest load.

Fig. 9 reports the macro-BS association ratio ρ_0 . MEET-

COBRA serves almost all vehicles through micro BSs at light loads and offloads more traffic to the macro BS as the load increases. From 21 to 27 Mbps, ρ_0 rises from 0.14% to 9.00%, while U decreases from 0.25% to 0.16%. This non-monotonic variation is consistent with GAP-HO redistributing traffic as micro-BS capacity constraints become limiting. At 29 Mbps, MEET-COBRA maintains $\rho_0 = 13.61\%$ with $L_{99} = 1.54$ ms, whereas Reactive-OBRA associates 27.13% of vehicles with the macro BS and has $L_{99} > 10^4$ ms. These results show the benefit of coordinating PHO, BF, and RA to serve traffic efficiently through the micro tier while balancing the traffic offloaded to the macro BS.

However, a smaller macro-BS association ratio alone does not imply better load coordination. At 29 Mbps, O-MAPPO-adapted has $\rho_0 = 7.74\%$, compared with 13.61% for MEET-COBRA, but its L_{99} reaches 364.76 ms. At this load, the average micro-BS RB utilization is 97.2% for O-MAPPO-adapted, compared with 79.8% for MEET-COBRA. Retaining more traffic in the micro tier reduces the use of higher-power macro-BS RBs, but leaves little unused micro-BS capacity to clear accumulated queues. Moreover, O-MAPPO-adapted cannot reassign a vehicle without an active HO trigger, whereas GAP-HO reoptimizes BS-vehicle associations in every frame as resource demands change.

The ablation results in Figs. 6–9 further distinguish the contributions of the three modules of MEET-COBRA. Under light loads, w/o GAP-HO consumes almost the same power as MEET-COBRA, indicating limited additional power benefit from global PHO optimization when micro-BS RB resources are sufficient. As the load increases, its power and violation probability diverge from MEET-COBRA, reaching 179.27 W and 5.67% at 27 Mbps. At this load, removing GAP-HO

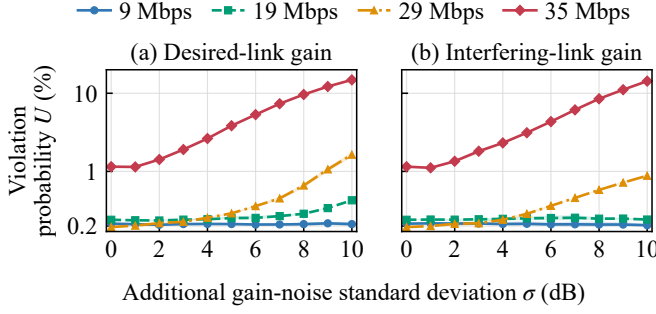


Fig. 10. Sensitivity of the latency constraint violation probability U to gain-prediction errors.

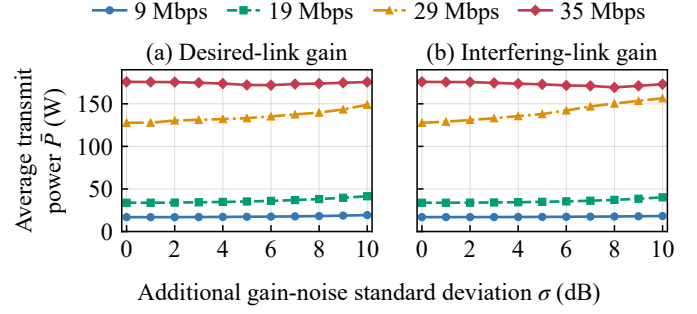


Fig. 11. Sensitivity of the average system transmit power $\bar{P}$ to gain-prediction errors.

causes the largest increases in both metrics among the three ablations. Since w/o GAP-HO retains PET-BF and OTR-RA, this comparison highlights the importance of jointly coordinating BS-vehicle associations under capacity constraints, rather than assigning vehicles sequentially.

Replacing PET-BF with random opportunistic beam search increases power throughout the evaluated range. PET-BF uses predicted beam-pair candidates to find a sufficiently strong beam pair and limits beam-search overhead through early termination. At 33 Mbps, w/o PET-BF has $U = 1.17\%$ and $L_{99} = 36.41$ ms, compared with 0.52% and 5.00 ms for MEET-COBRA. These differences occur despite similar macro-BS association ratios (17.87% and 17.96%) and $L_{90} < 1.3$ ms for both schemes, showing how efficient BF complements PHO in maintaining a small L_{99} .

Removing OTR-RA causes the largest power increase among the ablations at light loads. Its replacement assumes full RB utilization at interfering BSs and uses stochastic rounding, whereas OTR-RA accounts for estimated RB usage and queue urgency. At 7 Mbps, w/o OTR-RA uses 21.92 W, compared with 13.95 W for MEET-COBRA. Although the additional RB allocation can reduce latency under light loads, it leaves fewer RBs available as traffic increases. At 33 Mbps, w/o OTR-RA has $L_{90} = 1.13$ ms and $\rho_0 = 17.02\%$, close to the corresponding values of 1.24 ms and 17.96% for MEET-COBRA, but U increases from 0.52% to 4.40% and L_{99} from 5.00 to 365.26 ms. By rounding up RB allocations for urgent queues and rounding down those for nonurgent queues, OTR-RA reduces prolonged queue accumulation while limiting unnecessary RB use.

E. Impact of Gain-Prediction Errors

This subsection evaluates how gain-prediction errors affect the system transmit power and latency performance of MEET-COBRA. We add independent zero-mean Gaussian noise with standard deviation $\sigma \in \{0, 1, \dots, 10\}$ dB to either the original desired-link or interfering-link gain predictions. Keeping the trained models, beam-pair candidates, actual channels and traffic arrivals unchanged, we rerun MEET-COBRA at $\lambda \in \{9, 19, 29, 35\}$ Mbps with three paired random seeds.

Fig. 10 shows that U changes little at 9 and 19 Mbps for $\sigma \leq 5$ dB. With $\sigma = 5$ dB, U at 29 Mbps increases from 0.19% to 0.29% when noise is added to either type of gain

prediction. At 35 Mbps, however, adding noise with $\sigma = 5$ dB raises U from 1.15% to 3.84% and 3.13% for desired-link and interfering-link gain predictions, respectively.

Fig. 11 shows that larger gain-prediction errors generally increase transmit power at the three lower traffic loads. At 19 Mbps, adding noise with $\sigma = 10$ dB to the interfering-link gain predictions increases power from 33.80 to 40.15 W, while U remains approximately 0.24%. At 35 Mbps, however, power remains close to the full-load level while U rises sharply.

We further study the influence of gain-prediction errors on HO and beam probing. At 29 Mbps, adding noise with $\sigma = 5$ dB to the desired-link gain predictions increases the mean number of HOs per frame from 3.70 to 20.01 and the mean number of probes per beam search from 1.06 to 1.95. These additional HOs and beam probes reduce the time available for data transmission. At high loads, limited remaining RB resources make it harder to compensate for this loss of transmission time through additional RB allocation.

VI. CONCLUSION

This paper proposed MEET-COBRA, a ML-based energy-efficient proactive coordination framework for jointly optimizing PHO, BF and RA in HetNets. By leveraging ML-based prediction of beam-pair candidates and channel gains, MEET-COBRA provides a low-complexity online approach to efficiently address the two-timescale SINLP for COBRA.

Simulations driven by SUMO mobility traces and Sionna RT channel data validate the effectiveness of MEET-COBRA. The predicted beam-pair and channel gain information enables efficient global PHO optimization, reduces beam-search overhead, and mitigates RB wastage and latency constraint violations caused by suboptimal BF and RA decisions, thereby addressing the computational challenges of the two-timescale SINLP in practical deployments. Notably, MEET-COBRA effectively balances system transmit power and latency performance. At an illustrative medium-to-high load operating point ($\lambda = 27$ Mbps), it reduces the average system transmit power by approximately 49% compared with the Reactive-COBRA baseline, while decreasing the latency constraint violation probability from 9.84% to 0.16%. Compared with the baselines adapted from [22] and [23], MEET-COBRA uses less transmit power at light and medium loads and achieves lower latency constraint violation probabilities at high loads.

It also maintains low violation probabilities under moderate additional gain-prediction errors at light and medium loads.

Potential future work includes extending MEET-COBRA to multi-macro BS deployments, modeling HO failure recovery and detailed signaling procedures, and developing online adaptation mechanisms for the prediction models and algorithm hyper-parameters to enhance robustness under non-stationary traffic and radio environments.

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MEET-COBRA: Machine-learning-based Energy-Efficient proactive Coordination of handOver, Beamforming and Resource Allocation

Bowen Xie, Sheng Zhou, and Zhisheng Niu

Abstract—Energy-efficient coordination of handover, beamforming and resource allocation (COBRA) in heterogeneous networks (HetNets) remains underexplored, particularly for high-mobility vehicular scenarios under latency constraints. We formulate a stochastic integer non-linear program (SINLP) for COBRA to minimize long-term system transmit power subject to a queue-length-based latency constraint. However, the NP-hardness of the resulting SINLP, together with the high overhead of acquiring timely and accurate per-link channel state information (CSI), makes online global COBRA optimization computationally prohibitive. To address this challenge, we propose the MEET-COBRA framework, which uses machine learning (ML) to predict next-frame beam-pair candidates and channel gains from low-overhead historical CSI measurements, and translates these predictions into per-link resource block (RB) demand estimates for globally optimized proactive handover. Meanwhile, each cell performs prediction-guided beamforming and queue-aware resource allocation to keep the realized link efficiencies consistent with the RB-demand estimates, thereby meeting the latency constraint with low transmit power. Simulation results show that, at a medium-to-high traffic load, MEET-COBRA lowers the average transmit power by about 49% and reduces the latency constraint violation probability from 9.84% to 0.16% compared with the Reactive-COBRA baseline, with only lightweight ML inference overhead.

Index Terms—Heterogeneous networks, proactive handover, beamforming, resource allocation, machine learning, energy efficiency.

I. INTRODUCTION

As modern vehicles evolve into intelligent mobile terminals, their efficient communication with infrastructure, other vehicles, and pedestrians becomes essential, which leads to the emergence of connected autonomous vehicles (CAVs) and Vehicle-to-Everything (V2X) communications [1]. CAV applications demand high throughput for perception and infotainment and low latency for driving safety. These communication quality of service (QoS) requirements are increasingly considered in 3GPP enhanced V2X service specifications and the broader new radio (NR)-V2X ecosystem [2], [3].

In practice, heterogeneous network (HetNet) deployments integrating macro and micro base stations (BSs) across different frequency bands provide a promising evolutionary path for NR-V2X systems to accommodate these requirements [4]. Typically, macro BSs operate in lower-frequency bands to ensure broad and reliable coverage, whereas micro BSs utilize higher-frequency bands to enhance network capacity [5]. Such HetNet deployments can also reduce system transmit power

through traffic-aware resource allocation. Realizing these benefits requires balancing transmit power against timely traffic service. Increasing resource usage can reduce delay but also raise power consumption [6].

High mobility makes this balance more difficult. Fast-moving vehicles trigger frequent handovers (HOs), increasing signaling overhead and degrading link performance [7], [8]. Conventional reactive HO mechanisms based on thresholds and time-to-trigger further incur preparation and interruption latency and increase the risk of HO failures and ping-pong events [9]. Meanwhile, beam training and tracking become more error-prone under rapid topology changes and intermittent blockage, leading to excessive alignment overhead and reliability loss, especially near cell edges [10], [11].

Recent studies have mainly explored three directions. First, proactive handover (PHO) prepares HO in advance by forecasting traffic flows, vehicle trajectories, or channel states, thereby reducing failures and ping-pong events [9], [12]. With machine learning (ML), historical channel state information (CSI) can be leveraged to predict future positions [13], channel conditions [14], and target BSs [15], which further enhances PHO. Second, ML-assisted beamforming (BF) exploits side information such as out-of-band CSI [11] and context information [16] to infer blockage and narrow beam search space while maintaining robustness and throughput [17]. Reinforcement learning can also adapt beam training and tracking policies [18]. Third, queue-aware resource allocation (RA) prioritizes users by urgency or deadlines and opportunistically harnesses favorable channels to improve throughput and robustness under power budgets [19], [20].

Several studies also coordinate two or more of these decisions. Khosravi et al. [21] use deep Q-learning to coordinate HO and beam tracking, accounting for beam-training overhead in the effective throughput. Wang et al. [22] combine multi-agent reinforcement learning for HO triggering with optimization of target BS and beam selection and bandwidth allocation to improve throughput and delay performance. Heydarian et al. [23] coordinate user association, hybrid BF, and queue-aware user scheduling on multiple timescales. However, these studies primarily target throughput and delay performance, and do not explicitly address the joint coordination of PHO, BF, and RA for reducing the system transmit power while meeting latency requirements.

PHO and BF jointly determine the serving link gain and beam-training overhead. In HetNets, these quantities also depend on cross-tier differences in carrier frequencies, antenna configurations, and propagation conditions. Together with differences in bandwidth and transmit power across tiers, these factors affect the number of resource blocks (RBs)

The authors are with the Beijing National Research Center for Information Science and Technology, Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: xbw22@mails.tsinghua.edu.cn; {sheng.zhou, niuzhs}@tsinghua.edu.cn).

required to serve a given traffic demand and the resulting power consumption, while the available RB budgets constrain service feasibility. Moreover, HO changes the traffic loads of the source and target BSs, which can alter their resource usage and intra-tier co-channel interference, thereby affecting the resource demands of other links. This motivates energy-efficient Coordination of handOver, Beamforming and Resource Allocation (COBRA) in vehicular HetNets.

This paper extends our earlier conference work on joint PHO and RA [13]. Unlike the conference version, this paper incorporates MIMO channels, beam-pair selection, and beam-training overhead into the joint optimization. We also change the ML prediction targets from vehicle positions to beam-pair indices and link gains. Moreover, the simulation dataset uses ray-traced microcell MIMO channels instead of the statistical channel model adopted in the conference version.

We consider a two-tier vehicular HetNet, and formulate a COBRA optimization problem to minimize the long-term system transmit power subject to a queue-length-based latency constraint. This problem is a two-timescale stochastic integer non-linear program (SINLP). The frame-level PHO decisions are tightly coupled with slot-level BF and RA; the channel states and traffic arrivals are time-varying and only causally observable; and the decision space grows exponentially with the numbers of BSs, users, RBs, and BF codebook sizes. The NP-hardness of the resulting SINLP, together with the high overhead of acquiring timely and accurate per-link CSI, makes online global COBRA optimization computationally prohibitive.

To address these challenges, we develop MEET-COBRA, a prediction-driven coordination framework that jointly optimizes PHO, BF, and RA via a two-timescale closed loop. The key idea is to leverage low-overhead historical CSI measurements to predict next-frame beam-pair candidates and channel gains, and translate these predictions into per-link RB-demand estimates for globally optimized frame-level PHO. At the slot level, each BS performs prediction-guided BF and queue-aware RA to keep the realized link efficiencies consistent with these RB-demand estimates, thereby meeting the latency constraint with low transmit power.

To evaluate the proposed MEET-COBRA framework, we build a simulation platform that combines realistic vehicular mobility traces generated by SUMO [24] with ray-tracing-based channel data produced by Sionna RT [25]. Simulation results show that MEET-COBRA effectively reduces system transmit power and latency constraint violations across different traffic loads.

The rest of the paper is organized as follows: Section II details the system model; Section III formulates the optimization problem; Section IV presents the MEET-COBRA framework; Section V reports simulation results; and Section VI concludes the paper. The main notation is summarized in Table I.

II. SYSTEM MODEL

We consider a vehicular communication scenario covered by a two-tier HetNet adopting a control-data-plane separation architecture. The network consists of one macro BS and

TABLE I
MAIN NOTATION

Symbol	Description
m, v	Indices of BSs and vehicles
n, x, i	Indices of slots, frames, and within-frame slots
$\mathcal{M}, \mathcal{V}$	Sets of BSs and active vehicles
$N_{\text{HO}}, \Delta t$	Number of slots per frame and slot duration (s)
$\tau_{\text{HO}}, L_{\text{HO}}$	HO interruption duration (s) and number of HO interruption slots
δ_v, s_v	Binary indicators of HO execution and data-service availability
$c_{m,v}$	HO-aware RB-demand correction factor
K_m, B_m, p_m	Number of available RBs, bandwidth per RB (Hz), and transmit power per RB (W)
$u_{m,v}$	Binary indicator of BS-vehicle association
$k_{m,v}$	Number of allocated RBs
$\mathbf{H}_{m,v}$	MIMO channel matrix
$M_{\text{T}}, M_{\text{R}}$	Numbers of transmit/receive antennas
$\mathcal{C}_{\text{T}}, \mathcal{C}_{\text{R}}$	Transmit/receive beam codebooks
$\mathbf{w}_{\text{T}m,v}, \mathbf{w}_{\text{R}m,v}$	Transmit precoding/receive combining vectors
$g_{m,v}, g_{\text{I},m',v,v'}$	Instantaneous desired-/interfering-link gains
$G_{m,v}, G_{\text{I},m,v}$	Frame-level desired-/interfering-link gains
$\tilde{\mathcal{P}}_{m,v}$	Predicted probability distribution over beam pairs
N_{P}	Number of pilots per frame
M_{P}	Number of candidate beam pairs
η	Beam-search overhead factor
q_v, a_v	Amounts of queued and newly arriving data (bit)
λ_v	Mean data arrival rate (bps)
r_v	Effective receive rate (bps)
τ_v^{th}	Latency threshold (s)
U	Probability of queue-length constraint violation
$\bar{P}$	Time-averaged system transmit power (W)

Time indices are omitted for brevity. Symbols with tildes denote predicted or estimated quantities, which are not listed separately when their unmarked counterparts are included.

M micro BSs. The macro BS serves as the control-plane anchor and makes frame-level PHO decisions, while the micro BSs execute the corresponding data-plane re-association and perform slot-level BF and RA. The BS set is denoted by $\mathcal{M} = \{0, 1, \dots, M\}$, where index $m = 0$ represents the macro BS and $m > 0$ represents a micro BS. BS- m is allocated K_m orthogonal RBs in the frequency band f_m , where each RB has a bandwidth of B_m and a per-RB transmit power of p_m . For simplicity, all micro BSs are assumed to have identical frequency configurations, i.e., $K_m = K_\mu$, $f_m = f_\mu$ and $B_m = B_\mu$ for all $m > 0$, while the macro BS has distinct parameters K_0, f_0, B_0 .

We model the system in discrete time, indexed by slots $n \in \{0, 1, \dots\}$ of duration Δt . At any time slot n , the set of vehicles is $\mathcal{V}(n)$, with cardinality $V(n) = |\mathcal{V}(n)|$. The data-plane connection between BS- m and vehicle- $v \in \mathcal{V}(n)$ is represented by a binary indicator $u_{m,v}(n)$, where $u_{m,v}(n) = 1$ signifies the selected data-plane association and $u_{m,v}(n) = 0$ otherwise. Each vehicle can be associated with at most one BS on the data plane at a time, which imposes the constraint:

$$\sum_{m=0}^M u_{m,v}(n) \leq 1, \quad \forall v \in \mathcal{V}(n). \quad (1)$$

Note that the control-plane connection to the macro BS is

always maintained and thus is not explicitly modeled.

The data-plane connection state $u_{m,v}(n)$ is managed by a central PHO server deployed at the macro BS and updated periodically over a time window of N_{HO} slots, which define a PHO frame. The handover configuration for the x -th frame is determined during the preceding $(x-1)$ -th frame and remains static throughout frame x . Therefore, we introduce a frame-based index. Let $\{\cdot\}^{[x]} \triangleq \{\cdot\}(xN_{\text{HO}})$ denote a variable at the start of the x -th frame, and let $\{\cdot\}^{[x]}(i) \triangleq \{\cdot\}(xN_{\text{HO}} + i)$ be the variable at the i -th slot within that frame, where $i \in \mathcal{N}_{\text{HO}} \triangleq \{0, 1, \dots, N_{\text{HO}} - 1\}$. Consequently, the connection state is constant within any frame, which is expressed as:

$$u_{m,v}^{[x]}(i) = u_{m,v}^{[x]}, \quad \forall i \in \mathcal{N}_{\text{HO}}. \quad (2)$$

We further assume the set of vehicles is static throughout each frame, denoted as $\mathcal{V}^{[x]}$.

After a PHO decision is made, the signaling exchanges required for HO preparation are assumed to finish before the current frame ends, without occupying data-service resources. A vehicle switching BS is assumed to experience a data-service interruption of duration $\tau_{\text{HO}} = L_{\text{HO}}\Delta t$ caused by HO execution at the start of the next frame, where L_{HO} is a nonnegative integer less than N_{HO} . We assume successful HO execution for all PHO decisions. The macro control-plane connection and periodic CSI probing remain available during this data-plane HO interruption.

Let $\delta_v^{[x]} = 1$ if vehicle- v changes its serving BS from frame- $(x-1)$ to frame- x , and $\delta_v^{[x]} = 0$ otherwise. Its slot-level data-service availability is

$$s_v^{[x]}(i) = 1 - \delta_v^{[x]} \mathbf{1}\{i < L_{\text{HO}}\}, \quad (3)$$

where $\mathbf{1}\{\cdot\}$ is the indicator function.

A. Channel Model

We specify the channel models used to calculate the instantaneous desired-link gains $g_{m,v}^{[x]}(i)$ for both macro-to-vehicle ($m = 0$) and micro-to-vehicle ($m > 0$) links, as well as the **interfering-link** gains when applicable.

1) *Channel model for macro BS*: The macro BS operates in a lower-frequency band (e.g., the microwave band) to provide wide-area coverage. The frame-averaged channel gain between the macro BS and vehicle- v during the x -th frame, denoted as $G_{0,v}^{[x]}$, is modeled as

$$G_{0,v}^{[x]} = \alpha \left(d_{0,v}^{[x]}\right)^{-\beta} 10^{S_v^{[x]}/10}, \quad (4)$$

where α is a power gain factor, β is the path loss exponent, and $d_{m,v}^{[x]}$ is the distance between BS- m and vehicle- v in the x -th frame. The term $S_v^{[x]}$ captures the shadow fading effect, which is modeled as an independent and identically distributed (i.i.d.) zero-mean Gaussian random variable with variance σ_S^2 .

Based on the frame-averaged channel gain $G_{0,v}^{[x]}$, the instantaneous channel gain $g_{0,v}^{[x]}(i)$ in the i -th time slot of the x -th frame is modeled by Rician fading as

$$g_{0,v}^{[x]}(i) = G_{0,v}^{[x]} \left\| \sqrt{\frac{K_R}{K_R + 1}} + \sqrt{\frac{1}{K_R + 1}} h_{0,v}^{[x]}(i) \right\|^2, \quad (5)$$

where K_R is the Rician K-factor and $h_{m,v}^{[x]}(i) \sim \mathcal{CN}(0, 1)$ is an i.i.d. standard complex Gaussian random variable.

2) *Channel model for micro BS*: Each micro BS operates in a higher-frequency band (e.g., the mmWave band) and is equipped with a uniform linear array (ULA) of M_T antennas, while each vehicle has M_R antennas at this band. For the micro-to-vehicle links, we adopt a geometric stochastic MIMO channel model [26]. The channel between micro BS- m and vehicle- v during the x -th frame is represented by the frame-averaged channel matrix $\mathbf{H}_{m,v}^{[x]} \in \mathbb{C}^{M_R \times M_T}$, which is modeled as a sum of L distinct propagation paths:

$$\mathbf{H}_{m,v}^{[x]} = \sqrt{M_T M_R} \sum_{l=1}^L \gamma_l \mathbf{a}_R(\theta_l) \mathbf{a}_T(\psi_l)^\dagger, \quad (6)$$

where γ_l , θ_l , and ψ_l denote the complex impulse response, angle of arrival (AoA), and angle of departure (AoD) of the l -th path, respectively. The vectors $\mathbf{a}_R(\cdot) \in \mathbb{C}^{M_R}$ and $\mathbf{a}_T(\cdot) \in \mathbb{C}^{M_T}$ are the receive and transmit array response vectors. For the ULAs, the array response vector is given by

$$[\mathbf{a}(\phi)]_i = \frac{1}{\sqrt{N}} e^{-j \frac{2\pi d}{\lambda} i \sin \phi}, \quad i = 0, \dots, N-1, \quad (7)$$

where N is the number of antennas (M_T or M_R), d is the inter-element spacing (typically $d = \lambda/2$), λ is the carrier wavelength, and ϕ is the AoA or AoD (θ_l or ψ_l). The instantaneous channel matrix in the i -th slot of frame- x is modeled by applying small-scale Rician fading as

$$\mathbf{H}_{m,v}^{[x]}(i) = \left(\sqrt{\frac{K_R}{K_R + 1}} + \sqrt{\frac{1}{K_R + 1}} h_{m,v}^{[x]}(i) \right) \cdot \mathbf{H}_{m,v}^{[x]}. \quad (8)$$

Beamforming is employed for the MIMO links by selecting precoding and combining vectors from predefined codebooks. Specifically, we use discrete Fourier transform (DFT) codebooks. A DFT codebook of size N consists of N orthogonal vectors, $\mathcal{C} = \{\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{N-1}\}$, where each codeword $\mathbf{c}_n \in \mathbb{C}^{N \times 1}$ corresponds to a spatial direction. The i -th element of the n -th codeword is

$$[\mathbf{c}_n]_i = \frac{1}{\sqrt{N}} e^{-j \frac{2\pi n i}{N}}, \quad \forall n, i \in \{0, 1, \dots, N-1\}. \quad (9)$$

The transmit codebook $\mathcal{C}_T$ is a DFT codebook of size M_T , and the receive codebook $\mathcal{C}_R$ is a DFT codebook of size M_R .

When micro BS- m serves vehicle- v in the x -th frame, it selects a transmit precoding vector $\mathbf{w}_{T,m,v}^{[x]}(i) \in \mathcal{C}_T$, and vehicle- v selects a receive combining vector $\mathbf{w}_{R,m,v}^{[x]}(i) \in \mathcal{C}_R$. The resulting instantaneous desired-link channel gain is

$$g_{m,v}^{[x]}(i) = \left| \left(\mathbf{w}_{R,m,v}^{[x]}(i) \right)^\dagger \mathbf{H}_{m,v}^{[x]}(i) \mathbf{w}_{T,m,v}^{[x]}(i) \right|^2. \quad (10)$$

When another micro BS- m' serves vehicle- v' , the interfering-link gain from BS- m' to vehicle- v is

$$g_{I,m',v,v'}^{[x]}(i) = \left| \left(\mathbf{w}_{R,m',v'}^{[x]}(i) \right)^\dagger \mathbf{H}_{m',v}^{[x]}(i) \mathbf{w}_{T,m',v'}^{[x]}(i) \right|^2. \quad (11)$$

B. Resource Allocation Model

Let $k_{m,v}^{[x]}(i) \in \{0, 1, \dots, K_m\}$ denote the number of RBs allocated by BS- m to vehicle- v in the i -th time slot of the x -th frame. Since RBs can only be allocated to connected users without HO interruption, this variable must satisfy

$$k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} s_v^{[x]}(i) k_{m,v}^{[x]}(i), \quad \forall i \in \mathcal{N}_{\text{HO}}. \quad (12)$$

Each BS employs orthogonal frequency-division multiple access (OFDMA). Accordingly, the number of RBs utilized by BS- m , $k_m^{[x]}(i)$, is constrained by its total RB capacity K_m :

$$k_m^{[x]}(i) \triangleq \sum_{v \in \mathcal{V}^{[x]}} k_{m,v}^{[x]}(i) \leq K_m. \quad (13)$$

The instantaneous transmit power of BS- m is then given by

$$P_m^{[x]}(i) = p_m k_m^{[x]}(i). \quad (14)$$

The time-averaged system transmit power is the sum of the average power of all BSs:

$$\begin{aligned} \bar{P} &= \sum_{m=0}^M \bar{P}_m = \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{x=0}^{X-1} \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} \sum_{m=0}^M P_m^{[x]}(i) \\ &= \lim_{X \rightarrow \infty} \frac{1}{X N_{\text{HO}}} \sum_{x=0}^{X-1} \sum_{i \in \mathcal{N}_{\text{HO}}} \sum_{v \in \mathcal{V}^{[x]}} \sum_{m=0}^M p_m k_{m,v}^{[x]}(i). \end{aligned} \quad (15)$$

For a link with $k_{m,v}^{[x]}(i) > 0$, the signal-to-interference-plus-noise ratio (SINR) is modeled as

$$\text{SINR}_{m,v}^{[x]}(i) = \frac{p_{m,v}^{[x]}(i) g_{m,v}^{[x]}(i)}{F_m N_0 W_{m,v}^{[x]}(i) + I_{m,v}^{[x]}(i)}, \quad (16)$$

where $p_{m,v}^{[x]}(i) = k_{m,v}^{[x]}(i) p_m$ is the power allocated to the vehicle, $W_{m,v}^{[x]}(i) = k_{m,v}^{[x]}(i) B_m$ is the allocated bandwidth, N_0 is the noise power spectral density, F_m is the noise figure, and $I_{m,v}^{[x]}(i)$ is the inter-cell interference. For the macro BS, $I_{0,v}^{[x]}(i) = 0$ since its frequency band is orthogonal to those of micro BSs. Given the allocated RB counts, each micro BS assigns mutually disjoint RB sets to its associated vehicles uniformly at random, independently of the other micro BSs. The expected number of overlapping RBs between links (m, v) and (m', v') , with $m' \neq m$, is therefore $k_{m,v}^{[x]}(i) k_{m',v'}^{[x]}(i) / K_\mu$. Accordingly, the interference power averaged over the random RB assignments is given by

$$\begin{aligned} I_{m,v}^{[x]}(i) &= \begin{cases} \frac{k_{m,v}^{[x]}(i)}{K_\mu} \sum_{\substack{m' \neq 0, m \\ v' \in \mathcal{V}^{[x]}}} p_{m'} k_{m',v'}^{[x]}(i) g_{1,m',v,v'}^{[x]}(i), & m > 0, \\ 0, & m = 0. \end{cases} \end{aligned} \quad (17)$$

Under this average interference approximation, the link SINR is approximated as

$$\begin{aligned} \text{SINR}_{m,v}^{[x]}(i) &= \begin{cases} \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m + \sum_{\substack{m' \neq 0, m \\ v' \in \mathcal{V}^{[x]}}} \frac{p_{m'} k_{m',v'}^{[x]}(i)}{K_\mu} g_{1,m',v,v'}^{[x]}(i)}, & m > 0, \\ \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m}, & m = 0. \end{cases} \end{aligned} \quad (18)$$

The corresponding link rate is modeled using the Shannon formula as

$$R_{m,v}^{[x]}(i) = W_{m,v}^{[x]}(i) \log_2 \left(1 + \text{SINR}_{m,v}^{[x]}(i) \right). \quad (19)$$

When $k_{m,v}^{[x]}(i) = 0$, no data are transmitted on the link, and thus $R_{m,v}^{[x]}(i) = 0$. Considering the beam-searching overhead, the instantaneous receive rate of vehicle- v is given by

$$r_v^{[x]}(i) = \sum_{m=0}^M (1 - \eta_{m,v}^{[x]}(i)) R_{m,v}^{[x]}(i), \quad (20)$$

where $\eta_{m,v}^{[x]}(i) \in [0, 1)$ represents the throughput loss factor resulting from beam alignment. Specifically,

$$\eta_{m,v}^{[x]}(i) = \eta N_{m,v}^{[x]}(i), \quad (21)$$

with $N_{m,v}^{[x]}(i)$ denoting the number of beam-search attempts conducted by BS- m and vehicle- v in the i -th slot of frame- x , and η denoting the fraction of OFDM symbols in a slot required to probe one beam pair. Besides, the frame-averaged receive rate of vehicle- v is

$$\bar{r}_v^{[x]} \triangleq \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} r_v^{[x]}(i), \quad (22)$$

C. Traffic Model

We model the data traffic for each vehicle- v with a backlog queue, where $q_v(n)$ denotes its length at the beginning of time slot n . The queue length evolves as

$$q_v(n+1) = \max \{q_v(n) - r_v(n) \Delta t, 0\} + a_v(n), \quad (23)$$

where $a_v(n)$ is the amount of data arriving for vehicle- v during time slot- n . This model assumes that the existing backlog $q_v(n)$ is served first at a rate of $r_v(n)$, and new arrivals $a_v(n)$ are enqueued for service in subsequent slots. We denote the average data arrival rate for vehicle- v as λ_v , thus we have $\mathbb{E}[a_v(n)] = \lambda_v \Delta t$. We consider Poisson arrivals that are independent across vehicles and slots, with $a_v(n) \sim \text{Poisson}(\lambda_v \Delta t)$. This model does not capture temporally correlated traffic bursts.

Queueing delay is an important component of end-to-end latency, alongside transmission, propagation, and processing delays. We focus on queueing delay to assess how HO, BF, and RA jointly affect traffic accumulation through the available service rates. Motivated by Little's law [27], we define $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. Requiring this

proxy not to exceed the latency threshold τ_v^{th} of vehicle- v gives the queue-length-based latency constraint:

$$q_v(n) \leq \lambda_v \tau_v^{\text{th}}. \quad (24)$$

To evaluate the system performance with respect to this latency constraint, we define the latency constraint violation probability U following the approach in [28] as:

$$U = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \frac{\sum_{v \in \mathcal{V}(n)} U_v(n)}{V(n)}, \quad (25)$$

where $U_v(n) = 1$ if $q_v(n) > \lambda_v \tau_v^{\text{th}}$, and $U_v(n) = 0$ otherwise. Specifically, U measures the probability that the queueing-delay proxy $\tau_v(n)$ exceeds its threshold τ_v^{th} .

III. PROBLEM FORMULATION

Our goal is to minimize the long-term system transmit power while satisfying the latency-constrained QoS requirements for all vehicles. This joint optimization of HO, BF and RA over different time scales is formulated as the following problem

$$\begin{aligned} \mathbf{P1}: \quad & \min_{\mathbf{u}, \mathbf{k}, \mathbf{W}} \bar{P} \\ \text{s.t.} \quad & \text{C1: } q_v^{[x]}(i) \leq \lambda_v \tau_v^{\text{th}}, \\ & \text{C2: } \sum_{m=0}^M u_{m,v}^{[x]} \leq 1, \\ & \text{C3: } \sum_{v \in \mathcal{V}^{[x]}} k_{m,v}^{[x]}(i) \leq K_m, \\ & \text{C4: } k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} s_v^{[x]}(i) k_{m,v}^{[x]}(i), \\ & \text{C5: } u_{m,v}^{[x]} \in \{0, 1\}, \\ & \text{C6: } k_{m,v}^{[x]}(i) \in \{0, 1, \dots, K_m\}, \\ & \text{C7: } \mathbf{w}_{T,m,v}^{[x]}(i) \in \mathcal{C}_T, \mathbf{w}_{R,m,v}^{[x]}(i) \in \mathcal{C}_R. \end{aligned} \quad (26)$$

In **P1**, the decision variables are the sets of data-plane connection states $\mathbf{u} \triangleq \{u_{m,v}^{[x]}\}$, RB allocation numbers $\mathbf{k} \triangleq \{k_{m,v}^{[x]}(i)\}$, and BF vectors $\mathbf{W} \triangleq \{\mathbf{w}_{T,m,v}^{[x]}(i), \mathbf{w}_{R,m,v}^{[x]}(i)\}$. In our system, $\mathbf{u}$ is decided per frame, while $\mathbf{k}$ and $\mathbf{W}$ are decided per time slot. They are subject to several constraints. C1 is the queue-length-based latency constraint equivalent to Eq.(24). The HO constraints C2 and C5 enforce that each vehicle connects to at most one BS in the data-plane. The RA constraints C3, C4, and C6 limit the total RBs per BS, ensure RBs are only assigned to connected vehicles **without HO interruption**, and define the integer nature of $k_{m,v}^{[x]}(i)$. Finally, C7 restricts the BF vectors to be chosen from the predefined finite codebooks, which makes the BF decision a combinatorial integer optimization component.

P1 is a two-timescale SINLP. Obtaining its optimal *online* solution is computationally prohibitive due to the following challenges:

1. **Stochastic and causally observed dynamics:** Future system states, including the vehicle set $\mathcal{V}^{[x]}$, the channel matrices $\mathbf{H}_{m,v}^{[x]}(i)$, and the traffic arrivals $a_v(n)$, are time-varying and only causally observable. Moreover, current

decisions have lasting impacts on future power consumption and latency performance. Therefore, the long-horizon online optimization of **P1** is fundamentally challenging.

2. **Combinatorial decision space and NP-hardness:** All decision variables in **P1** are discrete and jointly define a high-dimensional decision space over BS-vehicle association, beam-pair selection, and RB allocation, further complicated by inter-cell interference coupling through the SINR expressions. Consequently, **P1** is an NP-hard integer nonconvex optimization problem whose feasible set grows exponentially with the numbers of BSs, vehicles, RBs, and BF codebook sizes, rendering exhaustive search or exact global optimization impractical.
3. **Coupled two-timescale optimization:** Frame-level HO decisions determine the association topology and constrain the slot-level BF and RA decisions within that frame, while the chosen BF vectors directly affect the desired-link gains, thereby influencing RB allocation and queue evolution. Such cross-timescale dependencies imply that optimizing PHO, BF, and RA in isolation can lead to persistently suboptimal performance, and call for low-complexity joint coordination mechanisms.
4. **Prohibitive CSI measurement overhead:** An online global optimizer would require timely and accurate per-link CSI for all BS-vehicle links and beam-pair options to evaluate channel gains and interference. In high-mobility vehicular scenarios, acquiring such per-link CSI via frequent channel measurements, beam training, and feedback incurs excessive overhead and latency, thus impractical for realization.

In summary, the NP-hardness of **P1**, together with stochastic dynamics, two-timescale coupling, and the prohibitive overhead of real-time per-link CSI acquisition, makes the online global optimization for **P1** computationally prohibitive.

IV. ALGORITHM DESIGN

To solve **P1** with an efficient and computationally feasible online approach, we propose a novel algorithm framework named Machine-learning-based Energy-Efficient proactive COBRA (MEET-COBRA). It leverages ML-predicted beam-pair candidates and channel gains to enable proactive, two-timescale COBRA optimization. The central PHO server determines the next-frame PHO configuration globally, while each BS performs slot-level BF and RA locally to serve its associated vehicles. The following subsections introduce the ML-based prediction module, the *predictive, frame-level* PHO algorithm, the *real-time, slot-level* BF and RA algorithms, and the overall MEET-COBRA procedure.

A. ML-Based Prediction

A neural network (NN) model is developed to predict, for each vehicle, the next-frame **best beam-pair indices**, **their corresponding desired-link gains**, and **interfering-link gains** to all micro BSs. To reduce CSI feedback overhead and avoid centralized NN inference latency that grows with the number of vehicles, each vehicle performs NN inference locally based on its historical CSI measurements obtained from

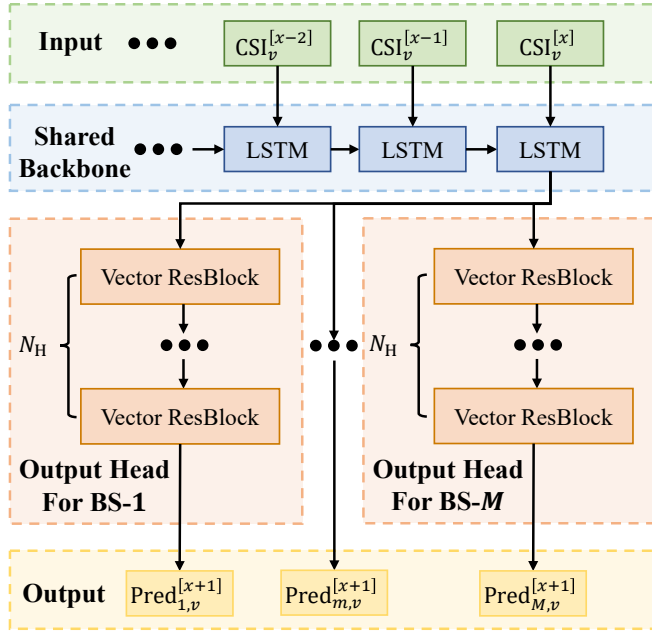


Fig. 1. Neural network architecture for predicting the next-frame best beam-pair index, its corresponding desired-link gain, and the interfering-link gain.

a low-overhead pilot probing procedure, and reports only the prediction results to the macro BS. The site-specific NN model is shared among vehicles by the macro BS over the control channel. **Deployment in a different propagation environment or BS layout requires retraining the prediction models using channel data from the target site.** As shown in Fig. 1, the NN consists of an input preprocessing layer, a shared feature-extraction backbone, and M micro-BS-specific output heads.

1) Input Layer: The NN input is a sequence of historical CSI measurements of vehicle- v . In each frame, the CSI is measured by a low-overhead pilot probing mechanism based on *pilot superposition*. Specifically, at the beginning of frame- x , all micro BSs simultaneously transmit a series of N_P pilot beams uniformly selected from the transmit codebook $\mathcal{C}_T$. In the p -th pilot probing ($p = 1, \dots, N_P$), the received signal at vehicle- v is the superposition of pilots from all micro BSs, i.e.,

$$\mathbf{y}_v^{[x]}(p) = \sum_{m=1}^M \mathbf{H}_{m,v}^{[x]} \mathbf{w}_{T,m}(p) s(p) + \mathbf{n}_v^{[x]}(p), \quad (27)$$

where $\mathbf{w}_{T,m}(p) \in \mathcal{C}_T$ is the transmit beam used by micro BS- m in the p -th probing, $s(p)$ is the pilot symbol, and $\mathbf{n}_v^{[x]}(p)$ is the received noise. Compared with exhaustive per-BS per-beam pilot sweeping with overhead $O(MM_T)$, pilot superposition reduces the measurement overhead to $O(N_P)$. Accordingly, the CSI measured by vehicle- v in frame- x is defined as

$$\text{CSI}_v^{[x]} \triangleq [\mathbf{y}_v^{[x]}(1)^T, \mathbf{y}_v^{[x]}(2)^T, \dots, \mathbf{y}_v^{[x]}(N_P)^T]^T \in \mathbb{C}^{M_R N_P}, \quad (28)$$

where $\mathbf{y}_v^{[x]}(p) \in \mathbb{C}^{M_R}$ is given in (27). Before being fed into the NN, $\text{CSI}_v^{[x]}$ is converted to a real-valued feature vector by normalizing amplitudes (in dB) and phases, and concatenating

them as $\mathbf{z}_{\text{in},v}^{[x]} \in \mathbb{R}^{2M_R N_P}$. This preprocessing operation is denoted as f_{prep} :

$$\mathbf{z}_{\text{in},v}^{[x]} = f_{\text{prep}}(\text{CSI}_v^{[x]}). \quad (29)$$

2) Shared Backbone: The shared backbone processes the historical CSI sequence of each vehicle to extract a compact representation of its mobility pattern and surrounding radio environment. It consists of a Long Short-Term Memory (LSTM) unit [29]. The LSTM unit, with N_C hidden cells, captures the temporal dependencies in the CSI by maintaining a hidden state $\mathbf{h}_v \in \mathbb{R}^{N_C}$ and a cell state $\mathbf{c}_v \in \mathbb{R}^{N_C}$. In frame- x , the LSTM unit updates these states based on the current CSI feature vector $\mathbf{z}_{\text{in},v}^{[x]}$ and the previous states ($\mathbf{h}_v^{[x-1]}, \mathbf{c}_v^{[x-1]}$):

$$(\mathbf{h}_v^{[x]}, \mathbf{c}_v^{[x]}) = f_{\text{LSTM}}(\mathbf{z}_{\text{in},v}^{[x]}, \mathbf{h}_v^{[x-1]}, \mathbf{c}_v^{[x-1]}; \theta_{\text{LSTM}}), \quad (30)$$

where f_{LSTM} denotes the LSTM transformations and θ_{LSTM} denotes the corresponding weights. **The states are retained across frames for each vehicle, so online inference processes only the current CSI feature vector.**

3) Per-BS Output Heads: The backbone hidden state $\mathbf{h}_v^{[x]}$ is fed into M micro-BS-specific output heads, one for each micro BS:

$$\text{Pred}_{m,v}^{[x+1]} = f_{\text{hd}}(\mathbf{h}_v^{[x]}; \theta_{\text{hd},m}), \quad \forall m \in \{1, 2, \dots, M\}, \quad (31)$$

where f_{hd} represents the common head structure, $\theta_{\text{hd},m}$ denotes the weights of the m -th head, and $\text{Pred}_{m,v}^{[x+1]}$ is the prediction for the link between micro BS- m and vehicle- v in frame- $(x+1)$. Each head comprises N_H stacked vector ResBlock units [30] with feature dimension N_C , followed by a task-specific final layer. For beam-pair prediction, the final layer has dimension $M_T M_R$, whereas for channel-gain prediction it has dimension 1.

4) Output Prediction: **Beam-pair and channel-gain prediction are implemented using independently trained NN models with a common backbone architecture and task-specific output layers:**

- 1. Beam Prediction Model:** **Beam-pair** prediction is formulated as a multi-class classification task. The final layer of each output head employs a softmax activation function to generate a predicted probability distribution $\hat{\mathcal{P}}_{m,v}^{[x+1]}$ over the $M_T M_R$ **beam pairs**. The model is trained with a cross-entropy loss, using a one-hot label whose non-zero entry indicates the index of the optimal **beam pair** that maximizes the **frame-averaged** desired-link gain in frame $(x+1)$.
- 2. Gain Prediction Models:** **Channel-gain** prediction is formulated as a regression task. We train two separate NN models with the same architecture but independent weights, one for the optimal desired-link gain and the other for the beam-averaged interfering-link gain. Each model has one output head per micro BS. In the desired-link gain model, each head outputs a scalar $\tilde{G}_{m,v}^{[x+1]}$ to approximate the **maximum frame-averaged** desired-link gain

$$G_{\text{opt},m,v}^{[x+1]} \triangleq \max_{\mathbf{w}_{T,m,v}^{[x+1]}, \mathbf{w}_{R,m,v}^{[x+1]}} G_{m,v}^{[x+1]}, \quad (32)$$

where $G_{m,v}^{[x+1]}$ is obtained by averaging the instantaneous desired-link gains in Eq. (10) over all slots in frame $(x+1)$. The interfering-link gain in (11) depends on the receive beam of the desired link and the transmit beam used by the interfering BS. To obtain a prediction target independent of these beam selections, we adopt the beam-averaged surrogate

$$\tilde{g}_{1,m',v}^{[x]} \triangleq \frac{1}{M_T M_R} \|\mathbf{H}_{m',v}^{[x]}(i)\|_F^2, \quad (33)$$

which equals the effective gain uniformly averaged over all beam pairs in the transmit and receive DFT codebooks. Each head of the interfering-link gain model outputs $\tilde{G}_{1,m',v}^{[x+1]}$, with training label $G_{1,m',v}^{[x+1]}$ defined as the frame average of $\tilde{g}_{1,m',v}^{[x+1]}(i)$. Both models are trained using a mean squared error (MSE) loss function, with predictions and labels represented in normalized dB.

For the macro BS ($m = 0$), whose low-frequency link does not involve beam alignment or inter-cell interference, only the frame-averaged desired-link gain $\tilde{G}_{0,v}^{[x+1]}$ is required. Owing to the slow variation of the macro-channel, this gain is approximated by the channel gain estimated from control-plane measurements in the current frame, without NN-based prediction.

B. Generalized Assignment-based Proactive Handover

This subsection proposes a generalized assignment-based proactive handover (GAP-HO) algorithm, detailed in Alg. 1, which determines the data-plane connection state for the upcoming frame. Its primary objective is to minimize the per-frame HetNet transmit power while satisfying both the latency-constrained QoS requirements of vehicles and the RB-capacity constraints of BSs.

To ensure queue stability, the expected frame-averaged receive rate $\mathbb{E}[\bar{r}_v^{[x]}]$ should be no less than the data arrival rate λ_v . The following theorem establishes a lower bound on the frame-averaged number of RBs required to satisfy this condition in the interference-free case.

Theorem 1. Consider vehicle- v associated with BS- m in the x -th frame. Given $G_{m,v}^{[x]}$, λ_v , B_m , p_m , F_m and N_0 , the frame-averaged number of allocated RBs, denoted by $\bar{k}_{m,v}^{[x]}$, required to ensure $\mathbb{E}[\bar{r}_v^{[x]}] \geq \lambda_v$ satisfies

$$\bar{k}_{m,v}^{[x]} \geq \lambda_v \left[B_m \log_2 \left(1 + \frac{p_m G_{m,v}^{[x]}}{F_m N_0 B_m} \right) \right]^{-1}, \quad (34)$$

where $\bar{k}_{m,v}^{[x]} \triangleq N_{\text{HO}}^{-1} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i)$.

Proof. Since vehicle- v is associated with BS- m in frame- x , we have $u_{m,v}^{[x]} = 1$ and $u_{m',v}^{[x]} = 0$ for all $m' \neq m$. Combining (12), (19), (20), and (22), the frame-averaged receive rate is

$$\bar{r}_v^{[x]} = \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i) B_m \log_2 \left(1 + \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m} \right). \quad (35)$$

For any constant $C > 0$, the function $f(z) = \log_2(1 + Cz)$ is concave on $z > 0$. By Jensen's inequality,

$$\mathbb{E} \left[\log_2 \left(1 + \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m} \right) \right] \leq \log_2 \left(1 + \frac{p_m \mathbb{E}[g_{m,v}^{[x]}(i)]}{F_m N_0 B_m} \right). \quad (36)$$

Moreover, by (10), $\mathbb{E}[g_{m,v}^{[x]}(i)] = G_{m,v}^{[x]}$. Taking expectation of $\bar{r}_v^{[x]}$ and using (36) yield

$$\begin{aligned} \mathbb{E}[\bar{r}_v^{[x]}] &= \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i) B_m \mathbb{E} \left[\log_2 \left(1 + \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m} \right) \right] \\ &\leq \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i) B_m \log_2 \left(1 + \frac{p_m G_{m,v}^{[x]}}{F_m N_0 B_m} \right) \\ &= \bar{k}_{m,v}^{[x]} B_m \log_2 \left(1 + \frac{p_m G_{m,v}^{[x]}}{F_m N_0 B_m} \right). \end{aligned} \quad (37)$$

Combining with $\mathbb{E}[\bar{r}_v^{[x]}] \geq \lambda_v$, we obtain the lower bound in (34). This completes the proof. $\square$

Based on Theorem 1, we now construct frame-averaged RB-demand estimates for each BS-vehicle pair- (m, v) in frame- $(x+1)$. For the macro BS ($m = 0$), whose low-frequency link is free of inter-cell interference and beam-alignment overhead, the required frame-averaged number of RBs for vehicle- v is estimated as

$$\tilde{k}_{0,v}^{[x+1]} = \lambda_v \left[B_0 \log_2 \left(1 + \frac{p_0 \tilde{G}_{0,v}^{[x+1]}}{F_0 N_0 B_0} \right) \right]^{-1}, \quad (38)$$

where $\tilde{G}_{0,v}^{[x+1]}$ is estimated from control-plane measurements in the current frame.

For each micro BS ($m > 0$), the lower bound in Theorem 1 is extended to account for inter-cell interference and beam-alignment overhead, while enhancing robustness against occasional large NN prediction errors. Given the NN predictions of the frame-averaged desired-link gains and interfering-link gain surrogates for frame- $(x+1)$, each gain is smoothed by weighted averaging with the corresponding gain measured in frame- x (when available); if no measurement exists, the NN prediction is used directly. With a slight abuse of notation, we still denote these robustness-enhanced gain estimates by $\tilde{G}_{m,v}^{[x+1]}$ and $\tilde{G}_{1,m',v}^{[x+1]}$. The effective SINR for vehicle- v served by BS- m in frame- $(x+1)$ is then approximated as

$$\widetilde{\text{SINR}}_{m,v}^{[x+1]} = \frac{p_m \tilde{G}_{m,v}^{[x+1]}}{F_m N_0 B_m + \sum_{m' \neq 0, m} \frac{\tilde{k}_{m',v}^{[x+1]}}{K_\mu} p_{m'} \tilde{G}_{1,m',v}^{[x+1]}}, \quad (39)$$

where $\tilde{k}_{m',v}^{[x+1]}$ denotes the estimated frame-averaged RB usage of BS- m' in frame- $(x+1)$ and K_μ is the RB capacity of each micro BS. To capture the beam-training overhead, we conservatively assume that at most M_P beam-search attempts are performed per slot, so that the effective data-transmission fraction per slot is at least $(1 - \eta M_P)$. Therefore, a conservative estimate of the required frame-averaged number of RBs is

$$\tilde{k}_{m,v}^{[x+1]} = \frac{\lambda_v}{(1 - \eta M_P) B_m \log_2(1 + \widetilde{\text{SINR}}_{m,v}^{[x+1]})}, \quad m > 0. \quad (40)$$

Algorithm 1 Generalized Assignment-based Proactive Handover (GAP-HO)

Input: $\tilde{G}_{m,v}^{[x+1]}, \tilde{G}_{1,m,v}^{[x+1]}, \lambda_v, p_m, B_m, F_m, K_m, N_0, K_\mu, N_{\text{iter}},$
 $M_P, \eta, \zeta, u_{m,v}^{[x]}, L_{\text{HO}}, N_{\text{HO}};$
Output: $u_{m,v}^{[x+1]},$

- 1: **Initialize** $\tilde{k}_m^{[x+1]} \leftarrow K_\mu$ for all $m > 0$;
 - 2: **Compute** $c_{m,v}^{[x+1]}$ for all $m \in \mathcal{M}$ and $v \in \mathcal{V}^{[x]}$ by (45);
 - 3: **Compute** $\tilde{k}_{0,v}^{[x+1]}$ for all $v \in \mathcal{V}^{[x]}$ by (38);
 - 4: **for** $n_{\text{iter}} = 1$ to N_{iter} **do**
 - 5: **Compute** $\tilde{k}_{m,v}^{[x+1]}$ for each $m > 0$ and $v \in \mathcal{V}^{[x]}$ by (40);
 - 6: **Form** **P2** with $\{\tilde{k}_{m,v}^{[x+1]}\}$ and $\{c_{m,v}^{[x+1]}\}$;
 - 7: **while** continuous relaxation of **P2** is infeasible **do**
 - 8: Multiply the right-hand side of (42) by ζ ;
 - 9: **end while**
 - 10: Apply the GAP approximation algorithm in [31] to solve **P2** and obtain a PHO schedule $\hat{u}_{m,v}^{[x+1]}$;
 - 11: **Compute** the **estimated** per-BS RB usage by $\hat{k}_m^{[x+1]} = \min \left\{ K_m, \sum_{v \in \mathcal{V}^{[x]}} \tilde{k}_{m,v}^{[x+1]} \hat{u}_{m,v}^{[x+1]} \right\}$ for all m ;
 - 12: $\tilde{k}_m^{[x+1]} \leftarrow \hat{k}_m^{[x+1]}$ for all m ;
 - 13: **end for**
 - 14: **while** $\hat{u}_{m,v}^{[x+1]}$ is not feasible **and** there exists a feasible reassignment **do**
 - 15: Execute the greedy feasibility-repair reassignment that yields the minimum power increase **subject to** (42);
 - 16: **end while**
 - 17: $u_{m,v}^{[x+1]} \leftarrow \hat{u}_{m,v}^{[x+1]}$;
 - 18: **return**
-

Since the interference term in (39) depends on the unknown per-BS RB usages $\{\tilde{k}_m^{[x+1]}\}$, we jointly compute the RB demands $\{\tilde{k}_{m,v}^{[x+1]}\}$ and the BS-vehicle associations $\{u_{m,v}^{[x+1]}\}$ via a fixed-point iteration over N_{iter} rounds, as summarized in Alg. 1. Specifically, we initialize $\tilde{k}_m^{[x+1]} = K_\mu$ for all $m > 0$, evaluate $\tilde{k}_{m,v}^{[x+1]}$ using (39) and (40), and then determine $u_{m,v}^{[x+1]}$ by solving the generalized assignment problem (GAP):

$$\text{P2:} \quad \min_{u_{m,v}^{[x+1]}} \sum_{m=0}^M \sum_{v \in \mathcal{V}^{[x]}} p_m \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \quad (41)$$

$$\text{s.t.} \quad \sum_{v \in \mathcal{V}^{[x]}} c_{m,v}^{[x+1]} \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \leq K_m, \quad (42)$$

$$\sum_{m=0}^M u_{m,v}^{[x+1]} = 1, \quad (43)$$

$$u_{m,v}^{[x+1]} \in \{0, 1\}, \quad (44)$$

where

$$c_{m,v}^{[x+1]} = \left[1 - \left(1 - u_{m,v}^{[x]} \right) \frac{L_{\text{HO}}}{N_{\text{HO}}} \right]^{-1}. \quad (45)$$

The factor $c_{m,v}^{[x+1]}$ accounts for the influence of HO interruption on the RB-capacity constraint. A vehicle switching BS receives no traffic data during the first L_{HO} slots of the next frame. It therefore requires more RBs per slot during the remaining $(N_{\text{HO}} - L_{\text{HO}})$ slots to serve the data accumulated during the HO interruption together with new traffic arrivals.

Since **P2** is NP-hard, we apply the polynomial-time approximation algorithm in [31], which applies a min-cost-matching-based rounding to the continuous relaxation solution of GAP, to obtain a near-optimal solution $\hat{u}_{m,v}^{[x+1]}$; if the continuous relaxation of **P2** is infeasible, we progressively relax (42) via the ζ -scaling procedure in Alg. 1 until feasibility is reached. Denoting $V \triangleq |\mathcal{V}^{[x]}|$, the computational complexity of this approximation algorithm is $O((M+V)V \log V)$. Moreover, as proved in [31], if **P2** is feasible and the optimal objective value is P^* , $\hat{u}_{m,v}^{[x+1]}$ satisfies an objective value no larger than P^* , while the RB-capacity constraints may be violated by at most a factor of two, i.e., $\sum_{v \in \mathcal{V}^{[x]}} c_{m,v}^{[x+1]} \tilde{k}_{m,v}^{[x+1]} \hat{u}_{m,v}^{[x+1]} \leq 2K_m$. We then update $\tilde{k}_m^{[x+1]} = \min \left\{ K_m, \sum_{v \in \mathcal{V}^{[x]}} \tilde{k}_{m,v}^{[x+1]} \hat{u}_{m,v}^{[x+1]} \right\}$ and **continue the fixed-point iteration until N_{iter} rounds are completed**. Finally, to strictly satisfy (42), a greedy feasibility-repair reassignment procedure offloads some vehicles from overloaded BSs (if exist) to underloaded BSs with minimum power increase; in the worst case, this reassignment incurs $O(MV^2)$ complexity. Therefore, the computational complexity of GAP-HO is $O(N_{\text{iter}}(M+V)V \log V + MV^2)$.

C. Predictive Early-Termination Beamforming

The predictive early-termination beamforming (PET-BF) algorithm in Alg. 2 is executed at the beginning of each time slot to select beam-pairs for the MIMO links between each micro BS and its associated vehicles. PET-BF exploits the NN-based predictions $\tilde{P}_{m,v}^{[x]}$ and $\tilde{G}_{m,v}^{[x]}$, obtained in the previous frame as described in Section IV-A, to guide a fast beam-training procedure in each slot of the current frame.

For each micro BS- m and its associated vehicle- v , PET-BF first selects the top- M_P beam-pairs with the highest probabilities from the predicted distribution $\tilde{P}_{m,v}^{[x+1]}$ over $M_T M_R$ candidates. This selection can be implemented via partial selection in $O(M_T M_R \log M_P)$ time. PET-BF then probes these candidates sequentially in descending order and applies an early-termination rule. Once the measured channel gain g_{meas} of a tested beam-pair exceeds the threshold $\tilde{G}_{m,v}^{[x]} - G_{\text{th}}$ in dB, that beam-pair is immediately selected and the search stops. If none of the M_P candidates satisfies the condition, the beam-pair with the largest measured gain among the probed candidates is selected. As a result, the probing-and-comparison cost is upper-bounded by $O(M_P)$ per link per slot.

This predict-then-refine strategy reduces the beam-training overhead $\eta_{m,v}^{[x]}(i)$ while tracking the instantaneous channel variations $g_{m,v}^{[x]}(i)$, thereby enabling robust and low-latency MIMO links for the considered V2X applications.

D. Opportunistic Tiered-Rounding Resource Allocation

For slot-level RA, we adopt an opportunistic tiered-rounding resource allocation (OTR-RA) algorithm, summarized in Alg. 3. In each slot, after PET-BF, each BS- m executes OTR-RA to allocate its RBs among the associated vehicles **without HO interruption** under the current channel conditions. The key idea is to exploit instantaneous RB efficiency for opportunistic scheduling, while using a queue-aware tiered-rounding rule for RB allocation.

Algorithm 2 Predictive Early-Termination Beamforming (PET-BF)

Input: $\tilde{\mathcal{P}}_{m,v}^{[x]}$, $\tilde{G}_{m,v}^{[x]}$, M_P , G_{th} , η ;

Output: $w_{T,m,v}^{[x]}(i)$, $w_{R,m,v}^{[x]}(i)$, $g_{m,v}^{[x]}(i)$, $\eta_{m,v}^{[x]}(i)$

- 1: Construct candidate beam-pair index set $\mathcal{W}$ by selecting the M_P largest entries of $\tilde{\mathcal{P}}_{m,v}^{[x]}$;
 - 2: **Initialize** $w_{best} \leftarrow \emptyset$, $g_{m,v}^{[x]}(i) \leftarrow -\infty$ and $\eta_{m,v}^{[x]}(i) \leftarrow 0$;
 - 3: **for all** $w \in \mathcal{W}$ (ordered by probability) **do**
 - 4: Probe beam-pair w to measure the instantaneous channel gain in dB: $g_{meas} \leftarrow \text{MeasureBeamGain}_{dB}(m, v, w)$;
 - 5: $\eta_{m,v}^{[x]}(i) \leftarrow \eta_{m,v}^{[x]}(i) + \eta$;
 - 6: **if** $g_{meas} > g_{m,v}^{[x]}(i)$ **then**
 - 7: $g_{m,v}^{[x]}(i) \leftarrow g_{meas}$; $w_{best} \leftarrow w$;
 - 8: **end if**
 - 9: **if** $g_{meas} > \tilde{G}_{m,v}^{[x]} - G_{th}$ **then**
 - 10: **break**;
 - 11: **end if**
 - 12: **end for**
 - 13: Map w_{best} to the corresponding $w_{T,m,v}^{[x]}(i)$ and $w_{R,m,v}^{[x]}(i)$;
 - 14: **return**
-

Algorithm 3 Opportunistic Tiered-Rounding Resource Allocation (OTR-RA)

Input: $u_{m,v}^{[x]}$, $s_v^{[x]}(i)$, $g_{m,v}^{[x]}(i)$, $\tilde{G}_{I,m',v}^{[x]}$, $\eta_{m,v}^{[x]}(i)$, $\tilde{k}_{m'}^{[x]}$, $q_v^{[x]}(i)$, K_m , B_m , Δt , p_m , F_m , N_0 , λ_v , τ_v^{th} ;

Output: $k_{m,v}^{[x]}(i)$;

- 1: **Initialize** $\bar{k}_{m,v}^{[x]}(i) \leftarrow 0$ for all $v \in \mathcal{V}^{[x]}$, $\bar{K}_m \leftarrow K_m$ and $\mathcal{V}_m^{[x]}(i) \leftarrow \{v \in \mathcal{V}^{[x]} \mid u_{m,v}^{[x]} s_v^{[x]}(i) = 1\}$;
 - 2: Compute $\Gamma_{m,v}^{[x]}(i)$ for all $v \in \mathcal{V}_m^{[x]}(i)$ by (46) and (47);
 - 3: Sort $\mathcal{V}_m^{[x]}(i)$ in descending order of $\Gamma_{m,v}^{[x]}(i)$ to obtain $\mathcal{S}_m^{[x]}$;
 - 4: **for all** $v \in \mathcal{S}_m^{[x]}$ **do**
 - 5: Compute $k_{m,v}^{[x]}(i)$ by (48) and (49);
 - 6: $\bar{K}_m \leftarrow \bar{K}_m - k_{m,v}^{[x]}(i)$;
 - 7: **end for**
 - 8: **return**
-

BS- m estimates the slot-level SINR using the measured desired-link gain, the predicted frame-averaged interfering-link gain, and the estimated RB usages obtained from GAP-HO:

$$\widetilde{\text{SINR}}_{m,v}^{[x]}(i) = \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m + \sum_{m' \neq m} \frac{\tilde{k}_{m'}^{[x]}}{\bar{K}_m} p_{m'} \tilde{G}_{I,m',v}^{[x]}}. \quad (46)$$

Here, $\tilde{G}_{I,m',v}^{[x]}$ is predicted in frame $(x-1)$ and used throughout frame- x . The corresponding RB efficiency is

$$\Gamma_{m,v}^{[x]}(i) \triangleq B_m \log_2(1 + \widetilde{\text{SINR}}_{m,v}^{[x]}(i)). \quad (47)$$

After computing $\Gamma_{m,v}^{[x]}(i)$ for all **available** associated vehicles, BS- m sorts them in descending order of $\Gamma_{m,v}^{[x]}(i)$ so that links with higher instantaneous RB efficiency are served first. The SINR and RB-efficiency evaluation requires $O(MV_m^{[x]})$ operations due to the summation over interfering micro BSs for each associated vehicle, and the sorting costs $O(V_m^{[x]} \log V_m^{[x]})$.

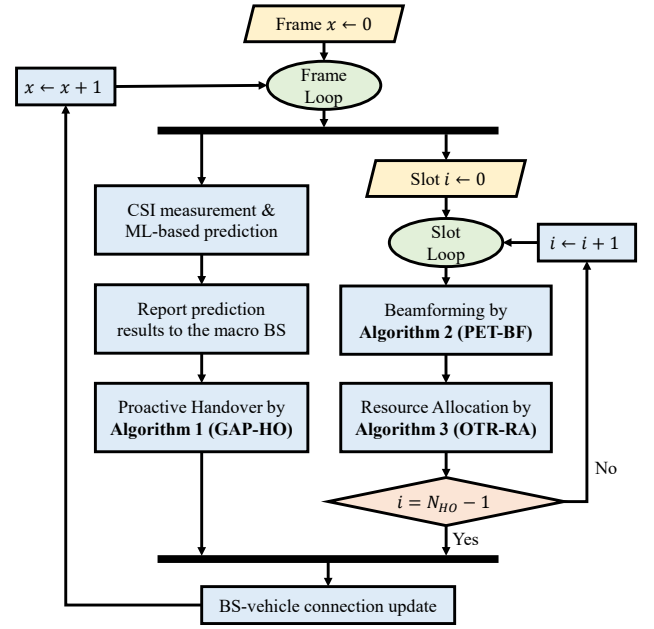


Fig. 2. Flowchart of the MEET-COBRA framework.

Given this ordered list, OTR-RA allocates RBs sequentially. For each vehicle- v , an ideal continuous RB demand to clear its backlog in slot- i is

$$\bar{k}_{m,v}^{[x]}(i) = \frac{q_v^{[x]}(i)}{(1 - \eta_{m,v}^{[x]}(i)) \Delta t \Gamma_{m,v}^{[x]}(i)}, \quad (48)$$

which is then mapped to an integer RA decision $k_{m,v}^{[x]}(i)$ via a queue-aware tiered-rounding rule, subject to the remaining RB budget $\bar{K}_m$:

$$k_{m,v}^{[x]}(i) = \begin{cases} \min\{\lceil \bar{k}_{m,v}^{[x]}(i) \rceil, \bar{K}_m\}, & \text{if } q_v^{[x]}(i) > \frac{\lambda_v \tau_v^{th}}{2}, \\ \min\{\lfloor \bar{k}_{m,v}^{[x]}(i) \rfloor, \bar{K}_m\}, & \text{otherwise.} \end{cases} \quad (49)$$

Then the budget updates as $\bar{K}_m \leftarrow \bar{K}_m - k_{m,v}^{[x]}(i)$. This tiered-rounding strategy avoids systematic RB waste that would occur if all $\bar{k}_{m,v}^{[x]}(i)$ were rounded up, while also preventing high-urgency users with $\bar{k}_{m,v}^{[x]}(i) < 1$ from repeatedly receiving no RBs under a purely rounding-down rule. The process continues until all the **available** associated vehicles have been scheduled or the RB budget is exhausted. Since each vehicle is processed at most once, the sequential allocation and budget updates cost $O(V_m^{[x]})$. Therefore, the overall per-slot computational complexity of OTR-RA at BS- m is $O(MV_m^{[x]} + V_m^{[x]} \log V_m^{[x]})$, making it a low-complexity RA scheme that balances RB efficiency and queue urgency under per-BS RB-capacity constraints.

E. MEET-COBRA

The proposed MEET-COBRA framework integrates the above modules into a unified two-timescale design, as illustrated in Fig. 2. Frame-level PHO decisions are computed

centrally at the macro BS, while slot-level BF and RA are executed locally at each BS.

At the beginning of frame- x , each vehicle performs local NN inference based on its historical CSI measurements to generate next-frame predictions $\{\hat{\mathcal{P}}_{m,v}^{[x+1]}, \hat{G}_{m,v}^{[x+1]}, \hat{G}_{l,m,v}^{[x+1]}\}$. Each vehicle reports only the top- M_P beam-pair indices in descending order of predicted probability and the two gain predictions for each micro BS to the macro BS through the control-plane uplink. Using these predictions, the macro BS executes GAP-HO to compute the PHO decisions $u_{m,v}^{[x+1]}$ for the next frame. The macro BS also forwards the interfering-link gain predictions and estimated RB usages required by OTR-RA to the corresponding serving BSs before frame- $(x+1)$ begins.

Within each slot- i of frame- x , each micro BS- m first applies PET-BF to probe the predicted top- M_P beam-pairs with early termination, thereby obtaining BF vectors $\{\mathbf{w}_{Tm,v}^{[x]}(i), \mathbf{w}_{Rm,v}^{[x]}(i)\}$ and instantaneous link gains $g_{m,v}^{[x]}(i)$ with limited beam-training overhead $\eta_{m,v}^{[x]}(i)$. After that, each BS applies OTR-RA to prioritize users by their RB efficiency $\Gamma_{m,v}^{[x]}(i)$ and perform queue-aware tiered-rounding to obtain integer RB allocations $k_{m,v}^{[x]}(i)$ under the RB-capacity constraints.

Remarkably, ML-based prediction serves as the coupling mechanism that aligns the modeling assumptions across HO, BF, and RA. The gain predictions enable GAP-HO to estimate the next-frame RB demands of each BS-vehicle pair and to solve **P2** based on a forward-looking characterization of the network state, while the beam-pair predictions guide PET-BF to realize high-quality beam alignment under reduced beam-training overhead. This improves the consistency between the link gains assumed in the RB-demand estimation of GAP-HO and the link gains realized during data transmission. Based on these realized link gains, OTR-RA further exploits opportunistic scheduling and queue-aware tiered rounding to enhance latency performance and reduce RB wastage, thereby narrowing the gap between the RB demands predicted by GAP-HO and the realized long-term RB usage. Consequently, MEET-COBRA provides a low-complexity online solution that effectively addresses the original problem **P1**.

Prediction errors can also affect this coordination. With other inputs fixed, overestimating the desired-link gain or underestimating the interfering-link gains lowers the estimated RB demand in (40), whereas errors in the opposite direction raise it. These changes affect the estimated power costs and RB requirements in GAP-HO, and can change BS-vehicle associations. In PET-BF, beam prediction errors affect the candidate beam-pair list and the search order, while desired-link gain errors affect the early-termination threshold. Both types of error can affect the gain of the selected beam pair and the beam-search overhead. OTR-RA uses measured desired-link gains, but errors in predicted interfering-link gains and estimated BS RB usages can affect RB-efficiency estimates, user ordering, and RB allocations. These effects can change transmit power and received rates, thereby influencing queue evolution and the violation probability U .

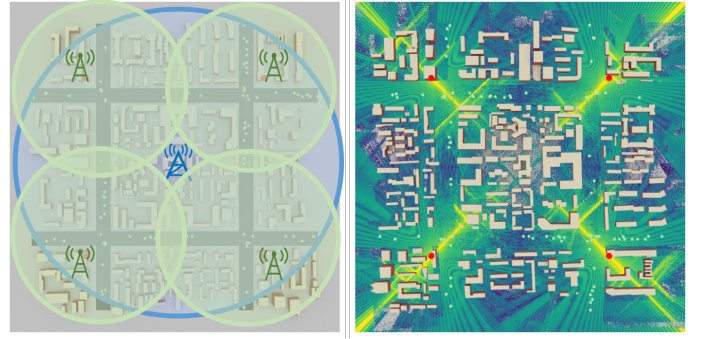


Fig. 3. Simulation scenario of the considered two-tier HetNet. Left: road layout, buildings, and locations and coverage regions of the macro BS (blue) and four micro BSs (green). Right: example Sionna RT ray-tracing-based received-power map at 28 GHz with vertically polarized antennas and micro-BS ULAs oriented toward the area center.

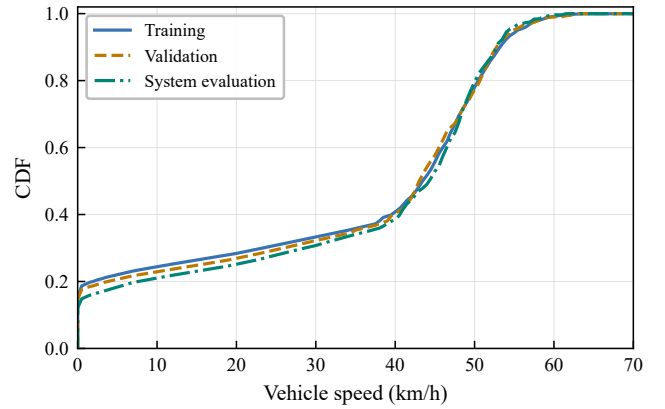


Fig. 4. Vehicle-speed CDFs over vehicle-frame samples from the training, validation, and system evaluation datasets.

V. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed MEET-COBRA framework using a simulation platform built on the SUMO [24] and the Sionna RT [25] simulators. We first describe the simulation setup and parameter configuration, and then present numerical results that compare MEET-COBRA with several baseline schemes in terms of power consumption and latency performance.

A. Simulation Setup

We consider a $1000 \text{ m} \times 1000 \text{ m}$ urban HetNet scenario shown in Fig. 3, where one macro BS is deployed at the center $(0, 0)$ and four micro BSs are located at $(\pm 300, \pm 300) \text{ m}$. The road topology consists of four intersections centered at $(\pm 250, \pm 250) \text{ m}$, and each road segment is bidirectional with three lanes per direction. Each micro BS employs a ULA oriented toward the scenario center. All antennas are vertically polarized. SUMO generates the vehicle trajectories, which are used by Sionna RT to obtain the time-varying MIMO channel matrices between vehicles and micro BSs. The large-scale channel gains between vehicles and the macro BS are generated using the 3GPP TR 38.901 urban macro model [32].

TABLE II
SIMULATION PARAMETERS

Global simulation parameters		Macro- and micro-BS parameters		
Parameter	Value	Parameter	Macro BS	Micro BS
Slot duration Δt	1 ms	Carrier frequency f_m	2.8 GHz	28 GHz
PHO frame length N_{HO}	100 slots	Number of RBs K_m	133	66
Pilots per frame N_P	8	RB bandwidth B_m	0.36 MHz	1.44 MHz
Rician factor K_R	3	Per-RB transmit power p_m	1 W	0.2 W
Beam-search overhead η	2/112	Noise figure F_m	5 dB	10 dB
Latency threshold τ_v^{th}	20 ms	Channel simulation	3GPP TR 38.901 UMa	Ray-tracing by Sionna RT [25]
HO interruption duration τ_{HO}	10 ms	Antenna number (Tx&Rx)	1&1	32&8
Number of GAP-HO iterations N_{iter}	2	Number of BSs	1	4
RB capacity relaxation factor ζ	1.1	Beamforming codebooks	Not applicable	DFT
Noise power spectral density N_0	-174 dBm/Hz	Number of candidate beam pairs M_P	Not applicable	5
System simulation duration per run	30 s	PET-BF gain margin G_{th}	Not applicable	5 dB

Fig. 4 shows the cumulative distribution functions (CDFs) of vehicle speed in the training, validation, and system evaluation datasets. The respective mean speeds are 33.56, 33.95, and 34.78 km/h. Samples with zero speed account for 14.97%, 14.64%, and 12.07% of the corresponding vehicle-frame observations. Vehicle speeds in the system evaluation dataset range from 0 to 61.53 km/h.

For the system-level performance comparisons in Section V-D, we vary the traffic arrival rate $\lambda_v = \lambda$ over $\{1, 3, \dots, 35\}$ Mbps and set $\tau_v^{\text{th}} = 20$ ms for all vehicles. For all schemes, receive rates are evaluated using actual RB overlaps and the selected transmit and receive beams. We set $N_{\text{iter}} = 2$ to balance performance and computational cost, and $\zeta = 1.1$ for 10% increments in the RB capacity bounds when the continuous relaxation of **P2** is infeasible. We use $M_P = 5$ and $G_{\text{th}} = 5$ dB to balance coverage of the optimal beam pair against beam search overhead, which occupies at most $\eta M_P = 8.93\%$ of each slot. Unless otherwise stated, the simulations use the parameter values listed in Table II.

B. Performance of ML-Based Prediction

We evaluate the NN models in Section IV-A for beam-pair and gain prediction. We set $N_C = 128$ for the LSTM unit and the vector ResBlock, with $N_H = 4$ vector ResBlocks per output head. The dataset is built from Sionna RT-generated channel traces spanning 6000 frames and contains 646,747 labeled samples from 706 vehicles. Each sample consists of the current-frame CSI of a vehicle, together with labels of the optimal beam-pair indices and the desired-link and beam-averaged interfering-link gains for the next frame. The dataset is randomly partitioned at the vehicle level, with 70% of the vehicles in the training set and the remaining 30% in the validation set.

All models are trained offline in two stages of 100 epochs each, using AdamW [33] with a weight decay of 10^{-4} . In the first stage, the models are randomly initialized and trained using the input windows of one to ten consecutive CSI frames at a learning rate of 10^{-3} . In the second stage, the models are initialized from the best first-stage checkpoints for stateful fine-tuning. This stage uses an initial learning rate of 10^{-4} and truncated backpropagation through time [34] with a truncation length of ten frames. Recurrent states $\mathbf{h}_v$ and $\mathbf{c}_v$ are carried across frames and reset only at the start of a new trajectory.

TABLE III
NN PARAMETER COUNTS AND INFERENCE FLOPS

Predictor	Parameters	Matrix FLOPs (10^6)
Beam-pair	1,060,864	2.10
Desired-link gain	533,524	1.05
Interfering-link gain	533,524	1.05
Total	2,127,912	4.20

Matrix FLOPs count LSTM and linear-layer matrix operations, with two FLOPs per multiply-accumulate.

Both training and validation use finite input windows in the first stage and process samples in time order along each vehicle trajectory in the second. In both stages, the beam checkpoint is selected by the highest top-1 validation accuracy and each gain checkpoint by the lowest validation MAE. The selected second-stage checkpoints are used in all subsequent evaluations.

Fig. 5 illustrates the training and validation performance of the beam and gain prediction models. Stateful fine-tuning increases the best top-1 validation accuracy of beam prediction from 86.68% to 89.16% and achieves a best top-3 validation accuracy of 98.65%. For the selected beam checkpoint, the validation top- M_P accuracy exceeds 99% for $M_P \geq 4$ and 99.9% for $M_P \geq 17$, indicating that a short candidate list contains the optimal beam pair with high probability. Stateful fine-tuning also reduces the best validation MAEs of desired-link and beam-averaged interfering-link gain prediction from 4.79 and 3.85 dB to 3.11 and 2.51 dB, respectively. These results show improved beam and gain prediction after stateful fine-tuning. We further assess the selected models across four speed groups of $[0, 1)$, $[1, 20)$, $[20, 40)$, and ≥ 40 km/h on the system evaluation dataset. From the lowest to the highest speed group, the desired-link gain MAEs are 1.13, 2.31, 3.15, and 3.73 dB, and the corresponding interfering-link gain MAEs are 1.19, 1.79, 2.32, and 3.07 dB. Top-5 beam prediction accuracy exceeds 99% in all four groups.

Table III summarizes the NN parameter counts and matrix FLOPs for single-vehicle stateful inference using the current-frame CSI and recurrent states. The three models require 8.51 MB of 32-bit floating-point (FP32) parameter storage. With FP32 inference on an AMD EPYC 7742 CPU using one thread, their combined inference latency has a median

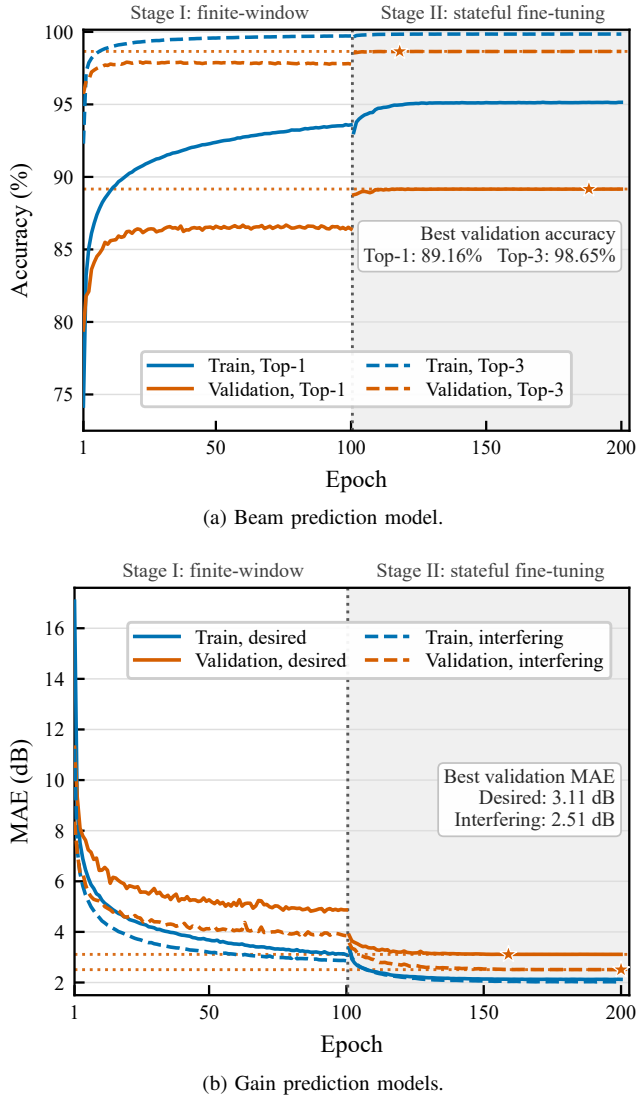


Fig. 5. Training and validation performance of the proposed NN models: (a) beam prediction model (top-1 and top-3 accuracies); (b) optimal desired-link and interfering-link gain prediction models (MAE). The vertical dotted line separates finite-window training (epochs 1–100) and stateful fine-tuning (epochs 101–200). Stars mark the best validation value of each metric.

of 5.38 ms and a 95th percentile of 5.72 ms. With $M = 4$, $M_P = 5$, and $M_T M_R = 256$, each vehicle reports, for each micro-BS link, M_P beam-pair indices in descending order of predicted probability, using 8 bits per index, together with one FP32 desired-link gain and one FP32 interfering-link gain. The reporting overhead is $M(8M_P + 2 \times 32) = 416$ bits per vehicle per frame (4.16 kbps). At the same reporting interval and 32-bit precision per real CSI component, this reduces reporting overhead by factors of 157.5 and 9.85 relative to full MIMO-CSI reporting and centralized inference using current-frame superposed-pilot observations, respectively.

C. Benchmark Schemes

To assess the performance of MEET-COBRA, we compare it with the following benchmark schemes:

1) **Oracle-MC**: MEET-COBRA with perfect beam and gain knowledge, where all ML predictions are replaced by

their ground-truth values from the simulator. It quantifies the performance loss due to prediction errors.

2) **Reactive-OBRA**: A reactive HetNet baseline without ML prediction or proactive coordination. Using current channel gains, it assigns vehicles sequentially to the BS with the lowest estimated transmit power cost among those with sufficient remaining RB capacity, falling back to the macro BS if none is available. BF uses a random opportunistic search inspired by [35]: after HO it probes M_P beam-pairs and keeps randomly probing thereafter, updating the serving beam only when a better one is found. RA estimates RB efficiency assuming full-load interfering micro BSs and applies stochastic rounding for integer RB allocation.

3) **Ablations**: *w/o GAP-HO*, *w/o PET-BF*, and *w/o OTR-RA*, where the corresponding module in MEET-COBRA is replaced by its Reactive-OBRA counterpart, while keeping the other two modules unchanged.

4) **O-MAPPO-adapted**: This baseline adapts the O-MAPPO framework in [22], which combines reinforcement learning with target BS optimization. A shared actor trained by multi-agent proximal policy optimization evaluates whether each vehicle should trigger HO after every 10 m of travel. For vehicles with active HO triggers, a mixed-integer optimizer selects target BSs using estimated RB demands, BS loads, and transmit power costs, with penalties for aggregate demands exceeding BS capacities. The actor and optimizer use the same predicted desired-link and interfering-link gains as MEET-COBRA. Association changes take effect in the next frame. Upon HO to a micro BS, hierarchical beam search starts after the HO interruption. It first probes all combinations of eight transmit and two receive sectors, then all combinations of four transmit and four receive DFT beams within the selected sectors, requiring 32 probes in total. In subsequent slots, local beam tracking probes the current beam pair and four adjacent pairs differing in only one beam index, and then retains the pair with the highest measured gain. Each BS applies OTR-RA in every slot.

5) **MTS-GS-HBF-adapted**: This baseline adapts the multi-timescale association and hybrid BF framework in [23]. Gale-Shapley matching updates BS–vehicle associations every five frames using the same predicted channel gains as MEET-COBRA. Matching preferences account for estimated rates, BS loads, queue backlogs, and energy and HO costs, while association decisions consider estimated RB demands and BS capacities. We replace the original hybrid BF with the hierarchical beam search and local tracking used in O-MAPPO-adapted. Unlike O-MAPPO-adapted, hierarchical beam search is repeated in the first available slot of every frame, after the HO interruption when applicable, followed by local beam tracking in the remaining slots. Each BS applies OTR-RA in every slot.

D. Performance Comparison

This subsection compares MEET-COBRA with the benchmark schemes in terms of transmit power and latency performance. At each data arrival rate, we run all schemes over the same 30 s mobility trace with three random seeds. Each

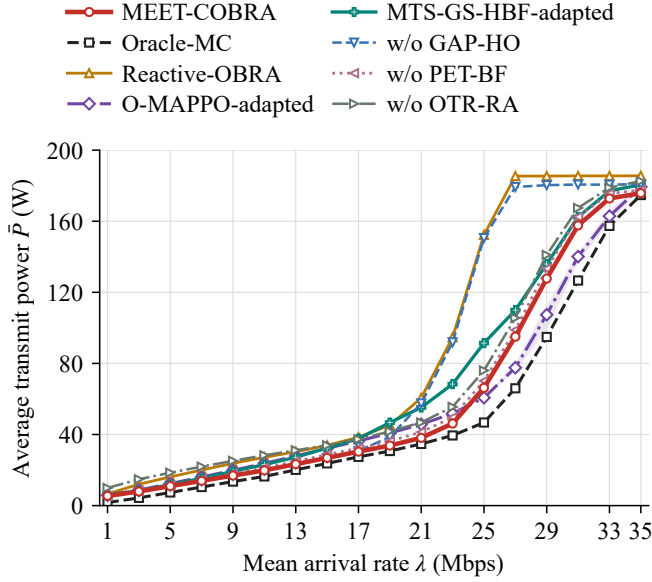


Fig. 6. Average system transmit power $\bar{P}$ versus the data arrival rate λ .

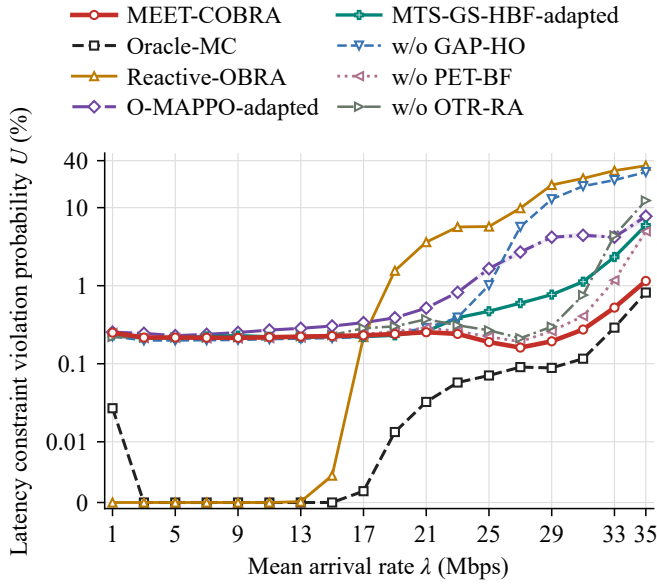


Fig. 7. Latency constraint violation probability U versus the data arrival rate λ . The vertical scale is linear below 0.01% and logarithmic above it.

seed generates a different set of traffic arrivals and small-scale fading samples, which are shared by all schemes. Curves show the mean of the three runs, and shaded bands span their minimum and maximum values.

Figs. 6 and 7 compare the average system transmit power $\bar{P}$ and the latency constraint violation probability U , respectively, at different data arrival rates λ . Under light loads, the transmit power of MEET-COBRA increases approximately linearly with λ , as additional traffic can largely be accommodated by allocating more RBs without changing serving links. At $\lambda = 19$ Mbps, MEET-COBRA uses 33.80 W with $U = 0.24\%$,

compared with 30.71 W and 0.013% for Oracle-MC. This relatively small power gap suggests that prediction errors have a limited impact on transmit power when RB resources are sufficient. The power gap widens at 27 Mbps, where the two schemes use 95.03 and 66.00 W, respectively. This larger power gap suggests that accurate RB-demand estimates become more important for coordinating PHO and RA when RB resources are limited.

Compared with Reactive-OBRA, MEET-COBRA achieves much lower transmit power and fewer latency violations under heavy traffic. At $\lambda = 27$ Mbps, Reactive-OBRA approaches the full-load power ceiling, using 185.39 W with $U = 9.84\%$, whereas MEET-COBRA uses 95.03 W with $U = 0.16\%$. These correspond to reductions of 48.7% in transmit power and 98.4% in violation probability. Further traffic increases mainly raise U for Reactive-OBRA because its RB resources are already nearly fully utilized. MEET-COBRA maintains a low violation probability over a wider range of data arrival rates, although U also increases as its RB resources approach full utilization. From 33 to 35 Mbps, its power increases only from 172.81 to 175.80 W, while U rises from 0.52% to 1.15%. Nevertheless, at 35 Mbps, the violation probability of MEET-COBRA remains well below the 34.46% observed for Reactive-OBRA.

Compared with O-MAPPO-adapted, MEET-COBRA uses less power over 1–23 Mbps and incurs fewer violations at every evaluated load. Over 25–33 Mbps, O-MAPPO-adapted uses less power but incurs substantially more violations. At 29 Mbps, O-MAPPO-adapted uses 107.40 W with $U = 4.18\%$, compared with 127.75 W and 0.19% for MEET-COBRA.

Compared with MTS-GS-HBF-adapted, MEET-COBRA has lower mean power at every evaluated load and fewer violations at $\lambda \geq 21$ Mbps. MTS-GS-HBF-adapted achieves slightly fewer violations at several lower loads. Its largest advantage in U occurs at 1 Mbps, with 0.23% compared with 0.25% for MEET-COBRA. At 29 Mbps, MTS-GS-HBF-adapted uses 135.91 W with $U = 0.77\%$, both higher than the corresponding values of 127.75 W and 0.19% for MEET-COBRA. MTS-GS-HBF-adapted updates associations only every five frames, which can delay traffic redistribution when link conditions and queue backlogs change. Its periodic hierarchical beam search and local tracking also require more probes than PET-BF in MEET-COBRA. The additional probing leaves less time for data transmission, increasing the RB demand for a given link gain and amount of data to transmit.

Fig. 8 reports L_{90} and L_{99} , the 90th and 99th percentiles of the queueing-delay proxy $\tau_v(n)$ over vehicle-slot samples within each run. As the arrival rate increases, L_{99} can rise sharply while L_{90} remains small. The mean L_{99} first exceeds 20 ms at 35 Mbps for MEET-COBRA, compared with 19 Mbps for Reactive-OBRA, 25 Mbps for O-MAPPO-adapted and w/o GAP-HO, 31 Mbps for MTS-GS-HBF-adapted, and 33 Mbps for the other two ablations. Oracle-MC remains below this threshold throughout the evaluated range. At 33 Mbps, MEET-COBRA and MTS-GS-HBF-adapted have similar L_{90} values (1.24 and 1.19 ms), but their L_{99} values are 5.00 and 267.54 ms, respectively. When evaluated separately

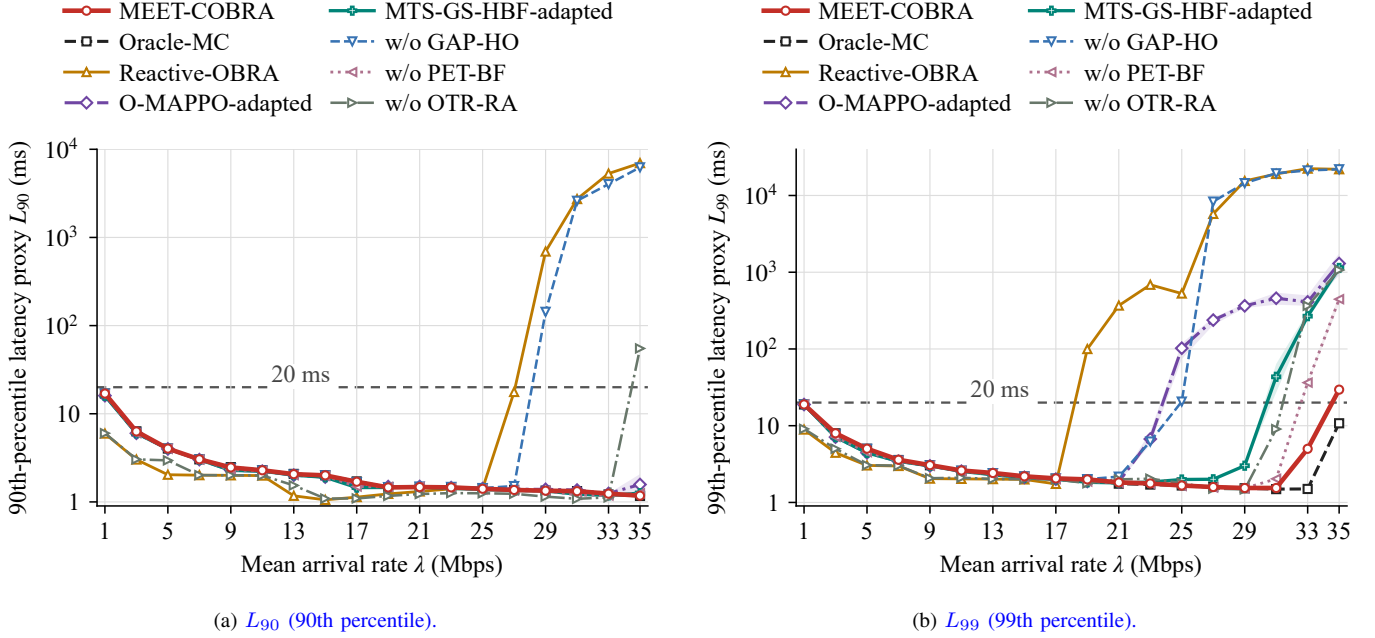


Fig. 8. The 90th and 99th percentiles of the queueing-delay proxy $\tau_v(n)$ versus the data arrival rate λ , where the horizontal dashed lines denote the latency threshold $\tau_v^{\text{th}} = 20$ ms.

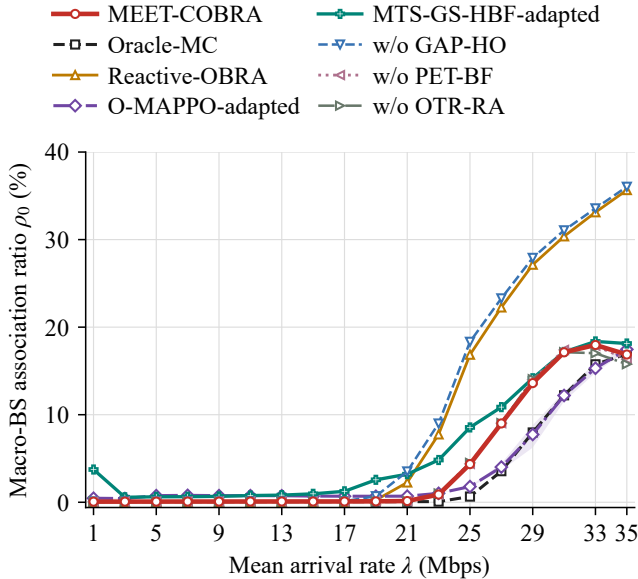


Fig. 9. Percentage of vehicles associated with the macro BS (BS-0) versus the data arrival rate λ .

for the speed groups $[0, 1)$, $[1, 20)$, $[20, 40)$, and ≥ 40 km/h, MEET-COBRA maintains a mean L_{99} below 20 ms in every group at all evaluated arrival rates up to 33 Mbps. At 35 Mbps, the corresponding values are 1.49, 11.19, 28.37, and 48.02 ms in increasing speed-group order. These results indicate that MEET-COBRA maintains low latency across the observed speed groups over most evaluated loads, while larger delays arise in the higher-speed groups at the highest load.

Fig. 9 reports the macro-BS association ratio ρ_0 . MEET-

COBRA serves almost all vehicles through micro BSs at light loads and offloads more traffic to the macro BS as the load increases. From 21 to 27 Mbps, ρ_0 rises from 0.14% to 9.00%, while U decreases from 0.25% to 0.16%. This non-monotonic variation is consistent with GAP-HO redistributing traffic as micro-BS capacity constraints become limiting. At 29 Mbps, MEET-COBRA maintains $\rho_0 = 13.61\%$ with $L_{99} = 1.54$ ms, whereas Reactive-OBRA associates 27.13% of vehicles with the macro BS and has $L_{99} > 10^4$ ms. These results show the benefit of coordinating PHO, BF, and RA to serve traffic efficiently through the micro tier while balancing the traffic offloaded to the macro BS.

However, a smaller macro-BS association ratio alone does not imply better load coordination. At 29 Mbps, O-MAPPO-adapted has $\rho_0 = 7.74\%$, compared with 13.61% for MEET-COBRA, but its L_{99} reaches 364.76 ms. At this load, the average micro-BS RB utilization is 97.2% for O-MAPPO-adapted, compared with 79.8% for MEET-COBRA. Retaining more traffic in the micro tier reduces the use of higher-power macro-BS RBs, but leaves little unused micro-BS capacity to clear accumulated queues. Moreover, O-MAPPO-adapted cannot reassign a vehicle without an active HO trigger, whereas GAP-HO reoptimizes BS-vehicle associations in every frame as resource demands change.

The ablation results in Figs. 6–9 further distinguish the contributions of the three modules of MEET-COBRA. Under light loads, w/o GAP-HO consumes almost the same power as MEET-COBRA, indicating limited additional power benefit from global PHO optimization when micro-BS RB resources are sufficient. As the load increases, its power and violation probability diverge from MEET-COBRA, reaching 179.27 W and 5.67% at 27 Mbps. At this load, removing GAP-HO

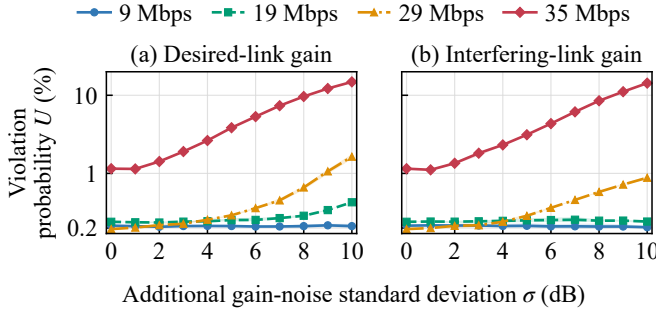


Fig. 10. Sensitivity of the latency constraint violation probability U to gain-prediction errors.

causes the largest increases in both metrics among the three ablations. Since w/o GAP-HO retains PET-BF and OTR-RA, this comparison highlights the importance of jointly coordinating BS-vehicle associations under capacity constraints, rather than assigning vehicles sequentially.

Replacing PET-BF with random opportunistic beam search increases power throughout the evaluated range. PET-BF uses predicted beam-pair candidates to find a sufficiently strong beam pair and limits beam-search overhead through early termination. At 33 Mbps, w/o PET-BF has $U = 1.17\%$ and $L_{99} = 36.41$ ms, compared with 0.52% and 5.00 ms for MEET-COBRA. These differences occur despite similar macro-BS association ratios (17.87% and 17.96%) and $L_{90} < 1.3$ ms for both schemes, showing how efficient BF complements PHO in maintaining a small L_{99} .

Removing OTR-RA causes the largest power increase among the ablations at light loads. Its replacement assumes full RB utilization at interfering BSs and uses stochastic rounding, whereas OTR-RA accounts for estimated RB usage and queue urgency. At 7 Mbps, w/o OTR-RA uses 21.92 W, compared with 13.95 W for MEET-COBRA. Although the additional RB allocation can reduce latency under light loads, it leaves fewer RBs available as traffic increases. At 33 Mbps, w/o OTR-RA has $L_{90} = 1.13$ ms and $\rho_0 = 17.02\%$, close to the corresponding values of 1.24 ms and 17.96% for MEET-COBRA, but U increases from 0.52% to 4.40% and L_{99} from 5.00 to 365.26 ms. By rounding up RB allocations for urgent queues and rounding down those for nonurgent queues, OTR-RA reduces prolonged queue accumulation while limiting unnecessary RB use.

E. Impact of Gain-Prediction Errors

This subsection evaluates how gain-prediction errors affect the system transmit power and latency performance of MEET-COBRA. We add independent zero-mean Gaussian noise with standard deviation $\sigma \in \{0, 1, \dots, 10\}$ dB to either the original desired-link or interfering-link gain predictions. Keeping the trained models, beam-pair candidates, actual channels and traffic arrivals unchanged, we rerun MEET-COBRA at $\lambda \in \{9, 19, 29, 35\}$ Mbps with three paired random seeds.

Fig. 10 shows that U changes little at 9 and 19 Mbps for $\sigma \leq 5$ dB. With $\sigma = 5$ dB, U at 29 Mbps increases from 0.19% to 0.29% when noise is added to either type of gain

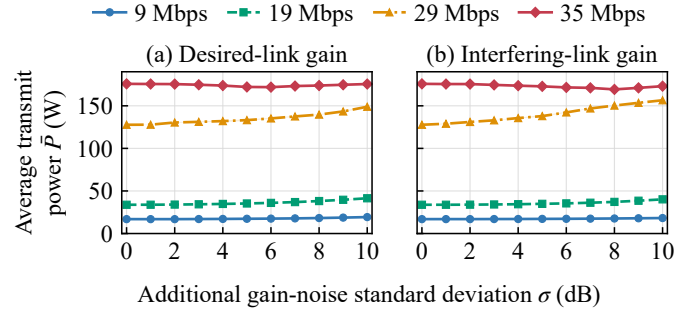


Fig. 11. Sensitivity of the average system transmit power $\bar{P}$ to gain-prediction errors.

prediction. At 35 Mbps, however, adding noise with $\sigma = 5$ dB raises U from 1.15% to 3.84% and 3.13% for desired-link and interfering-link gain predictions, respectively.

Fig. 11 shows that larger gain-prediction errors generally increase transmit power at the three lower traffic loads. At 19 Mbps, adding noise with $\sigma = 10$ dB to the interfering-link gain predictions increases power from 33.80 to 40.15 W, while U remains approximately 0.24%. At 35 Mbps, however, power remains close to the full-load level while U rises sharply.

We further study the influence of gain-prediction errors on HO and beam probing. At 29 Mbps, adding noise with $\sigma = 5$ dB to the desired-link gain predictions increases the mean number of HOs per frame from 3.70 to 20.01 and the mean number of probes per beam search from 1.06 to 1.95. These additional HOs and beam probes reduce the time available for data transmission. At high loads, limited remaining RB resources make it harder to compensate for this loss of transmission time through additional RB allocation.

VI. CONCLUSION

This paper proposed MEET-COBRA, a ML-based energy-efficient proactive coordination framework for jointly optimizing PHO, BF and RA in HetNets. By leveraging ML-based prediction of beam-pair candidates and channel gains, MEET-COBRA provides a low-complexity online approach to efficiently address the two-timescale SINLP for COBRA.

Simulations driven by SUMO mobility traces and Sionna RT channel data validate the effectiveness of MEET-COBRA. The predicted beam-pair and channel gain information enables efficient global PHO optimization, reduces beam-search overhead, and mitigates RB wastage and latency constraint violations caused by suboptimal BF and RA decisions, thereby addressing the computational challenges of the two-timescale SINLP in practical deployments. Notably, MEET-COBRA effectively balances system transmit power and latency performance. At an illustrative medium-to-high load operating point ($\lambda = 27$ Mbps), it reduces the average system transmit power by approximately 49% compared with the Reactive-COBRA baseline, while decreasing the latency constraint violation probability from 9.84% to 0.16%. Compared with the baselines adapted from [22] and [23], MEET-COBRA uses less transmit power at light and medium loads and achieves lower latency constraint violation probabilities at high loads.

It also maintains low violation probabilities under moderate additional gain-prediction errors at light and medium loads.

Potential future work includes extending MEET-COBRA to multi-macro BS deployments, modeling HO failure recovery and detailed signaling procedures, and developing online adaptation mechanisms for the prediction models and algorithm hyper-parameters to enhance robustness under non-stationary traffic and radio environments.

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Prior Publication Statement

MEET-COBRA: Machine-learning-based Energy-Efficient proactive Coordination of handOver, Beamforming and Resource Allocation

Bowen Xie, Sheng Zhou, and Zhisheng Niu

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Earlier publication

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Relationship and main extensions

The journal manuscript extends the above conference paper. Both works consider a two-tier vehicular heterogeneous network and use prediction-assisted proactive handover (PHO) and queue-aware resource allocation (RA) to reduce system transmit power under latency constraints. The main extensions are as follows.

1. **Joint coordination of PHO, beamforming, and RA.** The conference paper uses a scalar channel model with a prescribed beamforming gain and does not optimize beam selection. The journal manuscript introduces microcell multiple-input multiple-output (MIMO) channels, transmit and receive beam-pair selection, and beam-training overhead into the joint optimization. It develops prediction-guided beam search with early termination and coordinates it with frame-level PHO and slot-level RA. It also accounts for inter-cell interference and handover interruption when estimating resource demands and allocating resources.
2. **Prediction of beam pairs and channel gains.** The conference paper predicts vehicle positions from historical channel gains and estimates future link gains from the predicted positions. The journal manuscript instead uses historical superposed-pilot observations to predict next-frame beam-pair candidates, desired-link gains, and beam-averaged interfering-link gains. These predictions guide association decisions, beam search, and resource allocation without requiring full MIMO channel reporting.
3. **Ray-traced channels and broader evaluation.** The conference evaluation combines SUMO vehicle trajectories with statistically generated scalar channels. The journal evaluation uses SUMO mobility traces and Sionna Ray Tracing to generate microcell MIMO channels, and computes receive rates using the selected beams and explicit resource block assignments. It includes comparisons with two adapted joint handover and beamforming methods from the literature, module ablations, multiple random seeds, analyses of mobility and prediction errors, and measurements of neural-network complexity, inference latency, and reporting overhead.

The conference paper is cited as reference [13] in the journal manuscript. Section I, page 2, explicitly identifies the manuscript as an extension and summarizes the principal differences. A copy of the conference paper is provided separately with this statement.

AEPHORA: AI/ML-Based Energy-Efficient Proactive Handover and Resource Allocation

Bowen Xie*, Sheng Zhou*, Zhisheng Niu*, Hao Wu[†], Cong Shi[†]

*Beijing National Research Center for Information Science and Technology
Department of Electronic Engineering, Tsinghua University, Beijing 100084, China

[†]Standard Research Department, OPPO, Beijing, China

Email: xbw22@mails.tsinghua.edu.cn, {sheng.zhou, niuzhs}@tsinghua.edu.cn, {wuhao8, shicong}@oppo.com

Abstract—Future Vehicle-to-Everything (V2X) scenarios require high-speed, low-latency, and ultra-reliable communication services, particularly for applications such as autonomous driving and in-vehicle infotainment. Dense heterogeneous cellular networks, which incorporate both macro and micro base stations, can effectively address these demands. However, they introduce more frequent handovers and higher energy consumption. Proactive handover (PHO) mechanisms can significantly reduce handover delays and failure rates caused by frequent handovers, especially with the mobility prediction capability enhanced by artificial intelligence and machine learning (AI/ML) technologies. Nonetheless, the energy-efficient joint optimization of PHO and resource allocation (RA) remains underexplored. In this paper, we propose an AI/ML-based energy-efficient PHO and RA (AEPHORA) framework, which leverages AI/ML-based predictions of vehicular mobility to jointly optimize PHO and RA decisions. AEPHORA aims to minimize the time-averaged system transmit power while satisfying quality of service (QoS) constraints on communication delay and reliability. Simulation results demonstrate the effectiveness of the AEPHORA framework in balancing energy efficiency with QoS requirements in high-demand V2X environments.

I. INTRODUCTION

Modern vehicles are transforming into intelligent mobile terminals, necessitating real-time and efficient communication with their environments, other vehicles, and pedestrians. This evolution has led to the emergence of intelligent connected vehicles (ICVs) and Vehicle-to-Everything (V2X) communication [1], [2]. ICVs leverage V2X technology to connect with infrastructure, pedestrians, and the Internet, supporting applications such as autonomous driving, intelligent traffic management, and in-vehicle entertainment, all of which demand high quality of service (QoS). To meet the diverse communication QoS requirements in high-mobility scenarios, the heterogeneous networks (HetNets) with macro and micro base stations (BSs) is crucial for enhancing vehicular network performance [3]. This coordinated deployment strategy ensures large coverage in V2X scenarios while significantly lowering overall system energy consumption and data service latency [4], [5].

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However, V2X communication still faces several challenges. The high-speed movement of vehicles, coupled with the dense deployment of BSs, leads to frequent handovers (HOs), resulting in increased signaling overhead and degraded communication quality [6], [7]. Traditional HO schemes experience higher HO failure (HOF) rates and longer HO latency in dynamic environments, particularly in the millimeter-wave (mmWave) band [8]. Moreover, energy-efficient HO and resource allocation (RA) schemes are in urgent need [9].

Existing studies propose the proactive handover (PHO) strategy to address these issues [8], [10], [11], which involves forecasting traffic flow, vehicle trajectories, and wireless channel states to prepare HO in advance. By leveraging artificial intelligence and machine learning (AI/ML) technologies, it becomes feasible to estimate future vehicle positions or identify the optimal target BS based on historical channel state information [11], [12]. This predictive capability enhances PHO by reducing HO latency and HOF rates. However, these studies do not jointly optimize the PHO and RA strategies to minimize energy consumption while meeting the communication delay and reliability requirements.

This paper studies the joint PHO and RA problems in V2X Hetnets. We propose an AI/ML-based energy-efficient PHO and RA (AEPHORA) framework that jointly optimize PHO and RA decisions to minimize the time-averaged system transmit power while satisfying the ultra-reliable and low-latency communication (uRLLC) QoS constraints. Simulation results demonstrate the effectiveness of the proposed AEPHORA framework in achieving a good balance between system power and reliability.

II. SYSTEM MODEL

The paper considers a vehicular network downlink communication scenario with coordinated coverage by macro and micro BSs, as illustrated in Figure 1. In this scenario, there is one macro BS, M micro BSs (in Figure 1, $M = 4$) and a varying number of vehicles in the considered region $\mathcal{R}$. Each BS is equipped with N_{TX} transmit antennas, whereas each vehicle is equipped with a single receive antenna. The BS set is denoted

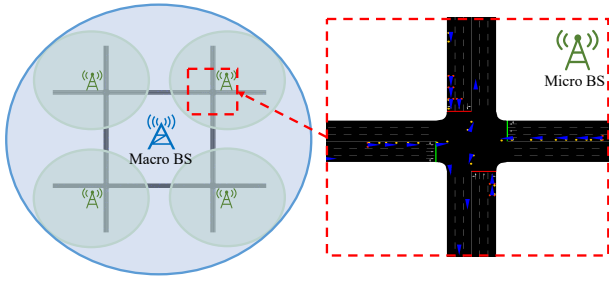


Fig. 1: The system model.

as $\mathcal{M} = \{0, 1, \dots, M\}$, where BS- m represents the macro BS for $m = 0$ and a micro BS for $m > 0$. BS- m operates in the frequency band f_m and is allocated K_m resource blocks (RBs), with each RB utilizing a bandwidth of Δf_m and a transmit power of p_m . It is assumed that all micro BSs share the same parameters, i.e., $f_m = f_1$, $K_m = K_1$, $\Delta f_m = \Delta f_1$, and $p_m = p_1$ for all $m > 0$.

The system operates in discrete time, with the time slot length denoted by Δt . In the n -th time slot, the set of vehicles in the network is denoted by $\mathcal{V}(n)$, and the number of vehicles is $V(n)$. The binary variable $u_{m,v}(n) \in \{0, 1\}$ represents the connection state between BS- m and vehicle- v in time slot- n , where $u_{m,v}(n) = 1$ denotes an active connection and $u_{m,v} = 0$ indicates no connection. Assume that each vehicle can only maintain a connection with one BS at a time, hence we have

$$\sum_{m=0}^M u_{m,v}(n) \leq 1. \quad (1)$$

The connection variable $u_{m,v}(n)$ is periodically updated by the PHO server at the macro BS, with an update cycle of N_{HO} time slots, referred to as a PHO frame. A PHO decision made in the $(x-1)$ -th PHO frame takes effect at the start of the x -th PHO frame, and the corresponding $u_{m,v}(n)$ remains unchanged throughout the x -th PHO frame. For the sake of notational simplicity, we denote $\{\cdot\}^{[x]} = \{\cdot\}(xN_{\text{HO}})$, $\{\cdot\}^{[x]}(i) = \{\cdot\}(xN_{\text{HO}} + i)$, $\mathcal{N}_{\text{HO}} = \{0, 1, \dots, N_{\text{HO}} - 1\}$. Thus, we have

$$u_{m,v}^{[x]}(i) = u_{m,v}^{[x]}, \forall i \in \mathcal{N}_{\text{HO}}. \quad (2)$$

The macro BS is deployed at the center of the vehicular network area to ensure comprehensive signal coverage across the entire region. In contrast, micro BSs are placed at high-traffic intersections or critical road segments to enhance network capacity and communication efficiency. With wired connections between the macro and micro BSs facilitating efficient data exchange and signaling, it is assumed that all BSs share channel and context information. This enables centralized PHO and RA decisions.

A. Communication Model

Define $k_{m,v}(n) \in \{0, 1, \dots, K_m\}$ as the number of RBs allocated by BS- m to vehicle- v in the n -th time slot, and let $k_m(n) \in \{0, 1, \dots, K_m\}$ represent the total number of RBs

used by BS- m in that time slot. BS- m can allocate RBs to vehicle- v only if their connection exists, hence

$$k_{m,v}(n) = u_{m,v}(n)k_m(n). \quad (3)$$

To avoid inter-vehicle interference, we assume that a BS cannot assign a specific RB to multiple vehicles at a time, leading to the constraint

$$k_m(n) = \sum_{v \in \mathcal{V}(n)} k_{m,v}(n) \leq K_m. \quad (4)$$

The transmit power of BS- m in time slot- n is based on its RB allocation and is given by

$$P_m(n) = p_m k_m(n). \quad (5)$$

The time-averaged system transmit power is

$$\bar{P} = \sum_{m=0}^M \bar{P}_m = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \sum_{m=0}^M \sum_{v \in \mathcal{V}(n)} p_m k_{m,v}(n). \quad (6)$$

Define $G_{m,v}^{[x]}$ as the average channel gain between BS- m and vehicle- v in the x -th PHO frame. Assume that

$$G_{m,v}^{[x]} = \alpha_m B_{m,v}^{[x]} \left(d_{m,v}^{[x]}\right)^{-\beta_m} 10^{X_{m,v}^{[x]}/10}, \quad (7)$$

where α_m is the power gain constant, β_m is the path loss exponent, $d_{m,v}^{[x]}$ is the distance, $B_{m,v}^{[x]}$ represents the beamforming gain and $X_{m,v}^{[x]}$ captures the shadow fading effects. $X_{m,v}^{[x]}$ is assumed to be an independent and identically distributed (i.i.d.) zero-mean Gaussian random variable with variance σ_X^2 . As beamforming is not the focus of this paper, we assume $B_{m,v}^{[x]} = \gamma_{\text{BF}} N_{\text{TX}}$, where $\gamma_{\text{BF}} \in (0, 1)$ represents the beamforming accuracy factor. In the i -th time slot of the x -th PHO frame, the instantaneous channel gain $g_{m,v}^{[x]}(i)$ between BS- m and vehicle- v is modeled by the Rician fading channel as

$$g_{m,v}^{[x]}(i) = G_{m,v}^{[x]} \left\| \sqrt{\frac{K_{\text{RIC}}}{K_{\text{RIC}} + 1}} + \sqrt{\frac{1}{K_{\text{RIC}} + 1}} h_{m,v}^{[x]}(i) \right\|^2, \quad (8)$$

where K_{RIC} is the Rician K-factor, $h_{m,v}^{[x]}(i) \sim \mathcal{CN}(0, 1)$ is an i.i.d. standard complex Gaussian random variable.

Given that $k_{m,v}(n) > 0$, the signal-to-noise ratio (SNR) from BS- m to vehicle v is expressed as

$$\text{SNR}_{m,v}(n) = \frac{p_{m,v}(n)g_{m,v}(n)}{N_0 W_{m,v}(n)}, \quad (9)$$

where N_0 denotes the noise power spectral density, $p_{m,v}(n) = k_{m,v}(n)p_m$ and $W_{m,v}(n) = k_{m,v}(n)\Delta f_m$. The transmission rate for downlink communication between BS- m and vehicle- v in time slot- n is given by

$$r_{m,v}(n) = W_{m,v}(n)\Delta t \log(1 + \text{SNR}_{m,v}(n)). \quad (10)$$

Therefore, the maximum amount of information received by vehicle- v in time slot- n is given by

$$r_v(n) = \sum_{m=0}^M k_{m,v}(n)\Delta f_m\Delta t \log\left(1 + \frac{p_m g_{m,v}(n)}{N_0 \Delta f_m}\right). \quad (11)$$

B. Traffic Model

Define $q_v(n)$ as the data backlog queue length of vehicle- v awaiting transmission by the BSs, which evolves as

$$q_v(n+1) = \max\{q_v(n) - r_v(n), 0\} + a_v(n), \quad (12)$$

where $a_v(n)$ is the amount of data arriving for vehicle- v during time slot- n . We assume that $a_v(n)$ is i.i.d. following a Poisson distribution with parameter $\lambda\Delta t$, where λ is the average data arrival rate. According to *Little's law* [13], the average queuing delay is proportional to the average queue length. This implies that the queuing delay $\tau_v(n)$ for vehicle- v in time slot- n can be approximated by $\tau_v(n) = \lambda^{-1}q_v(n)$. To meet the uRLLC QoS requirements for V2X applications, the queuing delay $\tau_v(n)$ should remain below the threshold τ^{th} , resulting in the QoS constraint

$$q_v(n) \leq \lambda\tau^{\text{th}}. \quad (13)$$

Define the QoS violation probability as

$$U = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \frac{\sum_{v \in \mathcal{V}(n)} U_v(n)}{V(n)}, \quad (14)$$

where $U_v(n) = 1$ if $q_v(n) > \lambda\tau^{\text{th}}$, and $U_v(n) = 0$ otherwise. The QoS violation probability U serves as an indicator of system reliability in meeting the QoS constraint (13), following the approach in [14].

III. PROBLEM FORMULATION

This paper aims to minimize the system energy consumption while adhering to QoS requirements and the constraints on HO and RA decisions. Specifically, the optimization problem is formulated as

$$\begin{aligned} \mathbf{P1}: \quad & \min_{\mathbf{u}_{m,v}, \mathbf{k}_{m,v}} \bar{P} \\ \text{s.t.} \quad & \text{Eqs. (1)-(4) and (13)} \\ & u_{m,v}(n) \in \{0, 1\}, \\ & k_{m,v}(n) \in \{0, 1, \dots, K_m\}. \end{aligned} \quad (15)$$

The decision variables $\mathbf{u}_{m,v}$ and $\mathbf{k}_{m,v}$ are based on the HO and RA strategies, respectively. Equations (1) and (2) represent the HO constraints, while Equations (3) and (4) represent the RA constraints. The QoS requirement is captured by Equation (13).

Problem **P1** is an integer stochastic optimization problem, where the stochasticity stems from the factors such as vehicle arrivals and departures within the considered region, changes in vehicle positions and speeds, random data traffic arrivals, and dynamic fluctuations in channel conditions. There are three key challenges for solving the problem:

- 1) Different decision timescales: HO and RA decisions operate on different timescales. While $\mathbf{u}_{m,v}$ updates every N_{HO} time slots, $\mathbf{k}_{m,v}$ varies in each time slot, and $k_{m,v}(n)$ can be positive only if $u_{m,v}(n) = 1$.
- 2) Environmental uncertainty: The randomness in traffic conditions, data arrivals, and channel states introduces uncertainty. Accurate future state information is difficult to obtain for making PHO and RA decisions.

- 3) High computational complexity: Problem **P1** is an NP-hard integer programming problem with exponential complexity.

Given that $\mathbf{u}_{m,v}$ and $\mathbf{k}_{m,v}$ operate on different timescales, problem **P1** is decomposed into sequential subproblems, each corresponding to a PHO frame. For each subproblem, the objective is to determine the optimal BS-vehicle connections and RB allocation within the PHO frame. The final state of each subproblem becomes the initial state for the next, forming a Markov decision process (MDP). The subproblem for the x -th PHO frame is formulated as

$$\mathbf{P2}: \quad \min_{\mathbf{u}_{m,v}^{[x]}, \mathbf{k}_{m,v}^{[x]}} \frac{1}{N_{\text{HO}}} \sum_{i=0}^{N_{\text{HO}}-1} \sum_{m=0}^M \sum_{v \in \mathcal{V}^{[x]}} p_m k_{m,v}^{[x]}(i) \quad (16)$$

$$\text{s.t.} \quad q_v^{[x]}(i) \leq \lambda\tau^{\text{th}}, \forall i \in \mathcal{N}_{\text{HO}}, \quad (17)$$

$$\sum_{m=0}^M u_{m,v}^{[x]} \leq 1, \quad (18)$$

$$\sum_{v \in \mathcal{V}^{[x]}} k_{m,v}^{[x]}(i) \leq K_m, \forall i \in \mathcal{N}_{\text{HO}}, \quad (19)$$

$$k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} k_{m,v}^{[x]}(i), \forall i \in \mathcal{N}_{\text{HO}}, \quad (20)$$

$$u_{m,v}^{[x]} \in \{0, 1\}, \quad (21)$$

$$k_{m,v}^{[x]}(i) \in \{0, 1, \dots, K_m\}, \forall i \in \mathcal{N}_{\text{HO}}. \quad (22)$$

Notably, in the PHO mechanism, the variable $u_{m,v}^{[x]}$ in problem **P2** for the x -th PHO frame must be determined in the $(x-1)$ -th PHO frame. This ensures BSs complete signaling and resource reservations in advance for seamless HO. To support the PHO mechanism, an AI/ML-based method is introduced in Section IV-A to predict future vehicle positions and channel states. Section IV-B presents an energy-efficient PHO algorithm to efficiently approximate the optimal $u_{m,v}^{[x]}$.

Given the BS-vehicle connection states $u_{m,v}^{[x]}$ in the x -th PHO frame, problem **P2** can be decoupled into $M+1$ single-BS RA subproblems as

$$\mathbf{P3}: \quad \min_{\mathbf{k}_{m,v}^{[x]}} \frac{1}{N_{\text{HO}}} \sum_{i=0}^{N_{\text{HO}}-1} \sum_{v \in \mathcal{V}_m^{[x]}} p_m k_{m,v}^{[x]}(i) \quad (23)$$

$$\text{s.t.} \quad q_v^{[x]}(i) \leq \lambda\tau^{\text{th}}, \forall i \in \mathcal{N}_{\text{HO}}, \quad (24)$$

$$\sum_{v \in \mathcal{V}_m^{[x]}} k_{m,v}^{[x]}(i) \leq K_m, \forall i \in \mathcal{N}_{\text{HO}}, \quad (25)$$

$$k_{m,v}^{[x]}(i) = 0, \forall v \notin \mathcal{V}_m^{[x]}, \forall i \in \mathcal{N}_{\text{HO}}, \quad (26)$$

$$k_{m,v}^{[x]}(i) \in \{0, 1, \dots, K_m\}, \forall i \in \mathcal{N}_{\text{HO}}, \quad (27)$$

where $\mathcal{V}_m^{[x]} = \{v \in \mathcal{V}^{[x]} | u_{m,v}^{[x]} = 1\}$ represents the set of vehicles connected to BS- m in the x -th PHO frame. Problem **P3** is still an integer stochastic optimization problem. Consequently, heuristic RA algorithms are proposed in Section IV-C to efficiently solve the problem.

IV. ALGORITHM DESIGN

In this section, we first present an AI/ML-based mobility prediction method to forecast the future positions and channel

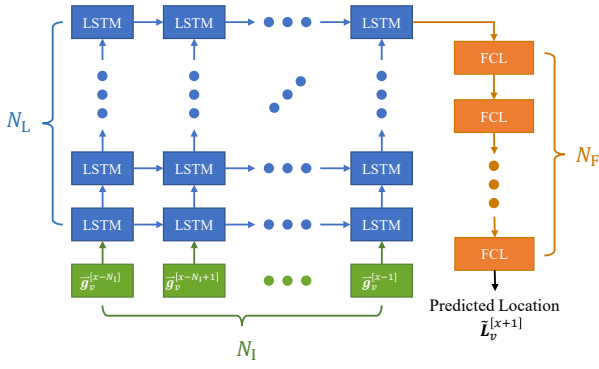


Fig. 2: The AI/ML-based mobility prediction model.

states for each vehicle. Subsequently, we propose energy-efficient PHO and heuristic RA strategies aimed at minimizing the system energy consumption while satisfying the QoS constraints. Finally, we introduce the AEPHORA framework, which integrates the aforementioned algorithms to facilitate energy-efficient PHO and RA decisions guided by the AI/ML-based prediction model.

A. AI/ML-Based Prediction

Figure 2 illustrates the architecture of the neural network (NN) model designed to predict the future location for each vehicle. The NN model comprises an input layer, N_L stacked Long Short-Term Memory (LSTM) layers and N_F stacked fully connected layers (FCLs) [15]. Each LSTM layer comprises 512 cells, while each FCL contains 64 neurons with ReLU activation, except for the final FCL, which has no activation function. Lower layers encode their inputs and pass the corresponding hidden states to subsequent layers. The model input consists of the historical channel gains between vehicle- v and all BSs over the past N_L PHO frames, represented as $\mathbf{g}_v^{[x-1]}, \dots, \mathbf{g}_v^{[x-N_L]} \in \mathbb{R}^{M+1}$, where $\mathbf{g}_v^{[x]} = [g_{0,v}^{[x]}, \dots, g_{M,v}^{[x]}]^T$. The output is the predicted location $\tilde{\mathbf{L}}_v^{[x+1]} \in \mathbb{R}^2$ for vehicle- v in the next PHO frame. Denote the NN model as $F(\cdot; \theta)$, where θ represents the model weights. Then we have

$$\tilde{\mathbf{L}}_v^{[x+1]} = F(\mathbf{g}_v^{[x-1]}, \dots, \mathbf{g}_v^{[x-N_L]}; \theta). \quad (28)$$

B. Energy-Efficient PHO

Given the predicted locations $\tilde{\mathbf{L}}_v^{[x+1]}$, the expected average channel gain is estimated by

$$\tilde{G}_{m,v}^{[x+1]} = \alpha_m \gamma_{\text{BF}} N_{\text{TX}} \left\| \tilde{\mathbf{L}}_v^{[x+1]} - \mathbf{L}_m^{\text{BS}} \right\|_2^{-\beta_m}. \quad (29)$$

Subsequently, we can estimate the minimum average number of RBs required to meet the QoS constraint for vehicle- v connected to BS- m in the $(x+1)$ -th frame, as follows:

$$\tilde{k}_{m,v}^{[x+1]} = \lambda \left[\Delta f_m \log \left(1 + \frac{p_m \tilde{G}_{m,v}^{[x+1]}}{N_0 \Delta f_m} \cdot 10^{-\frac{\sigma_x}{10}} \right) \right]^{-1}. \quad (30)$$

By reserving $\tilde{k}_{m,v}^{[x+1]}$ RBs for each vehicle- v connected to BS- m in the $(x+1)$ -th frame while ensuring that the total reserved RBs of each BS- m do not exceed its capacity K_m , the BS-vehicle association decision in problem **P2** can be approximated as a generalized assignment problem (GAP) as

$$\mathbf{P4}: \min_{u_{m,v}^{[x+1]}} \sum_{m=0}^M \sum_{v \in \mathcal{V}^{[x+1]}} p_m \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \quad (31)$$

$$\text{s.t.} \quad \sum_{v \in \mathcal{V}^{[x+1]}} \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \leq K_m, \forall m \in \mathcal{M}, \quad (32)$$

$$\sum_{m=0}^M u_{m,v}^{[x+1]} = 1, \forall v \in \mathcal{V}^{[x+1]} \quad (33)$$

$$u_{m,v}^{[x+1]} \in \{0, 1\}. \quad (34)$$

Since GAP is NP-hard, we employ the polynomial-time approximation algorithm for GAP as presented in [16] to develop an energy-efficient PHO schedule. This approximation algorithm can generate a schedule if and only if the linear programming (LP) relaxation of problem **P4** is feasible. In cases of significant network overload leading to infeasible LP relaxation, problem **P4** will be relaxed by repeatedly multiplying the right-hand side of constraint (32) by a relaxation factor $\zeta > 1$ until feasibility is achieved. In this paper, ζ is set to 1.1. It has been proved in [16] that if problem **P4** is feasible and its optimal solution yields an average power of P^* , the approximation algorithm in [16] can generate a solution that results in an average power not exceeding P^* . However, the approximate solution only satisfies $\sum_{v \in \mathcal{V}^{[x+1]}} \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \leq 2K_m$ instead of constraint (32), which may compromise the feasibility of the approximate solution.

The above strategy is called energy-efficient PHO (EPHO) algorithm, which is presented in **Alg. 1**.

Algorithm 1 Energy-Efficient PHO (EPHO)

Input: $\tilde{\mathbf{L}}_v^{[x+1]}, \mathbf{L}_m^{\text{BS}}, \alpha_m, \beta_m, p_m, \Delta f_m, \gamma_{\text{BF}}, N_{\text{TX}}, \zeta, N_0$;

Output: $\hat{u}_{m,v}^{[x+1]}$;

- 1: Calculate $\tilde{k}_{m,v}^{[x+1]}$ by Eqs. (29) and (30);
 - 2: **while** the LP relaxation of problem **P4** is not feasible **do**
 - 3: Multiply the right-hand side of constraint (32) by ζ ;
 - 4: **end while**
 - 5: Apply the GAP approximation algorithm in [16] to solve problem **P4** to generate a PHO schedule $\hat{u}_{m,v}^{[x+1]}$;
 - 6: **return**
-

C. Heuristic RA

A heuristic RA (HRA) strategy is detailed in **Alg. 2**, which guides each BS in allocating an appropriate number of RBs to connected vehicles in each time slot. The HRA strategy begins by sorting the connected vehicles in descending order based on their data backlog queue lengths. Subsequently, for each connected vehicle, the strategy estimates the average number

Algorithm 2 Heuristic RA (HRA)**Input:** $u_{m,v}^{[x]}, g_{m,v}^{[x]}, q_v^{[x]}(i), p_m, \Delta f_m, N_0, \lambda, \tau^{\text{th}};$ **Output:** $k_{m,v}^{[x]}(i);$

Initialization:

- 1: $k_{m,v}^{[x]}(i) \leftarrow 0, \bar{K}_m \leftarrow K_m, \mathcal{V}_m^{[x]} \leftarrow \{v \in \mathcal{V}^{[x]} | u_{m,v}^{[x]} = 1\};$
- 2: Sort the vehicles in $\mathcal{V}_m^{[x]}$ by their queue lengths $q_v^{[x]}(i)$ in descending order to obtain a prioritized sequence $\mathcal{S}_m^{[x]}$.
- 3: **for all** $v \in \mathcal{S}_m^{[x]}$ **do**
- 4: Calculate $\hat{k}_{m,v}^{[x]}(i)$ by Eqs. (35) and (36);
- 5: $k_{m,v}^{[x]}(i) \leftarrow \hat{k}_{m,v}^{[x]}(i);$
- 6: $\bar{K}_m \leftarrow \bar{K}_m - k_{m,v}^{[x]}(i);$
- 7: **end for**
- 8: **return**

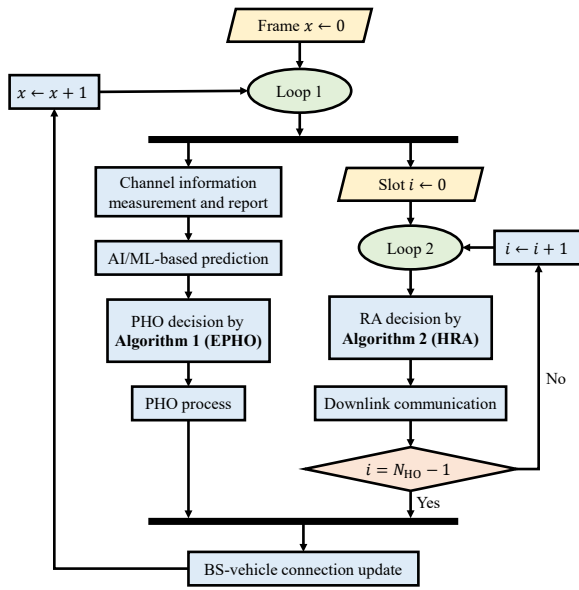


Fig. 3: Flowchart of the AEPHORA algorithm.

of RBs needed to eliminate the accumulated backlogged data by

$$\bar{k}_{m,v}^{[x]}(i) = q_v^{[x]}(i) \left[\Delta f_m \Delta t \log \left(1 + \frac{p_m g_{m,v}^{[x]}}{N_0 \Delta f_m} \right) \right]^{-1}. \quad (35)$$

Since the allocated RB number for each vehicle must be an integer and their sum cannot exceed the RB capacity K_m , the final allocated RB number is given by

$$\hat{k}_{m,v}^{[x]}(i) = \begin{cases} \min \left\{ \left\lceil \bar{k}_{m,v}^{[x]}(i) \right\rceil, \bar{K}_m \right\}, & \text{if } q_v^{[x]}(i) > \frac{\lambda \tau^{\text{th}}}{2}, \\ \min \left\{ \left\lfloor \bar{k}_{m,v}^{[x]}(i) \right\rfloor, \bar{K}_m \right\}, & \text{otherwise,} \end{cases} \quad (36)$$

where $\bar{K}_m$ denotes the number of the remaining RBs at BS- m .

D. AEPHORA

Algorithm 3 demonstrates the AI/ML-based energy-efficient PHO and RA optimization framework (AEPHORA). Within

Algorithm 3 AI/ML-Based Energy-Efficient Proactive Handover and Resource Allocation (AEPHORA)**Input:** $\theta, \mathcal{M}, N_0, N_{\text{HO}}, \gamma_{\text{BF}}, N_{\text{TX}}, \zeta, \lambda, \Delta t, \tau^{\text{th}}, K_m, p_m, \Delta f_m, \mathbf{L}_m^{\text{BS}}, \alpha_m, \beta_m, \mathcal{V}^{[x]};$ **Output:** $u_{m,v}^{[x]}, k_{m,v}^{[x]}(i);$

Initialization:

- 1: $\mathcal{V}^{[-1]} \leftarrow \emptyset, u_{m,v}^{[x]} \leftarrow 0, k_{m,v}^{[x]}(i) \leftarrow 0;$
- 2: **for all** $x = 0, 1, \dots$ **do**
- 3: Newly arrived vehicles are initially connected to the macro BS: $u_{0,v}^{[x]} \leftarrow 1, \forall v \in \mathcal{V}^{[x]} \setminus \mathcal{V}^{[x-1]};$
- 4: Newly left vehicles handover to another macro BS outside the region $\mathcal{R}$: $u_{m,v}^{[x]} \leftarrow 0, \forall m \in \mathcal{M}, \forall v \in \mathcal{V}^{[x-1]} \setminus \mathcal{V}^{[x]};$
- 5: Each vehicle measures and reports the instantaneous channel gains $g_{m,v}^{[x]}, \forall m \in \mathcal{M}, \forall v \in \mathcal{V}^{[x]};$
- 6: Based on historical channel state information, utilize the AI/ML-based prediction method to forecast the positions $\tilde{\mathbf{L}}_v^{[x+1]}$ of each vehicle in the next PHO frame, $\forall v \in \mathcal{V}^{[x]};$
- 7: **Algorithm 1** is applied to make the PHO decision $\hat{u}_{m,v}^{[x+1]}$ and initiate the PHO process for the next PHO frame;
- 8: **for all** $i = 0, 1, \dots, N_{\text{HO}} - 1$ **do**
- 9: **for all** $m \in \mathcal{M}$ **do**
- 10: **Algorithm 2** is applied for the RA decision $k_{m,v}^{[x]}(i);$
- 11: **end for**
- 12: Each BS conducts downlink communication and updates the data backlog queue $q_v^{[x]}(i+1)$ for all the connected vehicles based on Eqs. (11) and (12);
- 13: **end for**
- 14: Complete the PHO process for the next PHO frame and update the connection states accordingly: $u_{m,v}^{[x+1]} \leftarrow \hat{u}_{m,v}^{[x+1]}, \forall m \in \mathcal{M}, \forall v \in \mathcal{V}^{[x]};$
- 15: **end for**
- 16: **return**

each PHO frame, newly arrived vehicles are initially connected to the macro BS, while those departing the region are disconnected. Each vehicle measures and reports its instantaneous channel gains, providing real-time data for decision-making. Utilizing historical channel state information, the AEPHORA framework applies the AI/ML-based prediction strategy proposed in Section IV-A to forecast future vehicle positions, enabling PHO management. PHO decisions are derived from the outcomes of **Alg. 1**, while RA decisions are determined using **Alg. 2**. During each time slot in the current PHO frame, each BS conducts downlink communication and updates the data backlog queues for all connected vehicles. At the end of the current PHO frame, the PHO process for the subsequent frame is completed, and the BS-vehicle connection states are updated accordingly. We emphasize that the PHO process and the RA process occur in parallel, as illustrated in the flowchart of **Algorithm 3** in Fig. 3.

V. SIMULATION RESULTS

In this section, simulation results demonstrate the effectiveness of the proposed AEPHORA framework. The simulation scenario aligns with Figure 1, featuring one macro BS and four micro BSs positioned within an 800m by 800m area. The macro BS is located at the area center (0m, 0m), while the coordinates of the four micro BSs are $(\pm 250\text{m}, \pm 250\text{m})$. The centers of the four intersections are situated at $(\pm 200\text{m}, \pm 200\text{m})$. Each road consists of two directions, with three lanes allocated for each direction. The vehicular trajectories are generated using the road traffic simulator SUMO [17], providing realistic traffic dynamics for the simulations. The average number of vehicles in the considered region is about 220, with an average of 1.6 vehicles entering and 1.6 vehicles leaving the region per second. The channel gains for each vehicle and BS are randomly generated according to Eqs. (7) and (8). The parameter settings for simulation are summarized in Table I.

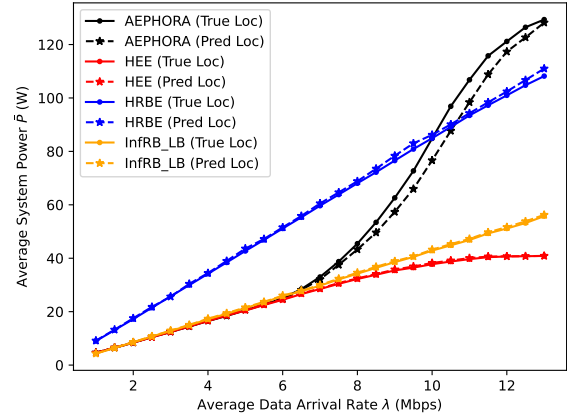
TABLE I: Simulation Parameter Settings

Parameter	Value		Parameter	Value
	Macro BS	Micro BS		
α_m	-55.4 dB	-72.0 dB	σ_X	4
β_m	2.2	3	Δt	1 ms
f_m	5.9 GHz	24 GHz	τ^{th}	20 ms
Δf_m	0.18 MHz	1.8 MHz	K_{RIC}	10
K_m	100	100	N_{HO}	100
p_m	30 dBm	20 dBm	N_{TX}	32
N_0	-173 dBm/Hz		γ_{BF}	0.5
Others	Simulation Duration = 30 s			
	Maximum driving velocity = 18.34 m/s			
	Average data arrival rates in [1.0, 1.5, 2.0, ..., 13.0] Mbps			

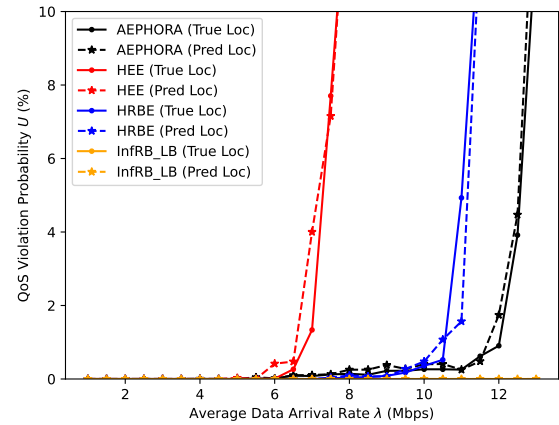
TABLE II: Model Training Parameters and Validation Results

N_L	N_F	Validation MAE		Training Parameters
		$\sigma_X = 0$	$\sigma_X = 4$	
1	1	2.967 m	18.925 m	Training Dataset Size: 1608415 Validation Dataset Size: 643366 Epoch Number: 100 Learning Rate: 10^{-3} Weight Decay: 10^{-5} Batch Size: 64 $N_I = 10$
	2	2.988 m	18.705 m	
	3	2.965 m	18.703 m	
2	1	2.971 m	18.791 m	
	2	2.985 m	18.668 m	
	3	2.980 m	18.714 m	
3	1	2.975 m	18.829 m	
	2	3.033 m	18.851 m	
	3	2.999 m	18.913 m	
4	1	3.054 m	18.849 m	
	2	3.063 m	18.852 m	
	3	3.183 m	19.073 m	

We trained the NN model using the Adam optimizer to minimize the mean squared error (MSE) between the predicted location $\tilde{\mathbf{L}}_v^{[x+1]}$ and the true location $\mathbf{L}_v^{[x+1]}$. The performance of the trained models is validated using the mean absolute error (MAE) as a key performance indicator. The parameter settings for model training, along with the results of the performance validation, are summarized in Table II. Increasing N_L and N_F does not guarantee improved performance. The optimal model parameters for the realistic environment with $\sigma_X = 4$ are $N_L = 2$ and $N_F = 2$, which will be utilized in the simulations.



(a) Average system power $\bar{P}$ versus average data arrival rate λ .



(b) QoS violation probability U versus average data arrival rate λ .

Fig. 4: Comparison between the proposed AEPHORA and other baseline schemes.

We compare the performance of the proposed AEPHORA scheme with three other baseline schemes listed as follows:

- 1) **HEE**: Each vehicle connects to the BS with the highest energy efficiency indicator, defined as the predicted transmission rate per unit of transmit power: $\frac{\Delta f_m}{p_m} \log \left(1 + p_m \frac{\tilde{G}_{m,v}^{[x+1]}}{N_o \Delta f_m} \right)$. The RA strategy is **Alg. 2**.
- 2) **HRBE**: Each vehicle connects to the BS with the highest RB efficiency indicator, defined as the predicted transmission rate per RB: $\Delta f_m \log \left(1 + p_m \frac{\tilde{G}_{m,v}^{[x+1]}}{N_o \Delta f_m} \right)$. The RA strategy is **Alg. 2**.
- 3) **InFRB_LB**: The PHO strategy aligns with HEE, but without considering integer and capacity constraints for RB allocation. Each vehicle is allocated $\bar{k}_{m,v}^{[x]}(i)$ RBs. While this RA strategy cannot be directly applied due to practical limitations, it establishes a lower bound on energy consumption necessary to satisfy the QoS constraint (13).

Additionally, to assess the impact of the AI/ML-based predic-

tion model on the system performance, each scheme is further divided into two versions: one employs vehicle positions predicted by the NN model, while the other utilizes actual vehicle positions.

The time-averaged system transmit power $\bar{P}$ versus average data arrival rate λ of these schemes is illustrated in Fig. 4a, while the QoS violation probability U is presented in Fig. 4b. Solid curves with point markers represent the schemes utilizing actual vehicle positions, while dashed curves with star markers correspond to the schemes employing NN-predicted locations. The solid and dashed lines for the same scheme are closely aligned, indicating that the algorithm performance loss due to NN prediction errors is acceptable.

In Figure 4, the simulation results can be divided into four stages based on the value of λ :

- 1) $\lambda \leq 6\text{Mbps}$: For all these four schemes, U is nearly zero, with $\bar{P}$ of AEPHORA and HEE closely aligning with the lower bound baseline InfRB_LB. HRBE exhibits a significantly higher $\bar{P}$ due to its prioritization of vehicular connections to the BS requiring the fewest RBs, thus sacrificing energy efficiency for increased system capacity.
- 2) $6\text{Mbps} < \lambda \leq 10\text{Mbps}$: $\bar{P}$ of AEPHORA rises more rapidly, approaching that of HRBE around $\lambda = 10\text{Mbps}$. This indicates that AEPHORA effectively balances the load by the adaptive handover of vehicles from overloaded micro BSs to the macro BS. While this results in an increase in $\bar{P}$, it ensures that U remains sufficiently low. In contrast, U for HEE rises sharply, and its $\bar{P}$ drops below that of InfRB_LB because it fails to offload vehicles from overloaded micro BSs to the macro BS, resulting in inadequate transmission rates.
- 3) $10\text{Mbps} < \lambda \leq 11.5\text{Mbps}$: AEPHORA surpasses HRBE in $\bar{P}$. However, U for AEPHORA remains near zero, while that for HRBE rises sharply. This suggests that AEPHORA maintains better uRLLC QoS through adaptive load balancing under high data rates compared to HRBE.
- 4) $\lambda > 11.5\text{Mbps}$: U for AEPHORA increases sharply, indicating that the traffic load exceeds the system capacity.

Specifically, AEPHORA supports a maximum data arrival rate with the QoS violation probability close to zero, nearly double that of HEE. Additionally, AEPHORA consumes only about half the system power of HRBE at low data arrival rates, while supporting higher maximum data arrival rates than HRBE.

VI. CONCLUSION

The proposed AEPHORA framework effectively balances system energy consumption with uRLLC QoS requirements, providing a significant advantage over the practical baselines HEE and HRBE. Under light traffic loads, AEPHORA approximates the lower bound of system power. With heavy traffic loads, AEPHORA adaptively offloads vehicles from overloaded micro BSs to the macro BS, ensuring QoS requirements at

the cost of a manageable increase in energy consumption. Consequently, AEPHORA achieves lower energy consumption while supporting heavier traffic loads, demonstrating its efficiency in real-world applications. However, the performance of AEPHORA depends on the accuracy of AI/ML-based predictions, as prediction errors can result in sub-optimal handover decisions. Better AI/ML-based mobility prediction models are required to further optimize the PHO process of AEPHORA.

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Responses to the Comments of the Editor and Reviewers

MEET-COBRA: Machine-learning-based Energy-Efficient proactive Coordination of handOver, Beamforming and Resource Allocation

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Bowen Xie, Sheng Zhou, and Zhisheng Niu

Beijing National Research Center for Information Science and Technology,

Department of Electronic Engineering, Tsinghua University, Beijing 100084, China

E-mail: xbw22@mails.tsinghua.edu.cn, {sheng.zhou, niuzhs}@tsinghua.edu.cn

Dear Editor and Reviewers,

We sincerely thank the Editor and the Reviewers for their careful evaluation of our manuscript and for their constructive comments. We summarize the major revisions below, followed by our detailed point-by-point responses. Changes in the marked-up manuscript are highlighted in blue.

Summary of Major Revisions

1. **Modeling and technical consistency.** Clarified the latency proxy and incorporated HO interruption with corresponding algorithm updates. Distinguished the online interference approximations from the directional interference calculations used in simulation and quantified the resulting bias in receive-rate estimates.
2. **Expanded evaluation.** Added two literature-based baselines for joint HO and BF. Reported vehicular speed distributions and analyzed how NN prediction accuracy and latency performance vary with vehicular speed. Examined the effects of gain-prediction errors on system transmit power and latency performance.
3. **Implementation overhead and presentation.** Quantified NN complexity, inference latency, and control-plane reporting overhead. Explained parameter choices, clarified the extension beyond the conference paper, expanded the discussion of related work, and added a notation table.

1 Editor

Comment 1 (Major revision decision and overall concerns)

While the work is regarded positively in general, the reviews have identified several areas that need major improvement, e.g., regarding problem formulation, technical rigor/consistency, evaluation, and presentation. Therefore, I invite you to respond to the reviewer(s)' comments and revise your manuscript.

It should be emphasized that a major revision decision does not imply eventual acceptance. A future version of the paper will only be accepted if the revision addresses the major concerns, and the reply is highly persuasive to the Editors and Reviewers.

Response from the Authors:

We thank the Editor for the opportunity to revise the manuscript and for highlighting the main concerns. We have revised the manuscript in the following three areas.

1. **Modeling and technical consistency.** We clarify the interpretation of the queueing-delay proxy and its violation probability, incorporate HO interruption into queue evolution, and update the PHO and RA algorithms accordingly. We also distinguish the interference approximations used for online decisions from the directional interference calculation with explicit RB assignments used in simulation. The combined effect of these approximations on estimated receive rates is quantified with associations, beam pairs, and RB counts held fixed.
2. **Expanded evaluation.** To broaden the comparison, we introduce two literature-based joint HO and BF baselines, O-MAPPO-adapted and MTS-GS-HBF-adapted. The former combines reinforcement learning with target-BS optimization, whereas the latter applies Gale–Shapley matching to BS–vehicle association and updates beamforming on a shorter timescale than association. We also report vehicular speed distributions and analyze how NN prediction accuracy and latency performance vary with vehicular speed. An additional sensitivity study examines how gain-prediction errors affect HO, beam probing, system transmit power, and latency performance.
3. **Implementation overhead and presentation.** To quantify implementation overhead, we report NN parameter counts, parameter storage, FLOPs, measured inference latency, and control-plane reporting overhead. We explain the algorithm parameter settings and clarify the contributions beyond the conference version and related joint optimization studies. We also add a notation table and revise terminology for consistency.

2 Reviewer 1

Comment 1 (Motivation and cross-tier heterogeneity)

The paper explains the coupling among handover, beamforming, and resource allocation in Het-Nets. It would be beneficial to further strengthen the motivation by more clearly articulating why this problem becomes fundamentally more unique and challenging in heterogeneous networks, beyond differences in system architecture. In particular, additional discussion on the new optimization challenges introduced by cross-tier heterogeneity could further improve the clarity of the problem formulation.

Response from the Authors:

We thank the Reviewer for this helpful suggestion. Cross-tier differences in carrier frequencies, antenna configurations, and propagation conditions affect the link gains and training overhead associated with PHO and BF decisions. Together with differences in bandwidth and transmit power, they change the RB requirements and power costs of serving the same traffic, while RB budgets constrain service feasibility. Thus, association cannot be decided from link gain alone. HO also changes the loads of the source and target BSs, which can alter their resource usage and intra-tier co-channel interference, affecting the resource demands of other links. The revised Introduction explains how these effects motivate joint coordination of PHO, BF, and RA to reduce system transmit power under latency constraints.

Changes in the Revised Manuscript:

The revised discussion reads:

PHO and BF jointly determine the serving link gain and beam-training overhead. In HetNets, these quantities also depend on cross-tier differences in carrier frequencies, antenna configurations, and propagation conditions. Together with differences in bandwidth and transmit power across tiers, these factors affect the number of resource blocks (RBs) required to serve a given traffic demand and the resulting power consumption, while the available RB budgets constrain service feasibility. Moreover, HO changes the traffic loads of the source and target BSs, which can alter their resource usage and intra-tier co-channel interference, thereby affecting the resource demands of other links. This motivates energy-efficient Coordination of handOver, Beamforming and Resource Allocation (COBRA) in vehicular HetNets.

Location in the Revised Manuscript: Pages 1–2, Section I (Introduction), paragraph beginning *PHO and BF jointly determine*.

Comment 2 (Queueing delay and end-to-end latency)

The latency formulation is currently based on queue-length constraints. The paper could be further strengthened by discussing the relationship between queueing delay and actual end-to-end latency in a two-tier vehicular HetNet. Incorporating or discussing additional latency components, such as handover interruption, beam training overhead, retransmissions, and signaling delay, would improve the realism of the latency evaluation.

Response from the Authors:

We thank the Reviewer for this suggestion. End-to-end latency includes queueing, transmission, propagation, and processing delays along the communication path. Our model focuses on queueing delay at the serving BS to assess how HO, BF, and RA jointly affect traffic accumulation through the available

service rates. As clarified in Section II-C, U measures threshold violations of the queueing-delay proxy $\tau_v(n) = q_v(n)/\lambda_v$, not a directly measured packet-delay tail probability.

We have now incorporated a first-order HO interruption model. A vehicle switching its serving BS receives no data service during the first L_{HO} slots of the next frame. Its traffic arrivals continue to accumulate in the queue during these slots, so the interruption affects the queue evolution underlying U . Beam-training overhead is already accounted for through the data-transmission fraction $1 - \eta_{m,v}^{[x]}(i)$ in the receive rate. Thus, both mechanisms affect the latency proxy through the available service rate. Their durations are therefore not added separately to the proxy.

The revised model distinguishes advance HO preparation from HO execution. HO preparation signaling is assumed to finish before the current frame ends without occupying data-service resources, whereas HO execution incurs the modeled data-service interruption. This work focuses on energy-efficient coordination of PHO, BF, and RA under queue-length-based latency constraints. To keep the modeling and analysis tractable, we assume successful handovers and do not separately model failure recovery, re-transmission delays, or other end-to-end latency components. Consistent with this scope, the manuscript evaluates a queueing-delay proxy rather than complete end-to-end latency. GAP-HO now corrects the estimated RB demand in its RB-capacity constraint to account for the slots lost to HO interruption. OTR-RA allocates no RBs to vehicles during the interruption.

Changes in the Revised Manuscript:

The explanation added before the queueing-delay proxy definition in Section II-C reads:

Queueing delay is an important component of end-to-end latency, alongside transmission, propagation, and processing delays. We focus on queueing delay to assess how HO, BF, and RA jointly affect traffic accumulation through the available service rates.

The HO assumptions added to Section II read:

After a PHO decision is made, the signaling exchanges required for HO preparation are assumed to finish before the current frame ends, without occupying data-service resources. A vehicle switching BS is assumed to experience a data-service interruption of duration $\tau_{HO} = L_{HO}\Delta t$ caused by HO execution at the start of the next frame, where L_{HO} is a nonnegative integer less than N_{HO} . We assume successful HO execution for all PHO decisions. The macro control-plane connection and periodic CSI probing remain available during this data-plane HO interruption.

The new indicator $\delta_v^{[x]}$ equals one when vehicle- v changes its serving BS between consecutive frames and zero otherwise. Equation (3) defines its data-service availability as

$$s_v^{[x]}(i) = 1 - \delta_v^{[x]} \mathbf{1}\{i < L_{HO}\}.$$

Equation (12) and constraint C4 of **P1** now enforce

$$k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} s_v^{[x]}(i) k_{m,v}^{[x]}(i).$$

This makes the service rate zero during HO interruption in the existing receive-rate and queue-update equations, Eqs. (20) and (23). The latency proxy and U remain as defined in Section II-C.

Location in the Revised Manuscript: Page 3, Section II, HO assumptions and Eq. (3); Page 4, Section II-B, Eq. (12) and receive-rate model; Pages 4–5, Section II-C, paragraph preceding Eq. (24); Page 5, Section III, constraint C4 of **P1**; Page 8, Section IV-B, HO-aware RB-capacity constraint; Pages 8–9,

Section IV-D, resource allocation during HO interruption.

Comment 3 (Terminology in Fig. 1)

In Fig. 1, the use of the term “Optimal” beam-pair and gain prediction may be slightly strong given that the neural network provides estimated outputs. The presentation could be made more precise by adopting terminology such as “Predicted” or “Estimated” beam-pair/gain.

Response from the Authors:

We thank the Reviewer for this suggestion. The original term *optimal* referred to the prediction targets: the beam pair that maximizes the frame-averaged desired-link gain within the prescribed codebooks and its corresponding desired-link gain. To avoid ambiguity about prediction accuracy, we have revised the caption of Fig. 1 and the opening sentence of Section IV-A to describe predicting the best beam-pair index and its corresponding gain.

Changes in the Revised Manuscript:

The revised Fig. 1 caption reads:

Neural network architecture for predicting the next-frame best beam-pair index, its corresponding desired-link gain, and the interfering-link gain.

The opening sentence of Section IV-A now identifies the prediction targets as *best beam-pair indices, their corresponding desired-link gains, and interfering-link gains.*

Location in the Revised Manuscript: Page 6, Fig. 1 caption; Page 5, opening sentence of Section IV-A; Pages 6–7, Section IV-A, Output Prediction.

Comment 4 (Propagation of prediction errors)

The paper presents promising prediction accuracy results. It would be valuable to further analyze how prediction errors propagate to higher-layer decisions, particularly proactive handover performance. Additional experiments quantifying the impact of prediction errors on HO reliability, association accuracy, or latency performance would provide deeper insight into the robustness of the proposed framework.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added an explanation of prediction error propagation in Section IV-E and a sensitivity study in Section V-E. The study quantifies the influence of prediction errors on HO, beam probing, system transmit power and latency performance.

Gain-prediction errors change the RB demand estimates in Eq. (40), and hence the estimated power costs and RB requirements used by GAP-HO to select BS–vehicle associations. PET-BF depends on the predicted beam-pair list and search order, while its early-termination threshold depends on the predicted desired-link gain. OTR-RA uses measured desired-link gains, but errors in the predicted interfering-link gains and estimated BS RB usages affect its RB-efficiency estimates and allocation decisions. These effects propagate to the queue evolution and violation probability U through the resulting receive rates.

To quantify the effect of gain-prediction errors, we add independent zero-mean Gaussian noise in dB, with standard deviation $\sigma \in \{0, 1, \dots, 10\}$ dB, to either the desired-link or interfering-link gain predictions, changing only one gain type at a time. The trained models, beam-pair candidates, and algorithm parameters remain fixed. For each traffic load and random seed, the actual channels and traffic arrivals are identical across noise settings. We rerun the MEET-COBRA framework at 9, 19, 29, and 35 Mbps

using three paired seeds, with each run covering 30 s. The case $\sigma = 0$ retains the original NN prediction errors.

At 29 Mbps, adding desired-link gain noise with $\sigma = 5$ dB increases the mean number of HOs per frame from 3.70 to 20.01 and the mean number of probes per beam search from 1.06 to 1.95. The latter statistic is averaged over the beam searches actually performed, excluding macro-BS associations and HO-interrupted slots. Additional HOs and beam probes reduce the time available for data transmission. At this operating point, U increases from 0.19% to 0.29%. These results show the influence of gain-prediction errors on MEET-COBRA, while MEET-COBRA still maintains a low violation probability at this load.

Figure 10 shows that U changes little at 9 and 19 Mbps for $\sigma \leq 5$ dB. At 35 Mbps, however, adding noise with $\sigma = 5$ dB raises U from 1.15% to 3.84% for desired-link gain errors and 3.13% for interfering-link gain errors. Figure 11 reports the corresponding power results. Larger errors generally increase power at lower loads, whereas power remains close to the full-load level at 35 Mbps as U increases. These results show that MEET-COBRA maintains low violation probabilities under moderate additional prediction errors at light and medium loads, although HO and beam-search overhead can increase.

Changes in the Revised Manuscript:

Section IV-E adds a paragraph tracing prediction errors through GAP-HO, PET-BF, and OTR-RA to the receive rates, queue evolution, and U . The new Section V-E, *Impact of Gain-Prediction Errors*, describes the noise-injection experiment and presents U and average system transmit power in Figs. 10 and 11, respectively. It also reports the changes in HO counts and probes per beam search and explains their effect on the time available for data transmission. The Conclusion summarizes the low violation probabilities maintained at light and medium loads under moderate additional gain-prediction errors.

Location in the Revised Manuscript: Page 10, Section IV-E, final paragraph; Page 15, Section V-E and Figs. 10–11; Pages 15–16, Section VI (Conclusion).

Comment 5 (Notation table)

The notation is relatively dense, and multiple related symbols are introduced throughout the paper. Adding a notation table summarizing the main variables and symbols would significantly improve readability and make the paper easier to follow.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added Table I (*Main Notation*) to summarize the main symbols used in the system model and algorithm design. The table note explains the omission of time indices and defines tildes as indicating predicted or estimated quantities; tilde-marked variants are not listed separately when their unmarked counterparts are included.

Location in the Revised Manuscript: Page 2, Table I (Main Notation), referenced at the end of Section I.

Comment 6 (Selection and sensitivity of heuristic parameters)

Several heuristic parameters, such as ζ , M_P , and N_{iter} , are introduced in the framework. Providing additional discussion on their selection criteria, practical interpretation, or parameter sensitivity would further improve the reproducibility and robustness evaluation of the proposed method.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added the values of N_{iter} , ζ , and M_P to Table II. Section V-A briefly explains the parameter choices and their practical roles, and identifies Table II as the source of the default simulation settings.

The parameter N_{iter} specifies the number of GAP-HO iterations in each PHO frame. The first round estimates interference assuming full RB utilization at all micro BSs, and the second uses the resulting BS loads to update the RB demand estimates and BS-vehicle associations. We evaluated the GAP-HO iteration settings. Oracle channel information was used to isolate the effect of iteration settings from NN prediction errors. All configurations retain the 10 ms HO interruption, its RB-demand correction in the capacity constraints of **P2**, and the physical BS capacity limits on estimated RB usage. We compared exactly two iterations with at most three or five iterations at traffic arrival rates of 21, 27, and 35 Mbps per vehicle, using three paired traffic seeds. The latter two configurations stopped early when the maximum absolute change in estimated BS load was at most $\varepsilon = 0.1$ RB. Only the iteration settings differed between these configurations. The table reports the three-seed means of average system transmit power $\bar{P}$ (W) and violation probability U (%).

Arrival rate (Mbps)	Fixed 2 rounds ($\bar{P}$, U)	Up to 3 rounds ($\bar{P}$, U)	Up to 5 rounds ($\bar{P}$, U)
21	(34.694, 0.03241)	(34.666, 0.03234)	(34.661, 0.03227)
27	(65.996, 0.09058)	(65.948, 0.09061)	(65.951, 0.09055)
35	(174.735, 0.81714)	(174.737, 0.81235)	(174.737, 0.81235)

Across these loads, the largest power reduction was below 0.10%, and the largest reduction in U was 0.0048 percentage points. The mean iteration counts increased from 2 to 2.02–3.00 with the three-round limit and 2.02–4.16 with the five-round limit. These small performance changes relative to the additional iterations support retaining $N_{\text{iter}} = 2$ as a practical balance between performance and computational cost.

Algorithm 1 now runs a fixed number of iterations, without the convergence test or its parameter ε .

The factor ζ controls the step size for relaxing the RB capacity bounds when the continuous relaxation of **P2** is infeasible. We use $\zeta = 1.1$ to increase these bounds by 10% per step, balancing the relaxation step size against the number of solver calls. This relaxation is used to obtain an assignment before the final procedure for restoring feasibility; it does not increase the physical RB capacities.

The number of candidate beam pairs M_P determines the tradeoff between coverage of the optimal beam pair and beam search overhead. For the selected beam predictor, the list of five candidates includes the optimal beam pair in more than 99% of the predictions on the validation set. With $\eta = 2/112$, this choice limits beam search overhead to 8.93% of each slot; early termination can reduce the actual overhead. Thus, its choice is supported by prediction coverage and the associated resource cost.

Changes in the Revised Manuscript:

Algorithm 1 now performs N_{iter} rounds without a test for convergence. Section IV-B specifies “a fixed-point iteration over N_{iter} rounds.” Table II lists the number of GAP-HO iterations $N_{\text{iter}} = 2$, the RB capacity relaxation factor $\zeta = 1.1$, and the number of candidate beam pairs $M_P = 5$, consistent with the definition of M_P in Table I. It also specifies the HO interruption duration $\tau_{\text{HO}} = 10$ ms, PET-BF gain margin $G_{\text{th}} = 5$ dB, and system simulation duration of 30 s per run. The parameter discussion at the end of Section V-A reads:

We set $N_{\text{iter}} = 2$ to balance performance and computational cost, and $\zeta = 1.1$ for 10% increments in the RB capacity bounds when the continuous relaxation of **P2** is infeasible.

We use $M_P = 5$ and $G_{th} = 5$ dB to balance coverage of the optimal beam pair against beam search overhead, which occupies at most $\eta M_P = 8.93\%$ of each slot. Unless otherwise stated, the simulations use the parameter values listed in Table II.

Location in the Revised Manuscript: Pages 7–8, Section IV-B and Algorithm 1; Page 11, Section V-A, Table II and the discussion of parameter settings.

Comment 7 (Neural-network complexity and inference overhead)

The paper emphasizes the lightweight nature of the ML inference process. Including quantitative information regarding neural network complexity, such as model size, inference latency, FLOPs, or runtime overhead, would further strengthen the practicality claims of the proposed framework.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added a quantitative assessment of NN computational complexity in Section V-B. Table III lists the parameter counts and matrix FLOPs for the three predictors. Together, they contain 2,127,912 parameters, requiring 8.51 MB of FP32 parameter storage, and perform 4.20 MFLOPs of matrix computation per vehicle for stateful inference using the current-frame CSI and recurrent states.

Section IV-A clarifies that online inference retains recurrent states across frames and processes the current-frame CSI, so its computation does not grow with the length of the vehicle trajectory. To support this operation, the models undergo 100 epochs of finite-window training followed by 100 epochs of stateful fine-tuning, using truncated backpropagation through time with a truncation length of ten frames. Section V-B describes the training and validation procedures and the checkpoint-selection criteria, and the updated Fig. 5 presents both training stages.

Using the selected second-stage checkpoints, we measured three rounds of stateful inference over 16 continuous vehicle trajectories on an AMD EPYC 7742 CPU with one thread, PyTorch 2.6.0, FP32, and batch size one. Across 14,166 timed vehicle-frame inferences after warmup, the combined latency has a median of 5.38 ms and a 95th percentile of 5.72 ms. Timing includes tensor conversion, the three forward passes, state retention, top-5 index selection, and gain denormalization, but excludes CSI acquisition, preprocessing, and transmission.

Changes in the Revised Manuscript:

Section IV-A now states:

The states are retained across frames for each vehicle, so online inference processes only the current CSI feature vector.

Section V-B describes finite-window training followed by stateful fine-tuning and specifies how the beam and gain checkpoints are selected. Figure 5 shows the training and validation curves for both stages, with stars identifying the best validation value of each metric. Table III (*NN Parameter Counts and Inference FLOPs*) and the following discussion quantify NN computation:

Table III summarizes the NN parameter counts and matrix FLOPs for single-vehicle stateful inference using the current-frame CSI and recurrent states. The three models require 8.51 MB of 32-bit floating-point (FP32) parameter storage. With FP32 inference on an AMD EPYC 7742 CPU using one thread, their combined inference latency has a median of 5.38 ms and a 95th percentile of 5.72 ms.

Location in the Revised Manuscript: Page 6, Section IV-A, stateful inference; Pages 11–12, Section V-B, two-stage training, Fig. 5, Table III, and the discussion of NN computation.

3 Reviewer 2

Overall Assessment and Strengths

The joint treatment of proactive handover, beamforming, and resource allocation is a real and underexplored gap, and using the per-link resource-block demand as the common currency that links prediction, association, and scheduling is clean and well motivated. The evaluation platform is credible, since combining SUMO traces with Sionna RT ray tracing is more realistic than the stochastic channel models most comparable papers use. The algorithmic engineering is careful, with per-module complexity analysis, a Shmoys–Tardos approximation for the assignment step with a stated two-factor capacity bound, and a reasonable ablation and oracle design. Theorem 1 gives a principled starting point for the demand estimate. Overall the work is competent and the writing is mostly clear.

Response from the Authors:

We thank the Reviewer for the careful assessment of our work and the constructive comments. In the following, we respond to each comment individually and explain the corresponding revisions to the manuscript.

Comment 1 (Interference-model consistency and validation)

The interference model is crude and inconsistent with the proposed algorithms. The interference-link gain in Eq. (10) uses a Frobenius-norm average, justified by random transmit and receive beam directions, and the resource-block overlap term in Eq. (16) assumes independent and uniformly random allocation across base stations. Neither holds in the proposed system, because every interfering micro base station runs the same deterministic PET-BF, directing beams at its own vehicles, and the same deterministic OTR-RA, which places resource blocks by efficiency rather than at random. Since the SINR in Eq. (17), and therefore the entire power and latency story, rests on this approximation, please quantify how far these surrogates deviate from directional interference, for example against a beam-aware interference computation from Sionna RT, or justify why the bias is negligible at the reported loads.

Response from the Authors:

We thank the Reviewer for raising this concern. All revised system-level simulations use explicit RB assignments and the actual transmit and receive beams to compute receive rates and update queues. The Frobenius-norm-based gain predictions and expected RB overlaps guide online decisions, but are not used to compute the realized receive rates in the simulator. We have clarified the two averaging operations and quantified their combined effect on receive-rate estimates.

The directional interfering-link gain in Eq. (11) depends on the receive beam of the desired link and the transmit beam of the interfering link. For the complete, normalized DFT codebooks, $\|H\|_F^2/(M_T M_R)$ equals the gain uniformly averaged over all beam pairs. Using this average to represent the gain under the selected beams is an approximation; it does not assume random beam selection by PET-BF. Separately, OTR-RA determines RB counts, whereas each micro BS maps those counts to disjoint RB sets uniformly at random, independently of other micro BSs. Thus, deterministic RB counts are compatible with the expected overlap $k_{m,v}k_{m',v'}/K_\mu$ in Eq. (17), which averages over RB indices. Using the resulting mean interference in the rate expression remains an approximation.

For online OTR-RA decisions, Eq. (46) uses the desired-link gain measured after PET-BF, the predicted

frame-averaged interfering-link gain, and RB usage estimates from GAP-HO; instantaneous interfering-link gains are not required.

To quantify the combined approximation error, we compare the two receive-rate calculations under identical associations, beam pairs, RB counts, and Sionna RT channels along each revised MEET-COBRA run. The diagnostic uses channel-derived beam averages from Eq. (33), rather than NN predictions, to isolate the averaging approximations from prediction errors. Both calculations include beam-search overhead and HO interruption; the approximate rates are recorded only for comparison and do not alter the queues. The comparison covers all 18 arrival rates $\{1, 3, \dots, 35\}$ Mbps, with 30 s per run.

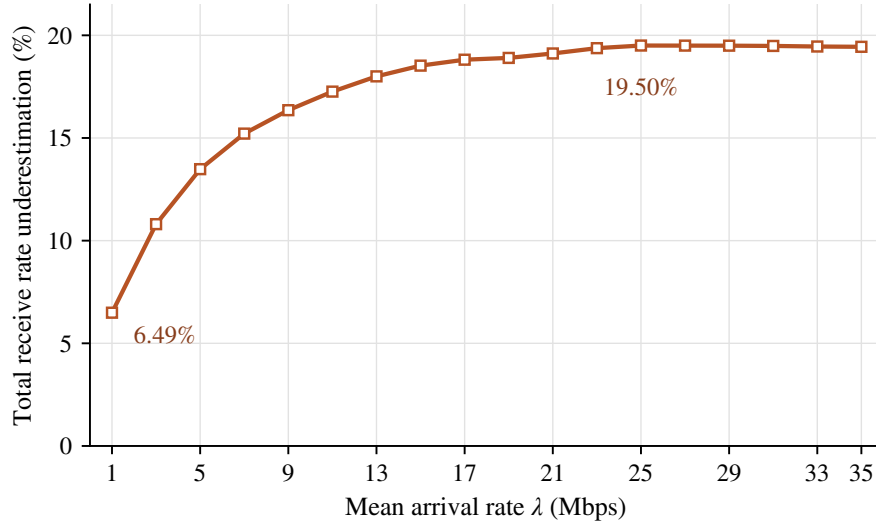


Fig. R1: Underestimation of the total receive rate of vehicles served by micro BSs when using beam-averaged gains and expected RB overlaps, relative to directional per-RB evaluation under identical decisions. Markers show means over three random seeds.

For each run, we sum receive rates over micro-BS users and average over slots. Figure R1 reports the mean underestimation across seeds at each load. The approximations underestimate this total receive rate by 6.5%–19.5% relative to directional per-RB evaluation under identical decisions, providing an empirically conservative aggregate rate estimate in the evaluated scenario.

Changes in the Revised Manuscript:

Sections II-A and II-B define the directional interfering-link gain in Eq. (11) and distinguish RB-count allocation from the random mapping to RB indices in Eqs. (17)–(18). Section IV-A defines the interfering-link gain prediction target as the frame average of the beam-averaged surrogate in Eq. (33).

Sections IV-B and IV-D specify the predicted surrogates used by GAP-HO and OTR-RA. Equation (46) and Algorithm 3 use measured desired-link gains, predicted frame-averaged interfering-link gains, and estimated RB usages. Section IV-E explains that the macro BS forwards the latter two quantities to the serving BSs before the frame begins.

Section V-A clarifies how receive rates are evaluated for all schemes:

For all schemes, receive rates are evaluated using actual RB overlaps and the selected transmit and receive beams.

Section V-D and Figs. 6–9 now report the updated system-level performance comparisons. The Abstract and Conclusion have been updated accordingly.

Location in the Revised Manuscript: Page 4, Section II-A, actual interfering-link gain and Eq. (11); Page 4, Section II-B, expected RB overlap and Eqs. (17)–(18); Page 7, Section IV-A, average gain approximation in Eq. (33); Page 7, Section IV-B, predicted frame-averaged gain surrogates; Page 9, Section IV-D, OTR-RA SINR estimate and Algorithm 3; Page 10, Section IV-E, prediction forwarding to the serving BSs; Page 11, Section V-A, receive-rate evaluation.

Comment 2 (Queue proxy versus delay guarantee)

The latency constraint is a mean-rate queue proxy, not a delay guarantee, yet the results are framed as uRLLC-grade reliability. Section II-C invokes Little’s law to write the per-slot constraint in Eq. (23). Little’s law is a long-run average identity, not a per-slot map from instantaneous backlog to instantaneous delay, so Eq. (23) bounds a scaled queue length rather than actual packet latency, and the metric U in Eq. (24) inherits this. Given the repeated uRLLC framing in the introduction, please either justify this rigorously.

Response from the Authors:

We thank the Reviewer for pointing out this distinction. We agree that the original wording did not adequately distinguish the queue-based metric from actual packet latency. Little’s law relates long-run average queue length and delay; it does not establish a per-slot equivalence between instantaneous backlog and packet delay. Section II-C now explicitly defines $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. Equation (24) constrains this proxy, while U in Eq. (25) averages its threshold violations over active vehicles and time slots. Thus, U is not a directly measured packet-delay tail probability. With the HO interruption model added in response to Comment 6, interrupted slots have zero data-service rate. The resulting backlog is included in the same queueing-delay proxy and violation metric, without changing their definitions. Figure 8 and its caption now explicitly identify L_{90} and L_{99} as percentiles of the queueing-delay proxy.

We have also removed the explicit uRLLC framing from the Introduction. In the Conclusion, *latency reliability* has been replaced with *latency performance* to keep the interpretation consistent with the defined metric.

Changes in the Revised Manuscript:

The interpretation preceding Eq. (24) now reads:

Motivated by Little’s law [27], we define $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. Requiring this proxy not to exceed the latency threshold τ_v^{th} of vehicle- v gives the queue-length-based latency constraint:

The sentence following the definition of $U_v(n)$ now reads:

Specifically, U measures the probability that the queueing-delay proxy $\tau_v(n)$ exceeds its threshold τ_v^{th} .

The corresponding sentence in the Conclusion reads:

Notably, MEET-COBRA effectively balances system transmit power and latency performance.

The explicit uRLLC framing in the Introduction has also been removed. Section V-D now states the percentile calculation, and Fig. 8 separates L_{90} and L_{99} .

Location in the Revised Manuscript: Page 1, opening paragraph of Section I; Pages 4–5, Section II-C, interpretation of Eqs. (24)–(25); Pages 13–15, Section V-D and Fig. 8, percentile calculation and results;

Pages 15–16, Section VI (Conclusion).

Comment 3 (Impact of gain-prediction error)

Gain-prediction error is not propagated into the performance metrics, and 4 dB MAE is large for mmWave link budgeting. Figure 4(b) reports validation MAE around 4.07 dB for the desired-link gain. At mmWave, 4 dB can move a link across the boundary between a usable and unusable beam, and it perturbs the demand estimate in Eq. (38) and hence U . Please add an error-propagation study, or at least plot U against injected gain-prediction noise, to show the accuracy achieved is actually sufficient.

Response from the Authors:

We thank the Reviewer for this comment. We have added an explanation of gain-prediction error propagation in Section IV-E and a sensitivity study in Section V-E. The study quantifies the effects on system transmit power and violation probability U . The revised gain predictors have validation MAEs of 3.11 and 2.51 dB for the desired-link and beam-averaged interfering-link gains, respectively.

The error-propagation discussion in Section IV-E explains the relevant decision paths. With other inputs fixed, overestimating the desired-link gain or underestimating the interfering-link gains lowers the RB demand estimate in Eq. (40), while errors in the opposite direction raise it. These estimates affect the power costs and RB requirements used by GAP-HO to select BS–vehicle associations. Desired-link gain errors also change the early-termination threshold in PET-BF. OTR-RA uses measured desired-link gains, but its RB-efficiency estimates depend on predicted interfering-link gains and estimated BS RB usages.

To quantify these effects, we add independent zero-mean Gaussian noise in dB, with standard deviation $\sigma \in \{0, 1, \dots, 10\}$ dB, to either the desired-link or interfering-link gain predictions. Only one gain type is affected by this noise at a time. The trained models, beam-pair candidates, and algorithm parameters remain fixed. For each traffic load and random seed, the actual channels and traffic arrivals are identical across noise settings. We rerun MEET-COBRA at 9, 19, 29, and 35 Mbps using three paired random seeds, with each run covering 30 s. The case $\sigma = 0$ retains the original NN prediction errors.

Figure 10 shows that U changes little at 9 and 19 Mbps for $\sigma \leq 5$ dB. At 29 Mbps, adding noise with $\sigma = 5$ dB increases U from 0.19% to 0.29% for either gain type. At 35 Mbps, the same noise level raises U from 1.15% to 3.84% and 3.13% for desired-link and interfering-link gain predictions, respectively. Figure 11 also quantifies the power cost. At 19 Mbps, adding interfering-link gain noise with $\sigma = 10$ dB increases power from 33.80 to 40.15 W while U remains approximately 0.24%. At 35 Mbps, power stays close to the full-load level while U rises. These results show that MEET-COBRA maintains low violation probabilities under moderate additional gain-prediction errors at light and medium loads.

Changes in the Revised Manuscript:

Section IV-E explains how gain errors affect RB demand estimates, PHO, beam probing, and resource allocation. Section V-E adds the separate desired-link and interfering-link gain-noise experiments. Figure 10 plots the latency constraint violation probability U , and Fig. 11 plots average system transmit power $\bar{P}$, against the standard deviation of the additional gain-prediction noise. The accompanying discussion compares the sensitivity across loads and relates it to HO interruption, beam-search overhead, and remaining RB resources. The Conclusion summarizes the low violation probabilities maintained at light and medium loads under moderate additional gain-prediction errors.

Location in the Revised Manuscript: Page 10, Section IV-E, final paragraph; Pages 11–12, Section V-B and Fig. 5(b), updated gain-prediction MAEs; Page 15, Section V-E and Figs. 10–11; Pages 15–16,

Section VI (Conclusion).

Comment 4 (Generalization of the learned models)

Generalization of the learned models is untested. The network is trained and validated on 6000 frames from a single map, a fixed four-micro-BS layout, and one traffic regime, with the split taken over non-overlapping time intervals of the same scenario, so all reported accuracy is in-distribution. There is no evaluation under a different topology, base-station placement, or arrival pattern, and the limitation is not acknowledged beyond a vague mention of online adaptation. Please add an out-of-scenario test, or state plainly that the model is site-specific and must be re-trained per deployment.

Response from the Authors:

We thank the Reviewer for highlighting this limitation. We agree that results from a single map and fixed BS layout do not establish generalization to other deployments. The revised NN training and validation sets contain disjoint vehicles from this same scenario. Following the Reviewer's suggested alternative, Section IV-A now explicitly states that deployment in a different propagation environment or BS layout requires retraining the site-specific prediction models using channel data from the target site. Generalization to other deployments has not been evaluated.

Changes in the Revised Manuscript:

Section IV-A adds the retraining requirement after the description of the site-specific model:

Deployment in a different propagation environment or BS layout requires retraining the prediction models using channel data from the target site.

Location in the Revised Manuscript: Pages 5–6, Section IV-A, opening paragraph; Page 11, Section V-B, vehicle-level training and validation split.

Comment 5 (Learning-based or optimization-based comparator)

The authors cite learning-based proactive handover and reinforcement-learning beam tracking, yet none appears as a baseline, and there is no conventional optimization comparator. A 32 percent gain over a deliberately weak baseline is not strong evidence of an advance. Please add at least one learning-based or optimization-based comparator.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added two literature-based baselines for coordinated HO and BF. One combines reinforcement learning with target BS optimization, while the other uses Gale–Shapley matching for BS–vehicle association and performs association and BF updates on different timescales.

O-MAPPO-adapted follows the reinforcement learning and optimization framework of Wang et al. [22]. A shared MAPPO actor evaluates whether each vehicle should trigger HO after every 10 m of travel. The actor and critic are initialized from models pretrained using ground-truth channel information. They are then fine-tuned using predicted channel gains across all evaluated loads to adapt the HO policy to prediction errors. Fine-tuning, validation-based checkpoint selection, and final testing use non-overlapping time intervals of the same scenario. A single checkpoint is selected based on validation performance and used at all test loads. For vehicles with active HO triggers, a mixed-integer optimizer then selects target BSs using estimated RB demands, BS loads, and transmit power costs, with penalties for aggregate demands exceeding BS capacities. The actor and optimizer use the same predicted desired-

link and interfering-link gains as MEET-COBRA. Unlike GAP-HO, it optimizes associations only for vehicles with active HO triggers. Upon HO to a micro BS, hierarchical beam search begins after the HO interruption. It first probes the 16 combinations of eight transmit and two receive sectors, then the 16 combinations of four transmit and four receive DFT beams within the selected sectors, requiring 32 probes in total. In subsequent slots, local beam tracking probes the current beam pair and four adjacent pairs differing in only one beam index, and retains the pair with the highest measured gain.

MTS-GS-HBF-adapted follows the framework of Heydarian et al. [23], which coordinates user association, analog BF, and user selection with digital BF on progressively shorter timescales. In our adaptation, Gale–Shapley matching updates BS–vehicle associations every five frames using the same predicted channel gains as MEET-COBRA. Matching preferences account for estimated rates, BS loads, queue backlogs, and energy and HO costs, while association decisions consider estimated RB demands and BS capacities. We replace the original hybrid BF with the 32-probe hierarchical beam search and five-probe local beam tracking used by O-MAPPO-adapted. Unlike O-MAPPO-adapted, hierarchical beam search is repeated in the first available slot of every frame, after the HO interruption when applicable, followed by local tracking in the remaining slots. This preserves the separation between association and BF timescales in [23].

Each BS applies OTR-RA in every slot in both baselines. Their respective beam-search overheads and the same 10 ms HO interruption are included in the evaluation.

Figs. 6–9 compare all eight schemes at arrival rates of $\{1, 3, \dots, 35\}$ Mbps per vehicle over the same 30 s mobility trace. MEET-COBRA has fewer violations than O-MAPPO-adapted at every evaluated load and lower mean power over 1–23 Mbps. O-MAPPO-adapted uses less power over 25–33 Mbps, but with substantially more violations. At 29 Mbps, MEET-COBRA uses 127.75 W with $U = 0.19\%$, whereas O-MAPPO-adapted uses 107.40 W with $U = 4.18\%$.

Compared with MTS-GS-HBF-adapted, MEET-COBRA has lower mean power at every evaluated load and fewer violations at $\lambda \geq 21$ Mbps. At 29 Mbps, MTS-GS-HBF-adapted uses 135.91 W with $U = 0.77\%$, both higher than the corresponding values for MEET-COBRA. MTS-GS-HBF-adapted has slightly fewer violations at several lower loads. Its largest advantage in mean U occurs at 1 Mbps, with 0.23% compared with 0.25% for MEET-COBRA.

Section V-D also examines the HO and BF mechanisms behind these comparisons. O-MAPPO-adapted retains more traffic in the micro tier at 29 Mbps, with micro-BS RB utilization of 97.2% compared with 79.8% for MEET-COBRA. This reduces the use of higher-power macro-BS RBs but leaves less micro-BS capacity to clear queues. The discussion of MTS-GS-HBF-adapted addresses its less frequent association updates and the data-transmission time consumed by probing. These comparisons extend the evaluation beyond Reactive-OBRA to alternative learning and optimization methods for coordinated HO and BF.

Changes in the Revised Manuscript:

Section V-C introduces both baselines, explaining their core ideas, adaptations, predicted channel inputs, and common use of OTR-RA. The descriptions specify the hierarchical beam search and local beam tracking procedures. Section V-D and Figs. 6–9 compare average system transmit power $\bar{P}$, latency constraint violation probability U , queueing-delay percentiles, and macro-BS association ratios across all evaluated loads. The discussion relates these results to association decisions, RB utilization, and beam-search overhead.

Location in the Revised Manuscript: Page 12, Section V-C, benchmark descriptions; Pages 12–15,

Section V-D and Figs. 6–9, comparisons across all evaluated loads and discussion.

Comment 6 (Inference, control-plane, and handover overhead)

The paper claims lightweight inference overhead but gives no inference latency, FLOPs, or control-plane bit budget. More importantly, the central selling point of proactive handover is reduced failures and interruption, yet handover signaling and interruption latency are deferred to future work and excluded from U. Please add a like-for-like overhead comparison and at least a first-order handover-latency model.

Response from the Authors:

We thank the Reviewer for this comment. Section V-B now quantifies NN computational complexity, inference latency, and the overhead of control-plane reporting. We have also added a first-order HO interruption model and updated the PHO and RA algorithms accordingly.

1) NN computational complexity and inference latency.

Table III reports the NN parameter counts and matrix FLOPs of the beam-pair, desired-link gain, and interfering-link gain predictors. Together, the three models contain 2,127,912 parameters, requiring 8.51 MB of 32-bit floating-point (FP32) parameter storage. Their combined matrix computation is 4.20 MFLOPs per vehicle per frame. This count covers the LSTM and linear-layer matrix operations, with two FLOPs per multiply–accumulate, as specified in the table note.

Online inference processes the current-frame CSI and retains recurrent states across frames, so its per-frame computational cost does not grow with trajectory length. With FP32 inference on an AMD EPYC 7742 CPU using one thread and batch size one, the three models have a combined inference latency of 5.38 ms at the median and 5.72 ms at the 95th percentile, corresponding to 5.38% and 5.72% of the 100 ms frame duration, respectively. Timing includes tensor conversion, the three forward passes, state retention, top-5 index selection, and gain denormalization. To measure NN inference separately, the timed interval starts with preprocessed CSI and ends with the prediction outputs, excluding CSI acquisition, input preprocessing, and report transmission. The reporting overhead is quantified below.

2) Control-plane reporting budget.

For each micro-BS link, the vehicle reports the top- M_P beam-pair indices in descending order of predicted probability and two FP32 gains, one for the desired link and one for the interfering link. The full probability distribution is not uploaded. With $M = 4$, $M_P = 5$, and $M_T M_R = 256$, each index requires 8 bits, giving $4(5 \times 8 + 2 \times 32) = 416$ bits per vehicle per frame, or 4.16 kbps for 100 ms frames.

To compare reporting overhead, we consider two alternative schemes that upload CSI instead of prediction results. At the same reporting interval and 32 bits per real CSI component, uploading all complex MIMO channel matrices requires $2MM_R M_T \times 32 = 65,536$ bits per vehicle per frame (655.36 kbps). Uploading the current-frame superposed-pilot observations for centralized NN inference requires $2M_R N_P \times 32 = 4,096$ bits per vehicle per frame (40.96 kbps) with $N_P = 8$. In this second alternative, the macro BS retains recurrent states, so previous observations need not be uploaded again. The reporting overheads of the two reference schemes are 157.5 and 9.85 times that of prediction reporting, respectively.

3) HO interruption model.

The revised model distinguishes advance HO preparation from HO execution. The signaling exchanges required for advance HO preparation are assumed to be completed before the end of the current frame without consuming data-service resources. When a vehicle changes its serving BS, HO execution at the start of the next frame interrupts data service for $\tau_{HO} = L_{HO}\Delta t$ seconds, where $0 \leq L_{HO} < N_{HO}$. The

control-plane connection to the macro BS and periodic CSI probing remain available throughout this interruption.

For frame-level PHO, the revised GAP-HO accounts for the shorter service period after switching BS by applying the HO-aware correction factor $c_{m,v}^{[x+1]}$ to the estimated RB demand in the RB-capacity constraint of **P2**. This factor is equal to one if the serving BS does not change and $N_{\text{HO}}/(N_{\text{HO}} - L_{\text{HO}})$ if it changes. The correction reflects the additional RBs needed per slot during the remaining $N_{\text{HO}} - L_{\text{HO}}$ slots to serve the data accumulated during HO interruption. For slot-level RA, the revised OTR-RA excludes HO-interrupted vehicles from scheduling and assigns them zero RBs. These zero RB allocations yield zero service in the queue-update equation, while traffic continues to arrive. The resulting backlog therefore affects the queueing-delay proxy $\tau_v(n) = q_v(n)/\lambda_v$ and its violation probability U . For tractable modeling and analysis, we assume successful HO execution and do not separately model HO failure recovery, retransmission delays, or other end-to-end latency components. All system-level comparisons in Figs. 6–9 have now been rerun with this model, using $\tau_{\text{HO}} = 10$ ms for every scheme.

Changes in the Revised Manuscript:

Section IV-A clarifies that recurrent states are retained across frames, so online inference processes only the current CSI feature vector. Section V-B adds Table III (*NN Parameter Counts and Inference FLOPs*) and reports the parameter storage and measured inference latency:

Table III summarizes the NN parameter counts and matrix FLOPs for single-vehicle stateful inference using the current-frame CSI and recurrent states. The three models require 8.51 MB of 32-bit floating-point (FP32) parameter storage. With FP32 inference on an AMD EPYC 7742 CPU using one thread, their combined inference latency has a median of 5.38 ms and a 95th percentile of 5.72 ms.

Section IV-E specifies that only the ranked top- M_{P} beam-pair indices and two gain predictions per micro BS are uploaded. Section V-B quantifies this reporting overhead and compares it with the two alternative CSI-reporting schemes:

With $M = 4$, $M_{\text{P}} = 5$, and $M_{\text{T}}M_{\text{R}} = 256$, each vehicle reports, for each micro-BS link, M_{P} beam-pair indices in descending order of predicted probability, using 8 bits per index, together with one FP32 desired-link gain and one FP32 interfering-link gain. The reporting overhead is $M(8M_{\text{P}} + 2 \times 32) = 416$ bits per vehicle per frame (4.16 kbps). At the same reporting interval and 32-bit precision per real CSI component, this reduces reporting overhead by factors of 157.5 and 9.85 relative to full MIMO-CSI reporting and centralized inference using current-frame superposed-pilot observations, respectively.

Section II introduces the first-order HO interruption model, distinguishing advance HO preparation from the data-service interruption during execution. With $\delta_v^{[x]} = 1$ for a vehicle changing BS and zero otherwise, Eq. (3) defines its slot-level data-service availability as

$$s_v^{[x]}(i) = 1 - \delta_v^{[x]} \mathbf{1}\{i < L_{\text{HO}}\}.$$

Equation (12) and constraint C4 of **P1** now enforce

$$k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} s_v^{[x]}(i) k_{m,v}^{[x]}(i).$$

For GAP-HO, Section IV-B revises the RB-capacity constraint of **P2**, Eq. (42), to

$$\sum_{v \in \mathcal{V}^{[x]}} c_{m,v}^{[x+1]} \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \leq K_m,$$

with the correction factor defined in Eq. (45) as

$$c_{m,v}^{[x+1]} = \left[1 - (1 - u_{m,v}^{[x]}) \frac{L_{\text{HO}}}{N_{\text{HO}}} \right]^{-1}.$$

The paragraph following Eq. (45) explains why vehicles switching BS require more RBs per slot after interruption. Algorithm 1 and its capacity bound are updated consistently.

For OTR-RA, Section IV-D and Algorithm 3 restrict scheduling to connected vehicles without HO interruption. Algorithm 3 initializes all $k_{m,v}^{[x]}(i)$ to zero and forms the eligible vehicle set as

$$\mathcal{V}_m^{[x]}(i) = \{v \in \mathcal{V}^{[x]} \mid u_{m,v}^{[x]} s_v^{[x]}(i) = 1\}.$$

Consequently, interrupted vehicles receive no data service in Eq. (20), while arrivals continue to accumulate through the queue update in Eq. (23). The resulting backlog affects the queueing-delay proxy and U defined in Section II-C. Table I includes the HO-related symbols. Table II specifies the HO interruption duration used in the simulations, $\tau_{\text{HO}} = 10$ ms.

Location in the Revised Manuscript: Page 2, Table I; Page 3, Section II, HO assumptions and Eq. (3); Page 4, Section II-B, Eqs. (12) and (20); Pages 4–5, Section II-C, queue evolution and latency metric; Page 5, Section III, constraint C4 of **P1**; Page 6, Section IV-A, stateful inference; Page 8, Section IV-B, Eqs. (42) and (45), and Algorithm 1; Pages 8–9, Section IV-D and Algorithm 3; Pages 9–10, Section IV-E, prediction reporting; Page 11, Section V-A, Table II, HO interruption duration; Pages 11–12, Section V-B, Table III, inference latency, and reporting budget; Pages 12–15, Section V-D and Figs. 6–9, system results with HO interruption.

Comment 7 (Traffic arrival process and heterogeneous QoS)

The traffic and QoS model is thinner than the motivation promises. The arrival process in Eq. (22) is characterized only by its mean, with no stated distribution, yet for a study whose headline metric is a violation tail, arrival burstiness dominates the result and U is hard to interpret without it. In addition, the introduction motivates heterogeneous QoS across eMBB, uRLLC, and energy, but the model carries a single backlog queue and a single latency class. Please specify the arrival process and either add a second QoS class or narrow the motivation.

Response from the Authors:

We thank the Reviewer for this comment. Section II-C now specifies Poisson arrivals with mean $\lambda_v \Delta t$, independent across vehicles and slots. This provides a reference model of independent random arrivals for comparing coordination schemes under controlled mean traffic loads. This arrival model was already used in the original simulations; the revision makes that existing assumption explicit. We also state that it does not capture temporally correlated traffic bursts. Although the algorithms do not require Poisson arrivals, performance under other arrival distributions has not been evaluated.

Section II-C now defines $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. In Eq. (25), U averages threshold violations of this proxy over active vehicles and time slots. Little's law relates long-run averages and does not imply a per-slot equivalence between this proxy and actual packet delay. Accordingly, U is not a directly measured packet-delay tail probability.

Following the Reviewer's suggestion, we have narrowed the motivation to reducing system transmit power under queue-length-based latency constraints, rather than introducing multiple QoS classes. The revised Introduction removes the explicit uRLLC framing and distinguishes the power objective from

application requirements. Although the formulation allows vehicle-specific thresholds τ_v^{th} , all simulations retain a common 20 ms threshold. Performance under heterogeneous latency requirements has not been evaluated.

Changes in the Revised Manuscript:

The arrival-process description at the end of the first paragraph of Section II-C reads:

We consider Poisson arrivals that are independent across vehicles and slots, with $a_v(n) \sim \text{Poisson}(\lambda_v \Delta t)$. This model does not capture temporally correlated traffic bursts.

The interpretation preceding Eq. (24) now reads:

Motivated by Little's law [27], we define $\tau_v(n) \triangleq \lambda_v^{-1} q_v(n)$ as a queueing-delay proxy. Requiring this proxy not to exceed the latency threshold τ_v^{th} of vehicle- v gives the queue-length-based latency constraint:

The sentence following the definition of $U_v(n)$ now reads:

Specifically, U measures the probability that the queueing-delay proxy $\tau_v(n)$ exceeds its threshold τ_v^{th} .

The latency threshold is now denoted by τ_v^{th} for vehicle- v . Accordingly, Eq. (24) becomes $q_v(n) \leq \lambda_v \tau_v^{\text{th}}$, and the tiered-rounding condition in Eq. (49) becomes $q_v^{[x]}(i) > \lambda_v \tau_v^{\text{th}}/2$. The corresponding notation in the violation indicator, constraint C1 of **P1**, algorithm inputs, tables, and simulation descriptions has been updated consistently.

The traffic configuration added to Section V-A reads:

For the system-level performance comparisons in Section V-D, we vary the traffic arrival rate $\lambda_v = \lambda$ over $\{1, 3, \dots, 35\}$ Mbps and set $\tau_v^{\text{th}} = 20$ ms for all vehicles.

The application-level motivation in the Introduction now reads:

CAV applications demand high throughput for perception and infotainment and low latency for driving safety.

The subsequent phrase ‘‘Such diverse communication quality of service (QoS) requirements’’ has been changed to ‘‘These communication quality of service (QoS) requirements.’’ The system transmit-power objective is discussed separately in the following HetNet paragraph.

Location in the Revised Manuscript: Page 1, Section I (Introduction), opening two paragraphs; Page 2, Table I; Pages 4–5, Section II-C (Traffic Model), arrival-process description and interpretation of Eqs. (24)–(25); Page 5, Section III, constraint C1 of **P1**; Page 9, Section IV-D, Algorithm 3 and Eq. (49); Pages 10–11, Section V-A, traffic configuration and Table II; Pages 12–15, Section V-D, latency-result descriptions.

Comment 8 (OTR-RA acronym in Fig. 2)

Figure 2 labels the resource-allocation block OUR-RA, while the text and Algorithm 3 use OTR-RA. Please fix the acronym.

Response from the Authors:

Thank you for pointing out this typographical error. We have corrected *OUR-RA* to *OTR-RA* in Fig. 2.

Location in the Revised Manuscript: Page 9, Fig. 2, resource-allocation block.

4 Reviewer 3

Summary

This manuscript proposes MEET-COBRA, a framework for jointly coordinating handover, beamforming, and resource allocation in a two-tier vehicular network, using machine learning (LSTM) to predict channel gains and beam pairs ahead of time. The authors argue that solving this joint problem exactly is computationally impractical and instead propose a two-timescale approach: predictions drive the handover decision once per frame, while beamforming and resource allocation adapt every slot based on those predictions and on real-time measurements. The simulation setup combines SUMO mobility traces with Sionna ray-tracing channel data and the evaluation includes both oracle bounds and per-module ablations that give a fairly complete picture of where the reported gains come from. The proposed scheme shows a meaningful reduction in transmit power and latency violations compared to a reactive baseline. I have a number of concerns detailed below that should be addressed in a revision.

Response from the Authors:

We thank the Reviewer for the careful summary of our work and the constructive comments. The responses below explain how we have revised the manuscript to address the specific concerns.

Major Comments

Comment 1 (Relationship to the authors' conference paper)

This work reads as an extension of the authors' own earlier conference paper, which focused on joint handover and resource allocation for a similar vehicular HetNet setup. The framework here appears to build on that by adding beamforming as a third coordinated decision, along with a more detailed channel model and simulation setup. Given this overlap, I would encourage the authors to state clearly, early in the paper, that this is an extended version of that prior work, and to briefly summarize what is specifically new here, since it would give readers a complete picture of what this paper contributes.

Response from the Authors:

We thank the Reviewer for this suggestion. We have added a paragraph in the Introduction that explicitly identifies this paper as an extension of our earlier conference work and summarizes the differences in channel modeling, beamforming optimization, ML prediction targets, and simulation channel data.

Changes in the Revised Manuscript:

The added paragraph reads:

This paper extends our earlier conference work on joint PHO and RA [13]. Unlike the conference version, this paper incorporates MIMO channels, beam-pair selection, and beam-training overhead into the joint optimization. We also change the ML prediction targets from vehicle positions to beam-pair indices and link gains. Moreover, the simulation dataset uses ray-traced microcell MIMO channels instead of the statistical channel model adopted in the conference version.

Location in the Revised Manuscript: Page 2, Section I (Introduction), paragraph beginning *This paper extends*.

Comment 2 (ML computation and reporting overhead versus CSI overhead)

One of the paper's central motivations is reducing the overhead of channel state acquisition, which is a reasonable goal. That said, in the proposed design, each vehicle needs to run three separate ML prediction models locally every frame, one each for beam-pair selection, desired-link gain, and interference-link gain, with all of it reported to the macro base station each frame. This introduces its own computational and reporting overhead, and it isn't clear from the paper how that overhead compares to the CSI overhead the framework is designed to avoid in the first place. A brief discussion of this would strengthen the paper's claim of achieving lower overhead with lightweight ML inference.

Response from the Authors:

We thank the Reviewer for this suggestion. We agree that the computational and reporting costs should be quantified alongside the reduction in CSI acquisition overhead. Section V-B now reports these costs and compares prediction reporting with alternative CSI-reporting schemes.

For local computation, the three predictors together contain 2,127,912 parameters, require 8.51 MB of 32-bit floating-point (FP32) parameter storage, and perform 4.20 MFLOPs of matrix computation per vehicle per frame, as reported in Table III. They retain recurrent states across frames and process only the current-frame CSI, so per-frame computational complexity does not grow with trajectory length. Using FP32 arithmetic, a batch size of one, and a single thread on an AMD EPYC 7742 CPU, the three models have a combined inference latency with a median of 5.38 ms and a 95th percentile of 5.72 ms, compared with the 100 ms frame duration.

For CSI acquisition, the pilot-superposition procedure in Section IV-A uses $N_p = 8$ probing rounds per frame, with all $M = 4$ micro BSs transmitting simultaneously in each round. Under the same receiver observation model, exhaustively sweeping the $M_T = 32$ transmit beams of each BS separately would require $MM_T = 128$ rounds. Thus, pilot superposition reduces the number of frame-level CSI probing rounds by a factor of $MM_T/N_p = 16$.

For reporting, each vehicle sends, for each micro-BS link, the top- M_p beam-pair indices in descending order of predicted probability and two FP32 gains. The full probability distribution is not uploaded. With $M = 4$, $M_p = 5$, and $M_T M_R = 256$, the reporting overhead is $M(8M_p + 2 \times 32) = 416$ bits per vehicle per frame, or 4.16 kbps. For comparison, two alternatives upload CSI instead of predictions. At the same reporting interval and 32 bits per real CSI component, full MIMO-CSI reporting requires $2MM_R M_T \times 32 = 65,536$ bits, whereas current-frame superposed-pilot observations for centralized inference require $2M_R N_p \times 32 = 4,096$ bits per vehicle per frame. In the latter case, the macro BS retains recurrent states, so past observations need not be uploaded again. The reporting overheads of the two reference schemes are 157.5 and 9.85 times that of prediction reporting, respectively.

Overall, MEET-COBRA reduces the overhead of frame-level CSI probing and uplink reporting at the cost of modest local computation per vehicle.

Changes in the Revised Manuscript:

Section IV-A describes pilot superposition and stateful inference. Table III in Section V-B gives the parameter counts and matrix FLOPs of the three predictors, followed by:

Table III summarizes the NN parameter counts and matrix FLOPs for single-vehicle stateful inference

using the current-frame CSI and recurrent states. The three models require 8.51 MB of 32-bit floating-point (FP32) parameter storage. With FP32 inference on an AMD EPYC 7742 CPU using one thread, their combined inference latency has a median of 5.38 ms and a 95th percentile of 5.72 ms.

Section IV-E specifies that only the ranked candidate beam-pair indices and two gain predictions per micro BS are uploaded. Section V-B adds the reporting-budget comparison:

With $M = 4$, $M_P = 5$, and $M_TM_R = 256$, each vehicle reports, for each micro-BS link, M_P beam-pair indices in descending order of predicted probability, using 8 bits per index, together with one FP32 desired-link gain and one FP32 interfering-link gain. The reporting overhead is $M(8M_P + 2 \times 32) = 416$ bits per vehicle per frame (4.16 kbps). At the same reporting interval and 32-bit precision per real CSI component, this reduces reporting overhead by factors of 157.5 and 9.85 relative to full MIMO-CSI reporting and centralized inference using current-frame superposed-pilot observations, respectively.

Location in the Revised Manuscript: Page 6, Section IV-A, Input Layer and Shared Backbone; Pages 9–10, Section IV-E, prediction reporting; Page 11, Table II, pilot and antenna configuration; Pages 11–12, Section V-B, Table III, inference latency, and reporting budget.

Comment 3 (Coordinated handover and beamforming baseline)

The comparison in Section V is against a reactive baseline, and that baseline is fairly weak on all three fronts at once: handover triggered by signal-strength thresholds, beam selection through random probing, and resource allocation under a worst-case interference assumption. The per-module ablation study shown in the paper is a nice complement to this, since it shows what each of the three proposed modules is individually contributing. It would also help to compare against a coordinated handover and beamforming baseline from existing literature.

Response from the Authors:

We thank the Reviewer for this suggestion. We have retained the module ablations and added two literature-based baselines for coordinated HO and BF. One combines reinforcement learning with target BS optimization, while the other uses Gale–Shapley matching for BS–vehicle association and performs association and BF updates on different timescales.

O-MAPPO-adapted follows the reinforcement learning and optimization framework of Wang et al. [22]. A shared MAPPO actor evaluates whether each vehicle should trigger HO after every 10 m of travel. The actor and critic are initialized from models pretrained using ground-truth channel information. They are then fine-tuned using predicted channel gains across all evaluated loads to adapt the HO policy to prediction errors. Fine-tuning, validation-based checkpoint selection, and final testing use non-overlapping time intervals of the same scenario. A single checkpoint is selected based on validation performance and used at all test loads. For vehicles with active HO triggers, a mixed-integer optimizer then selects target BSs using estimated RB demands, BS loads, and transmit power costs, with penalties for aggregate demands exceeding BS capacities. The actor and optimizer use the same predicted desired-link and interfering-link gains as MEET-COBRA. Unlike GAP-HO, it optimizes associations only for vehicles with active HO triggers. Upon HO to a micro BS, hierarchical beam search begins after the HO interruption. It first probes the 16 combinations of eight transmit and two receive sectors, then the 16 combinations of four transmit and four receive DFT beams within the selected sectors, requiring 32 probes in total. In subsequent slots, local beam tracking probes the current beam pair and four adjacent pairs differing in only one beam index, retaining the pair with the highest measured gain.

MTS-GS-HBF-adapted follows the framework of Heydarian et al. [23], which coordinates user association, analog BF, and user selection with digital BF on progressively shorter timescales. In our

adaptation, Gale–Shapley matching updates BS–vehicle associations every five frames using the same predicted channel gains as MEET-COBRA. Matching preferences account for estimated rates, BS loads, queue backlogs, and energy and HO costs, while association decisions consider estimated RB demands and BS capacities. We replace the original hybrid BF with the 32-probe hierarchical beam search and five-probe local beam tracking used by O-MAPPO-adapted, without using predicted beam-pair candidates or PET-BF. Unlike O-MAPPO-adapted, hierarchical beam search is repeated in the first available slot of every frame, after the HO interruption when applicable, followed by local tracking in the remaining slots. This preserves the separation between association and BF timescales in [23].

Each BS applies OTR-RA in every slot in both baselines. Their respective beam-search overheads and the same 10 ms HO interruption are included in the evaluation.

Figs. 6–9 compare all eight schemes at arrival rates of $\{1, 3, \dots, 35\}$ Mbps per vehicle over the same 30 s mobility trace. Across the evaluated range, MEET-COBRA has fewer violations than O-MAPPO-adapted at every load and lower mean power over 1–23 Mbps. O-MAPPO-adapted uses less power over 25–33 Mbps, but with substantially more violations. Compared with MTS-GS-HBF-adapted, MEET-COBRA has lower mean power at every load and fewer violations at $\lambda \geq 21$ Mbps. MTS-GS-HBF-adapted has slightly fewer violations at several lower loads. Its largest advantage in mean U occurs at 1 Mbps, with 0.23% compared with 0.25% for MEET-COBRA.

Section V-D also examines the HO and BF mechanisms behind these comparisons. O-MAPPO-adapted retains more traffic in the micro tier at 29 Mbps, with micro-BS RB utilization of 97.2% compared with 79.8% for MEET-COBRA. This reduces the use of higher-power macro-BS RBs but leaves less micro-BS capacity to clear queues. The discussion of MTS-GS-HBF-adapted addresses its less frequent association updates and the data-transmission time consumed by probing. The added comparisons therefore evaluate alternative HO and BF coordination methods beyond Reactive-OBRA.

Changes in the Revised Manuscript:

Section V-C introduces both baselines, explaining their core ideas, adaptations, predicted channel inputs, and common use of OTR-RA. The descriptions specify the hierarchical beam search and local beam tracking procedures. Section V-D and Figs. 6–9 compare average system transmit power $\bar{P}$, latency constraint violation probability U , queueing-delay percentiles, and macro-BS association ratios across all evaluated loads. The discussion relates these results to association decisions, RB utilization, and beam-search overhead.

Location in the Revised Manuscript: Page 12, Section V-C, O-MAPPO-adapted and MTS-GS-HBF-adapted; Pages 12–15, Section V-D and Figs. 6–9, multi-seed performance comparisons.

Comment 4 (Vehicle-speed distribution and mobility sensitivity)

Despite high-mobility vehicular scenarios being the paper’s central motivation, it does not report any information on vehicle speed anywhere in the manuscript. Speed is arguably one of the main factors driving how quickly channel conditions change from one frame to the next, so it directly affects how difficult the prediction problem actually is. The authors do state that training and validation samples are drawn from non-overlapping time intervals of the same simulated traffic, but the vehicle speed distribution within that dataset is never reported. It would help to see the speed distribution used, and a result showing how performance of the proposed framework changes across different mobility levels.

Response from the Authors:

We thank the Reviewer for this suggestion. We agree that vehicle speeds are needed to interpret both prediction accuracy and system performance. We have added the speed distributions of the training, validation, and system evaluation datasets, together with prediction and latency results for four speed groups.

1) Vehicle-speed distributions.

Figure 4 in Section V-A now reports the speed CDFs, reproduced below in Fig. R2. The statistics are calculated over vehicle-frame observations, with stationary samples defined by zero speed.

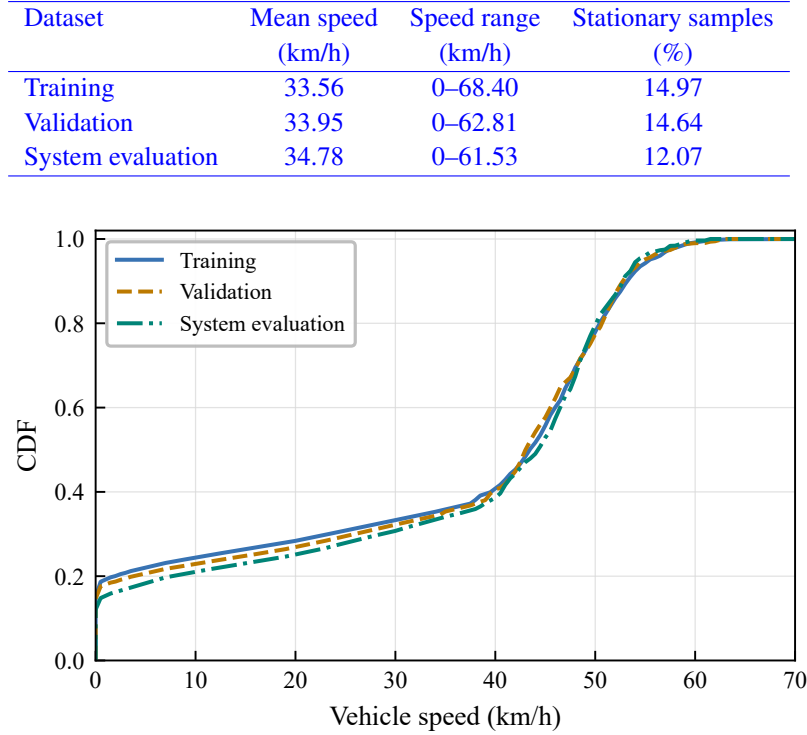


Fig. R2: Vehicle-speed CDFs over vehicle-frame samples from the training, validation, and system evaluation datasets.

2) Prediction performance across speed groups.

We use four predefined speed groups of $[0, 1)$, $[1, 20)$, $[20, 40)$, and ≥ 40 km/h. The first group includes stationary and nearly stationary vehicles. The same selected checkpoints are used for all groups, with recurrent states carried along each complete vehicle trajectory. Predictions are grouped afterward by vehicle speed in the target frame. The following results use the prediction outputs from the system evaluation dataset, without additional training. Metrics are averaged over the four micro-BS links and all samples in each group. The gain metrics refer to the optimal desired-link gain and the beam-averaged interfering-link gain.

Speed group (km/h)	Beam accuracy (%)		Gain MAE (dB)	
	Top-1	Top-5	Desired	Interfering
$[0, 1)$	96.06	99.74	1.13	1.19
$[1, 20)$	87.92	99.42	2.31	1.79
$[20, 40)$	87.23	99.18	3.15	2.32
≥ 40	87.61	99.57	3.73	3.07

Both gain MAEs increase across the speed groups. Top-1 beam accuracy is lower in the three higher-speed groups than in the $[0, 1)$ km/h group, but top-5 accuracy remains above 99% in every group. Thus, the list of five candidate beam pairs retains high coverage of the optimal beam pair despite the larger gain errors in the higher-speed groups. Section V-B now reports the four pairs of gain MAEs and the top-5 accuracy result.

3) System latency performance across speed groups.

We also analyzed MEET-COBRA and Oracle-MC at all 18 arrival rates $\{1, 3, \dots, 35\}$ Mbps with three random seeds over the same 30 s urban mobility trace. Grouping is applied after simulation, preserving the original trajectories and queue histories. Within each run and speed group, L_{99} is the 99th percentile of the vehicle-slot samples of the queueing-delay proxy $\tau_v(n)$. The reported values are averaged over the three runs.

For MEET-COBRA, the mean L_{99} remains below 20 ms in all four groups at every evaluated arrival rate up to 33 Mbps. At 35 Mbps, the four values are 1.49, 11.19, 28.37, and 48.02 ms in increasing speed-group order. Section V-D now reports these results, showing both the latency performance across the observed speed range and the larger delays at the highest load. In the ≥ 40 km/h group at this load, Oracle-MC also has $L_{99} = 26.61$ ms, so the observed latency violations cannot be attributed entirely to prediction errors.

Changes in the Revised Manuscript:

Section V-A adds Fig. 4 and reports the mean speed and stationary sample fraction for each dataset, together with the speed range of the system evaluation dataset. Section V-B adds the four speed groups and their gain MAEs, noting that top-5 beam prediction accuracy exceeds 99% in every group.

Section V-D adds the following analysis to the discussion of latency percentiles:

When evaluated separately for the speed groups $[0, 1)$, $[1, 20)$, $[20, 40)$, and ≥ 40 km/h, MEET-COBRA maintains a mean L_{99} below 20 ms in every group at all evaluated arrival rates up to 33 Mbps. At 35 Mbps, the corresponding values are 1.49, 11.19, 28.37, and 48.02 ms in increasing speed-group order. These results indicate that MEET-COBRA maintains low latency across the observed speed groups over most evaluated loads, while larger delays arise in the higher-speed groups at the highest load.

Location in the Revised Manuscript: Pages 10–11, Section V-A and Fig. 4, speed distributions; Page 11, Section V-B, prediction performance by speed group; Page 14, Section V-D, latency performance by speed group.

Minor Comments

Comment 5 (OTR-RA acronym in Fig. 2)

Fig. 2 labels the resource allocation block “OUR-RA,” while the surrounding text and Algorithm 3 consistently use “OTR-RA.”

Response from the Authors:

Thank you for pointing out this typographical error. We have corrected *OUR-RA* to *OTR-RA* in Fig. 2.

Location in the Revised Manuscript: Page 9, Fig. 2, resource-allocation block.

Comment 6 (Handover-algorithm parameters in the simulation table)

A few parameters used in the handover algorithm aren't listed in the main parameter table: N_{iter} (the maximum number of iterations in the fixed-point refinement step), ζ (the factor used to relax the capacity constraint when the relaxation is infeasible), and ε (the convergence tolerance for that iteration) in Algorithm 1. Worth adding for reproducibility.

Response from the Authors:

We thank the Reviewer for pointing out the missing parameter settings. Table II now lists $N_{\text{iter}} = 2$ and $\zeta = 1.1$, and Section V-A explains their use. Algorithm 1 now performs exactly N_{iter} rounds without a convergence test, so ε has been removed.

Changes in the Revised Manuscript:

Table II adds the number of GAP-HO iterations $N_{\text{iter}} = 2$ and the RB capacity relaxation factor $\zeta = 1.1$. Section IV-B and Algorithm 1 specify N_{iter} rounds and remove the convergence test and its parameter ε . Section V-A explains the parameter choices and states that Table II gives the default simulation settings.

Location in the Revised Manuscript: Pages 7–8, Section IV-B and Algorithm 1; Page 11, Section V-A, Table II and the discussion of parameter settings.

Comment 7 (Related work on joint optimization)

The related work section covers handover, beamforming, and resource allocation as separate lines of work, even though the paper's central claim rests on coordinating them. It would strengthen the positioning of the contribution to also discuss prior work that combines two or more of these problems.

Response from the Authors:

We thank the Reviewer for this helpful suggestion. We have added a paragraph to the Introduction reviewing three studies that coordinate HO with beam tracking, HO with beam selection and bandwidth allocation, and user association with hybrid BF and queue-aware scheduling, respectively. The discussion acknowledges these existing coordination approaches and distinguishes our objective of reducing system transmit power through joint PHO, BF, and RA while meeting latency requirements. To support this positioning, we have also clarified the energy-saving motivation in the preceding HetNet discussion by explaining the balance between transmit power and timely traffic service.

Changes in the Revised Manuscript:

The added discussion of joint optimization reads:

Several studies also coordinate two or more of these decisions. Khosravi et al. [21] use deep Q-learning to coordinate HO and beam tracking, accounting for beam-training overhead in the effective throughput. Wang et al. [22] combine multi-agent reinforcement learning for HO triggering with optimization of target BS and beam selection and bandwidth allocation to improve throughput and delay performance. Heydarian et al. [23] coordinate user association, hybrid BF, and queue-aware user scheduling on multiple timescales. However, these studies primarily target throughput and delay performance, and do not explicitly address the joint coordination of PHO, BF, and RA for reducing the system transmit power while meeting latency requirements.

The energy-saving motivation earlier in the Introduction now reads:

Such HetNet deployments can also reduce system transmit power through traffic-aware resource allocation. Realizing these benefits requires balancing transmit power against timely traffic service. Increasing resource usage can reduce delay but also raise power consumption [6].

The three newly added references are listed below, using their numbers in the revised manuscript:

- [21] S. Khosravi, H. S. Ghadikolaie, J. Zander, and M. Petrova, “Reinforcement Learning-based Joint Handover and Beam Tracking in Millimeter-wave Networks,” in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, 2023, pp. 1–6.
- [22] R. Wang, Y. Sun, C. Zhang, B. Yang, M. Imran, and L. Zhang, “A novel handover scheme for millimeter wave network: An approach of integrating reinforcement learning and optimization,” *Digital Commun. Netw.*, vol. 10, no. 5, pp. 1493–1502, 2024.
- [23] M. Heydarian, D. Colle, M. Pickavet, and W. Tavernier, “Multi-timescale scheme for cooperative user association and hybrid beamforming in mmWave MIMO systems,” *EURASIP J. Wireless Commun. Netw.*, vol. 2025, Art. no. 55, 2025.

Location in the Revised Manuscript: Page 1, Section I (Introduction), paragraph beginning *Several studies also coordinate*; Page 1, HetNet paragraph beginning *In practice*; references [21]–[23].

Yours sincerely,
Bowen Xie, Sheng Zhou, and Zhisheng Niu